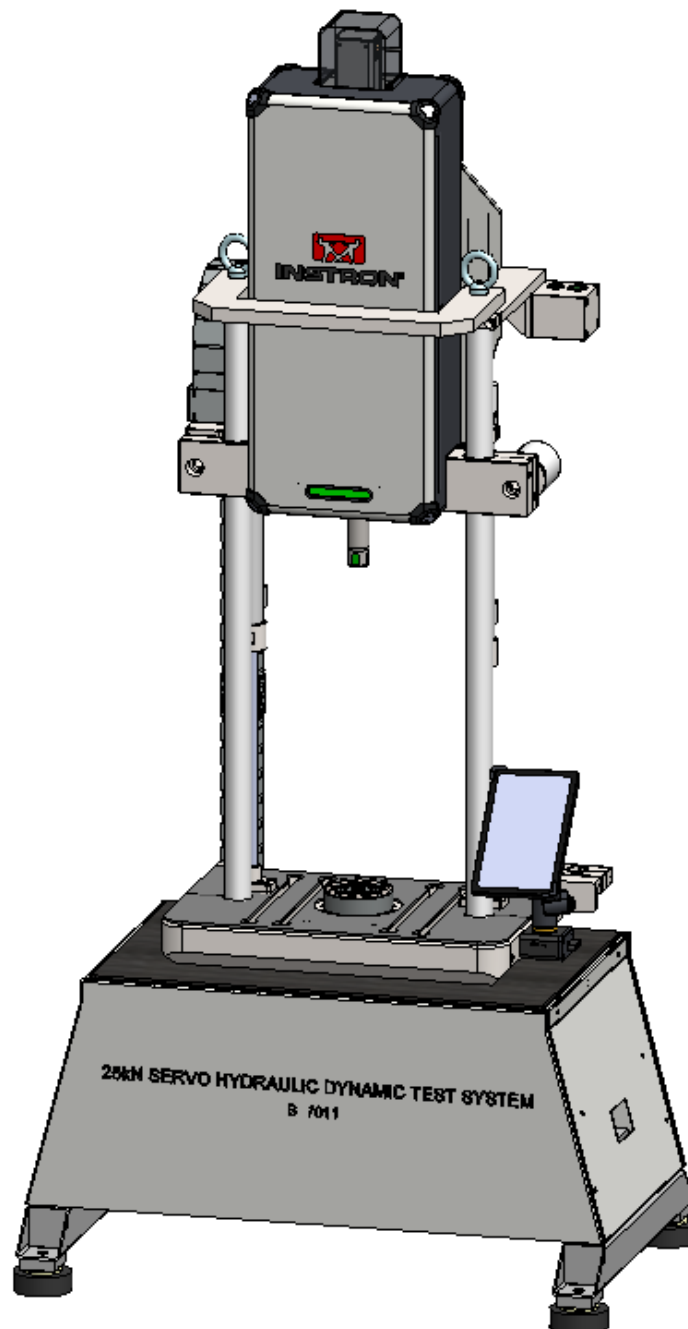


Operation Instruction Manual



115162 23100313





INSTRON®

MACHINE: 25 kN DYNAMIC RATED SERVO

HYDRAULIC FATIGUE TESTING MACHINE

CUSTOMER: IIT JAMMU, J & K

MANUAL COPY No: ITW/1123/UTM/40

Operation Instruction Manual



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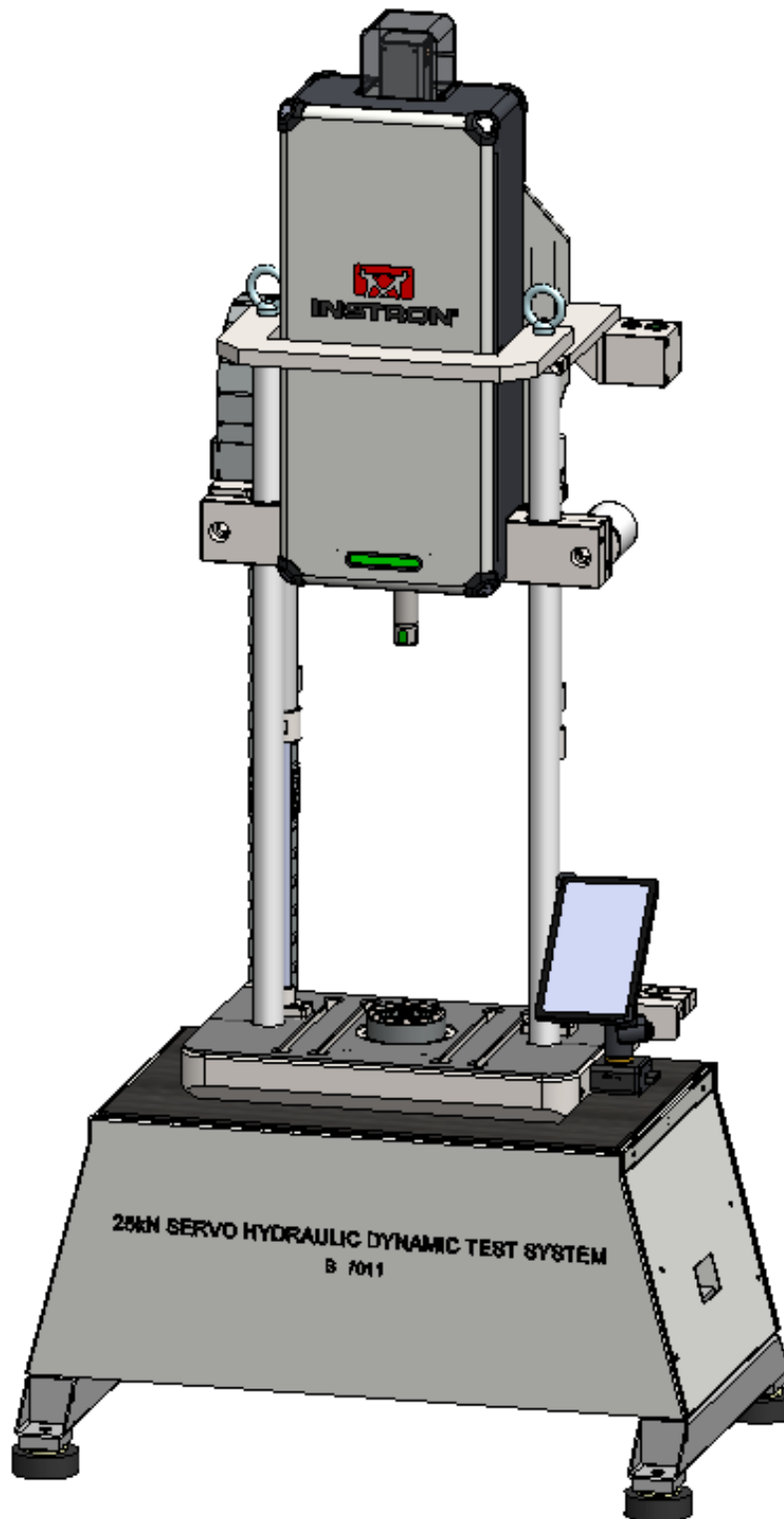
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sales.india@instron.com for Sales information. Sale, etc

Visit www.instron.com – for Product information

ITW India Private Limited systems are engineered and constructed with care in every phase of design and assembly. To ensure reliability, mechanical and electrical components have been selected from the best components and materials. It has been proven that a thorough understanding of the system, and especially the system software, will pay big dividends. You as our customer are our most valued asset. We take pride in our systems and are proud you have become an owner. We welcome your comments about our products and wish that you express them. It is the only way that we can continue to build the best available test systems to satisfy your needs. Thank you for your support.

Operation Instruction Manual



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1 General Safety Precautions



Testing systems are potentially hazardous.



Materials testing involves inherent hazards ranging from high forces, rapid motions, to stored energy. You must be aware of all moving and operating components that are potentially hazardous, particularly the actuator in a servo hydraulic testing system or the moving crosshead in a testing system.

Carefully read all relevant manuals and observe any and all Warnings and Cautions. The term Warning is used where a hazard may lead to an injury or even death. The term Caution is used where a hazard may lead to potential equipment damage and severe loss of valuable data.

ITW India Private Limited (BISS Division) products comply with a wide range of national and international safety standards, applicable to materials and structural testing. Our products are intrinsically designed to the strict ITW India Private Limited (BISS Division) Safety Standard. (available on request)

Due to the wide range of practical applications for our appliances' additional safety devices and necessary standard operating procedures are necessary to prevent unforeseen circumstances/accidents and upholding the safety norms.

We strongly advise using additional safety devices such as protective shielding, warning signs or methods of restricting equipment access, which are readily made available on request.

The following pages provide a detailed guide to various general warnings that you must always heed while using testing equipment. You will find more specific/niche Warnings and Cautions clearly mentioned whenever a potential hazard exists.

In order to adhere to the safety guidelines, it is mandatory for you to gain a thorough understanding of the equipment by reading instruction manuals and to always use good judgment.

We strongly recommend that you carry out your own product safety risk assessment.

Warnings



Hazard – Always ensure to press the Emergency Stop button whenever you consider that an environment is potentially unsafe

The Emergency Stop button shuts off hydraulic power or de-energizes electrical drive and avoids any unsafe actions. It does not isolate the system from electrical power, other means are provided to disconnect the electrical supply. Whenever you consider that safety is being compromised, stop the test using the Emergency Stop button. Investigate and resolve the situation yourself or contact an ITW India Private Limited (BISS Division) personnel before switching ON the machine.



Flying Debris Hazard - Make sure that the test specimens are installed correctly in grips or fixtures in order to eliminate stresses that can lead to breakage of grip jaws or fixture components.

Incorrect mounting of test specimens creates stresses in grip jaws or fixture components that can ultimately result in the breakage of these components. The high energies involved can cause the broken parts to be projected beyond the allocated test area. Ensure that you install specimens in the center of the grip jaws in line with the load path. Insert specimens into the jaws by at least the amount recommended in your grip documentation. This amount can vary between 66% to 100% insertion depths. (refer to supplied instructions for your specific grips) so we recommend you use the centering aid and alignment devices provided.



Hazard - Protect electrical cables from damage and inadvertent disconnection.

The loss of controlling and feedback signals due to a potentially disconnected or damaged cable causes an open loop condition that may drive the actuator or crosshead rapidly to its extremes positions. Protect all electrical cables, particularly transducer cables, from damage. Never route cables across the floor without protection or suspend cables overhead under excessive strain. Use protective padding to avoid inter cable frictions especially around corners or through wall openings.

Warnings



High/Low Temperature Hazard - Wear protective clothing when handling equipment under extreme temperatures.



Materials testing is often carried out at non-ambient temperatures using ovens, furnaces or cryogenic chambers. Extreme temperature implies an operating temperature exceeding 60 °C (140 °F) or below 0 °C (32 °F). You must use protective clothing, such as gloves, when handling equipment at these temperatures. Display a warning notice concerning low or high temperature operation whenever temperature control equipment is in use. You should note that the hazard from extreme temperature can extend beyond the immediate test area.



Crush Hazard - Take care when mounting or removing a specimen, assembly, structure, or load string component.

Mounting or removal of a specimen, assembly, structure, or load string component involves working inside the impact zone between the grips or fixtures. When working in this area, ensure that no other personnel can remotely access or operate any of the system controls. Always keep clear of the jaws of a grip or fixture. Keep clear of the hazard area between the grips or fixtures during actuator or crosshead movement. Ensure that all actuator or crosshead movements necessary for mounting or removal are slow and, at a low force setting whenever possible.



Hazard - Never place a testing system in off-line mode in the computer control. Ensure that no actuator or crosshead movement will occur upon switching over to manual control.

The actuator or crosshead will immediately respond to manual control settings when the system is placed off-line from computer control. Before transferring to manual control, make sure that the control settings are appropriately optimized to prevent any unexpected movements of the actuator or the crosshead.

Warnings



Hazard - Set appropriate limits before performing loop tuning or running waveforms or tests.

Operational limits are included within the testing system to suspend motion or shut off the system when upper and/or lower bounds of actuator or crosshead travel, or force or strain, are reached during testing. Correct setting of operational limits by the operator, prior to testing, will drastically reduce the risk of damage to the test article, system and consequently to the operator



Electrical Hazard - Disconnect the electrical power supply before removing the covers to electrical equipment.

Disconnect equipment from the electrical power supply before removing any electrical safety covers or replacing fuses. Do not reconnect the power source while the covers are removed. Refit covers immediately.



Rotating Machinery Hazard - Disconnect power supplies before removing the covers to rotating machinery.

Disconnect equipment from all power supplies before removing any cover which gives access to rotating machinery. Do not reconnect any power supply while the covers are removed unless you are specifically instructed to do so in the manual. If the equipment needs to be operated to perform

maintenance tasks with the covers removed, ensure that all loose clothing, long hair, etc. is tied back neatly and refit covers immediately.



Hazard - Shut down the hydraulic power supply and discharge hydraulic pressure before disconnection of any hydraulic fluid coupling.

Do not disconnect any hydraulic coupling without first shutting down the hydraulic power supply and discharging stored pressure to zero. Tie down or otherwise secure all pressurized hoses to prevent movement during system operation and to prevent the hose from whipping about in the event of a rupture.



Hazard - Shut off the supply of compressed gas and discharge residual gas pressure before you disconnect any compressed gas coupling.

Do not release gas connections without first disconnecting the gas supply and discharging any residual pressure to zero

Warnings



Explosion Hazard - Wear eye protection and use protective shields or screens whenever there is a possibility of a potential hazard from the failure of a specimen, assembly or structure under test.



Wear eye protection and use protective shields or screens whenever there is a risk of injury to operators and observers due to the failure of a test specimen, assembly or structure, especially where explosive disintegration may occur. Due to the wide range of specimen materials, assemblies or structures that may be tested, any hazard resulting from the failure of a test specimen, assembly or structure must be assessed on a priority basis and suitable precautionary procedures have to be taken.



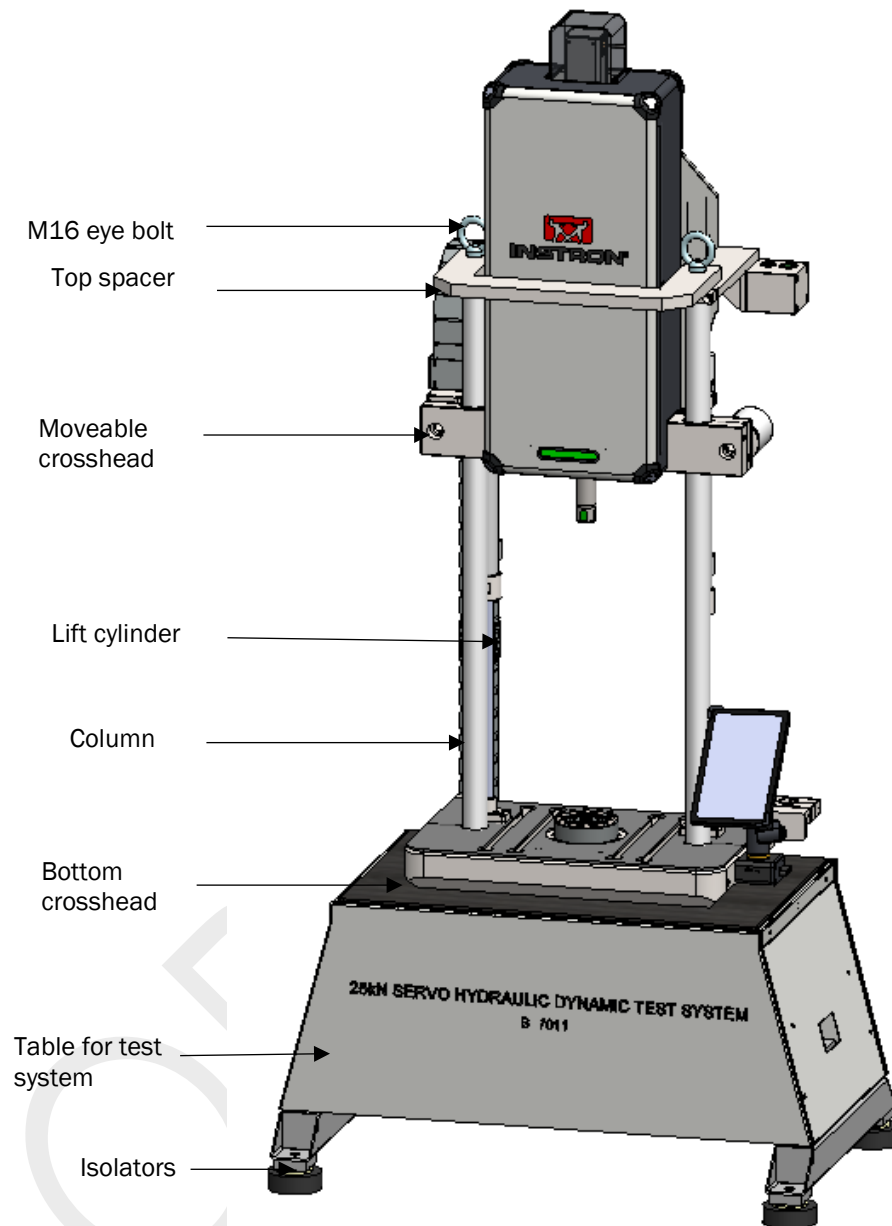
Hazard - Ensure components of the load string are correctly pre-loaded to minimize the risk of fatigue failure.

Dynamic systems, especially where load reversals through zero are occurring, are at a risk of fatigue cracks developing if components of the load string are not correctly pre-loaded to one another. Apply the specified torque to all load string fasteners and the correct setting to wedge washers or spiral washers. Visually inspect highly stressed components such as grips and threaded adapters prior to every fatigue test for signs of wear or fatigue damage.

2 System Components & Identification

SI No	Model No	Description
1	AC-01-0025	25kN Loadframe
2	AC-02-0025	25 kN Servo-hydraulic Actuator
3	AC-06-0035	25kN Load Cell
4	AC-02-0005	NG6 Actuator Control Manifold
5	AC-02-0145	4 LPM Servo Valve
6	AC-02-0405	Actuator Hold
7	AC-02-0505	0.5 L Accumulator
8	AC-02-E060	60 mm Digital Linear Encoder
9	AC-03-0004	4 LPM Hydraulic Power Pack
10	AC-05-8025	25 kN Universal Mechanical Grip
11	8025-LT-X	Adaptor for Threaded Specimen
12	AC-05-0XFX	Clevis grips
13	8025-LW-X	Adaptor for Flat and Round Specimens
14	8025-C-1	Compression plate - Dia 60mm
15	8025-3-1C	3-point bend test fixture
16	AC-07-1025	G.L: 12.5 mm T.L: 6.25/-3.1 mm Room Temperature Extensometer
17	AC-07-0015	G.L: 10 mm T.L: 4/-1 mm Crack Opening Displacement Gage
18	AC-04-2370-10	Single Station Digital Servo Controller
19	AC-08-0020	Test Builder Software
20	AC-08-0001	Tensile and Compression Testing Software (As per ASTM E8 and ASTM E9)
21	AC-08-0008	Low Cycle Fatigue Testing Software (As per ASTM E606)
22	AC-08-0004	VAFCP Application Software

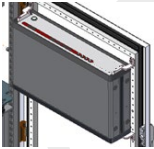
2.1 System control Components



3 System Control Components

Refer to the table below to identify control components of the system.

In order to ensure optimum system operation, it is important to be able to identify and understand the basic functions of these components.

Component	Description
Emergency stop 	Press this button to momentarily stop the actuator and all other associated controls. This action results in an immediate halt of all frame motions, disabling it further instructions.
Main power disconnect switch 	Provides main power control to the system. When the switch is off, all power to the load frame, Controller and HPS are shut off. Power to the computer system is unaffected as they have their own power supply.
Pump start 	Press this button to start the HPS.
Controllers 	The 2020 Series control unit houses control components that: receive and process data from various system transducers, communicate with the system's controlling software, and provide feedback to the system's servo valve to operate the frame as set up in the controlling software.
PLI control panel 	The controls on this panel allow the operator to manually adjust the actuator position, to start and stop tests and other testing controls/functions.

4 System Safety & Information Labelling

Label	Definition and Purpose
	Crush hazard - Indicates that a pinching or crush hazard exists from two objects coming together and instructs the user to read and understand the operator's manual prior to using the machine.
	Electrical hazard - Indicates that a hazard exists from high voltage or electrical current.
	High pressure hazard - Indicates that a hazard exists from high pressure. Do not adjust or reset any pressure settings until you have thoroughly read and understood the operator's manual as this can result in mass equipment damage. Pressure settings should only be adjusted by an Instron service engineer.
	Eye protection - Indicates that a flying debris hazard exists either from specimen failure or improper usage of system components. Wear eye protection and use protective shields or screens whenever appropriate. Be sure to read and understand the operator's manual before using the system.
	Read the manual - Read and understand the operator's manual before using the system.
	General hazard - Indicates a general hazard. Be sure to read and understand the operator's manual before you start using the system. Only authorized personnel should service the equipment.
	Ground stud - Indicates a ground stud. Connect to an appropriate ground/earth system.
	Protective earth - Indicates the protective earth terminal for the main power supply.

5 Product Documentation Guidelines

Installation Manual	Information about preparing your site for eventual system installation, system retrieval and system relocation.
Operating Instructions	How to use your system components and controls, and other frequently performed operating tasks.
Machine Manual	How to properly set up and use any accessories you have purchased, for example grips, fixtures, extensometers, transducers, hydraulic power units, actuators, and environmental chambers (if supplied).
Online Help	Software products come complete with context sensitive help, which provides detailed information on how to use all the software features.

6 System Usage Limits

Intended use and foreseeable misuse:	When equipped with appropriate fixtures, the system is intended to conduct static tension, compression, bend, flexure and shear tests on a variety of specimen shapes and material types. The material types must be limited to those disintegrated into no more than two pieces that will be retained in the fixtures or whose failure does not result in hazardous projectiles. The system can operate in servo control mode only through an integrated ITW India Pvt Ltd (BISS Division) Materials Testing Software package. The system must never be left unattended while in operation. The system should not be used for any purpose other than static testing of the material types specified above. It must not be used to test specimens of material types that result in hazardous projectiles unless an interlocked guard with an appropriate performance level is added. You must conduct your own risk assessment and take appropriate measures to prevent operator injury and guard the equipment. ITW India Pvt Ltd (BISS Division) offers safety rated guards for this purpose. Please contact your local ITW India Pvt Ltd (BISS Division) representative for more information. The system must not be used as a press. The system should not be operated by anyone who does not fit the criteria specified in the “Operator characteristics:” and “Operator training/experience:” sections of this table.
Intervention procedures:	The system is equipped with Emergency Stop controls
Use type:	Educational & Industrial
Operator characteristics:	Typical adult in good health with no disabilities that prevent safe operation of the testing system.
Operator training/experience:	Operators should be familiar with the operation of materials testing systems in general and with the inherent hazards of such testing. Operators should gain a thorough understanding of this equipment by reading these instructions and all other relevant documents provided. Initial basic safety and operational training is provided by ITW India Pvt Ltd (BISS Division) Service personnel during installation of the system. Maintenance personnel or technicians should have sufficient training and skills in order to safely perform all the procedures. Trainees and apprentices should only operate or maintain the system under direct supervision of qualified operator or maintenance personnel. The general public should not have access to the system.
Exposure of other persons to hazards:	Exposure to hazards is greatly reduced by the knowledge of qualified operators. Anyone who does not fit the criteria specified in the “ Operator characteristics: ” and “ Operator training/experience: ” sections of this table should not have access to the system.

7 HPS Standard Operating Procedure (SOP)

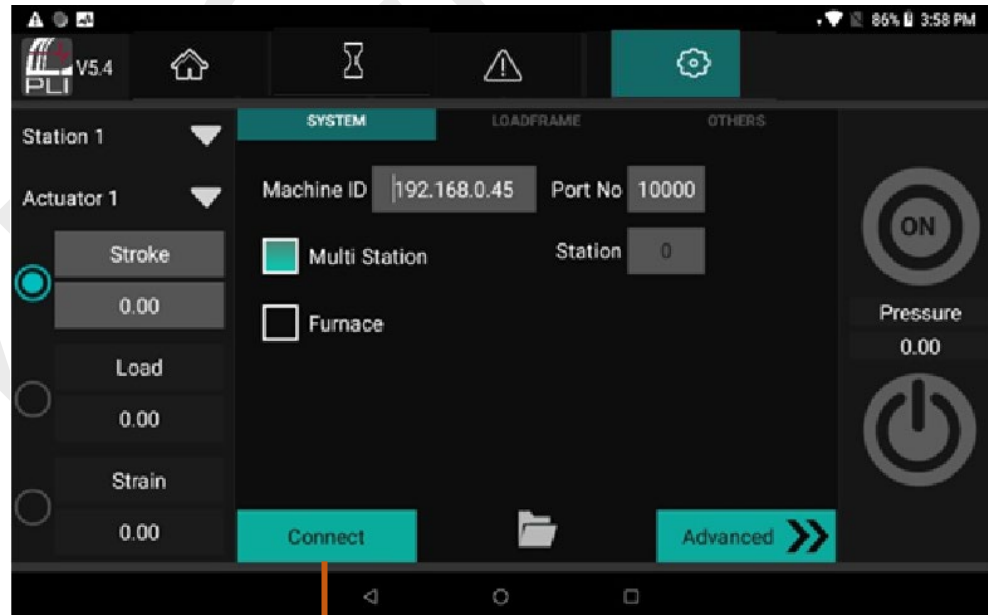
HPS is operated entirely by the PLI Tab, and by functions within the controlling software:

- PLI Tab
- MTL32 basic & application software

8 PLI Tab Operating Guidelines

Settings

To select a required language go to Settings→Others→Language



In PLI settings tab, verify Machine ID and port No, press connect button to establish communication between MTL32 and PLI (First time). Once Successfully connected, every time PLI application launches, it will try to connect with the same machine. connect button need not to press every time.

Home Page

The screenshot shows the PLI V5.4 Home Page interface. The top bar includes a home icon, a timer icon, a warning icon, and a settings icon. The main display area shows 'Station 1' selected, with a dropdown for 'Actuator' and a 'Displacement' of 0.000 mm. Below this, 'Load' is 0.044 kN and 'Extensometer' is 2.347 mm. The status 'Connected' is displayed. On the right, there are two large buttons: a yellow 'LO' button and a red power button. At the bottom, 'Elapsed Time: 00:01:43' and 'Machine Usage: 1496:09:17' are shown.

Select required station (Visible only if multi station option selected)

Select required Actuator

Select required control mode

Press On button to go to low pressure and after few seconds, press again to go to high pressure

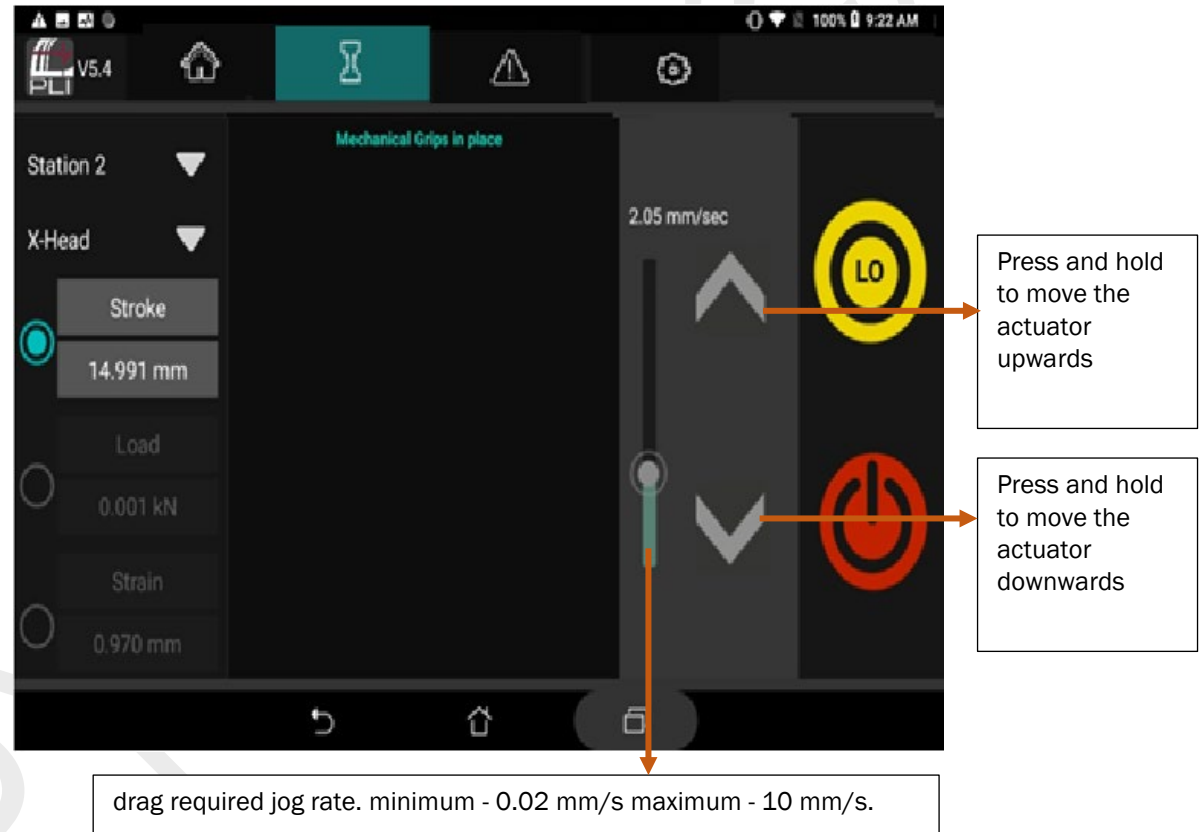
Press this button to switch off the drive

PLI application running time. This time will reset for every launch.

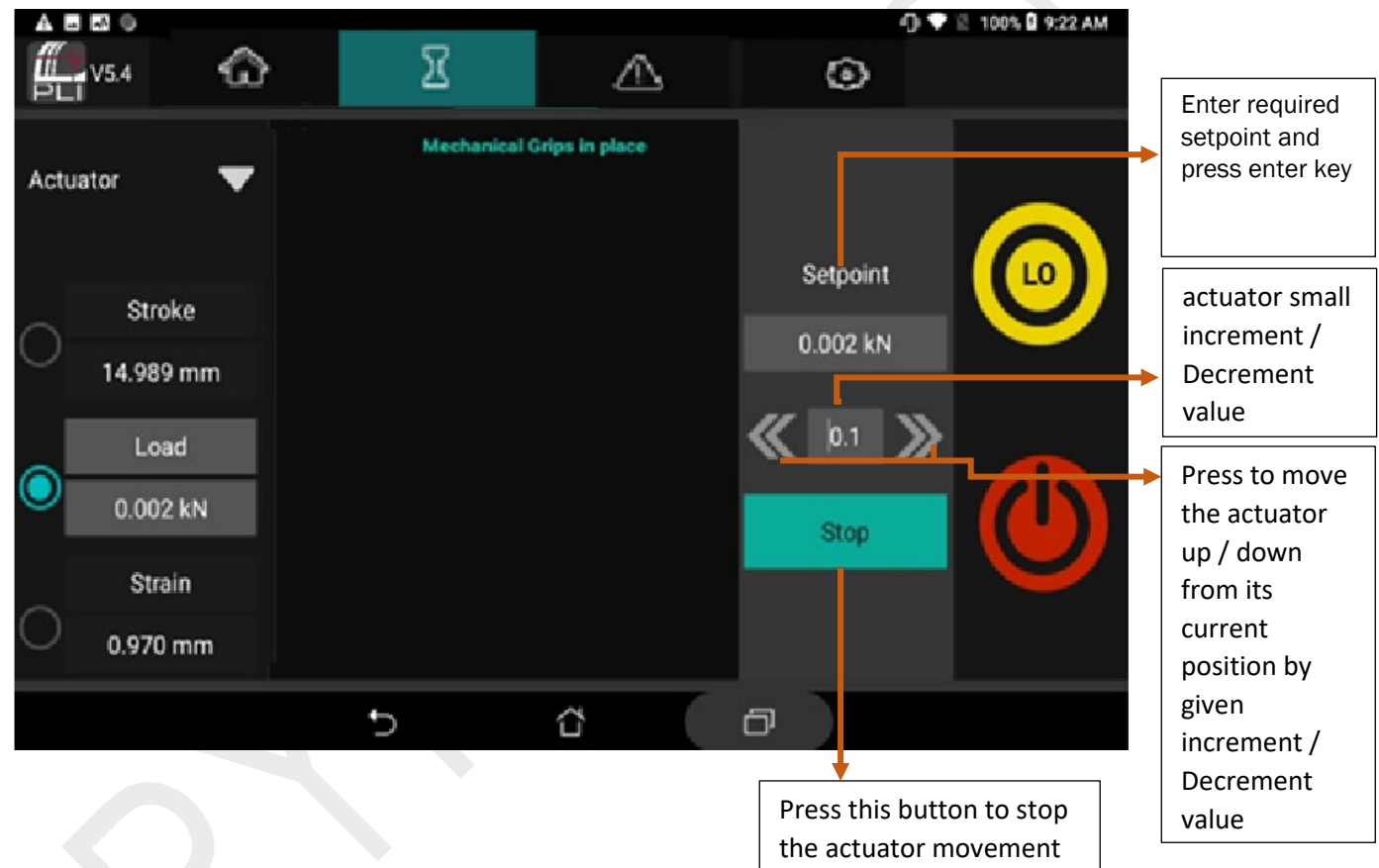
Machine usage hours with power on

Actuator Jog

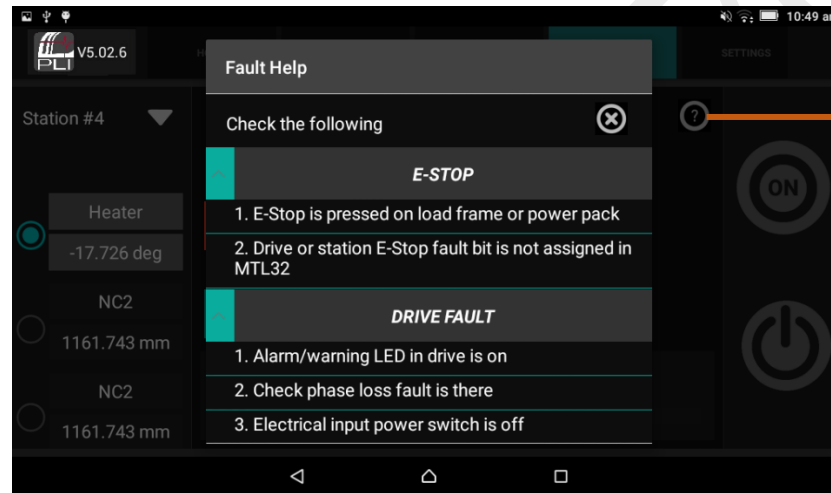
Actuator jogging is applicable only for displacement control mode.



By providing a setpoint, the operator is now able to move the actuator in load or strain control mode.



Fault panel



By pressing help button in fault page, operator can able to see the fault help message. Read the help and remove the fault.

9 Testing

- Tension
- Compression
- VAFCP
- J1c
- K1c
- Low Cycle Fatigue
- High Cycle Fatigue

9.1 Tension Testing (Static)

TITLE Standard test method for tension testing of metallic materials
TEST REFERENCE ASTM E8-08

Requirements

- Grips/Fixtures: Mechanical Wedge
- Wedges :

SI No	Available wedges	Specimen thickness/Dia
1	Flat (0 – 5mm)	Max - 5 mm (Width - 25mm)
2	Round (M8, M9, M10 Dia)	M8, M9, M10

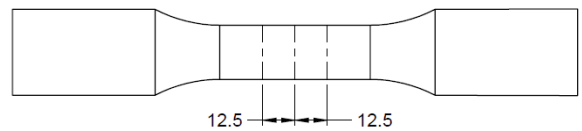
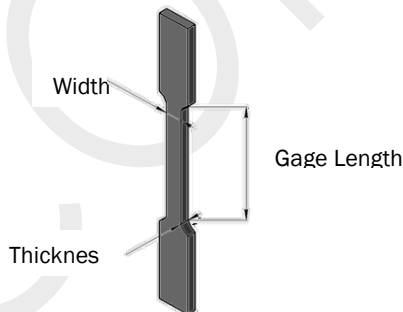
- Extensometer: Recommended 12.5 mm gage length extensometer.
- Torque Wrench (M10 socket), M5 Allen key, Wooden blocks

Displacement/strain-controlled test procedure

- Specimen geometry
- Powering up the system
- Test setup
 - Mount grips
 - Insert suitable wedges & mount specimen
 - Mount extensometer
- Run the test
- Export data

9.1.1 Specimen Geometry

- Measure specimen dimensions on the gauge section of specimen (3 times)
 - Instrument required - Vernier Calliper
- Take average value of 3 readings



- Mark the gauge length on the reduced section & centre of the specimen as shown in Fig 1

Specimen Parameters Table

Parameters	Software Configuration Options	Default Setting
Geometry	Standard Geometry options include Circular and Irregular. Irregular is chosen as the default so the user will be prompted to enter the cross-sectional Area (nominal) as per the required standard. Note: Dimensions reported in the Results Table must be updated if the geometry is changed.	Irregular (Area)
Specimen Length	Specimen Length should represent the gauge length of the specimen, not the total length. This is the distance between the grips for machined specimens and this length will be used to determine strain from actuator travel after extensometer removal.	-
Initial Gauge Marks	Initial Gauge Marks represents the distance between the marks that are placed on the specimen for purposes of measuring Elongation After Fracture. These marks are often made with a marker but can also be made other ways.	-

9.1.2 Powering up the system

a. Please ensure to switch ON

- System Mains

- Host Computer/Laptop



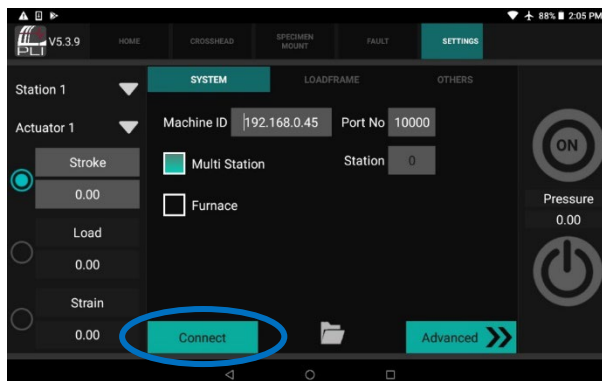
- b. Double click on MTL 32  icon on the desktop
- c. Click "Go-Online" button & wait for 15-30 seconds

9.1.3 Test Setup

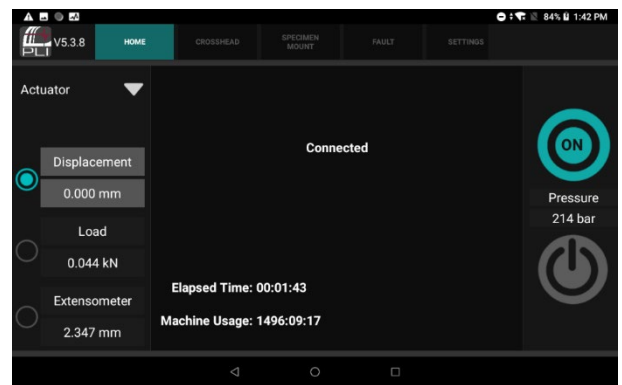
❖ Mount grips

Top grip on the actuator rod end and bottom grip to the load cell will already be mounted. Only accessories pin for respective tests needs to be mounted (refer to the Machine & Maintenance manual>> Universal Grips manual for more details on universal grips.) but before that ensure that the PLI application on tab is open.

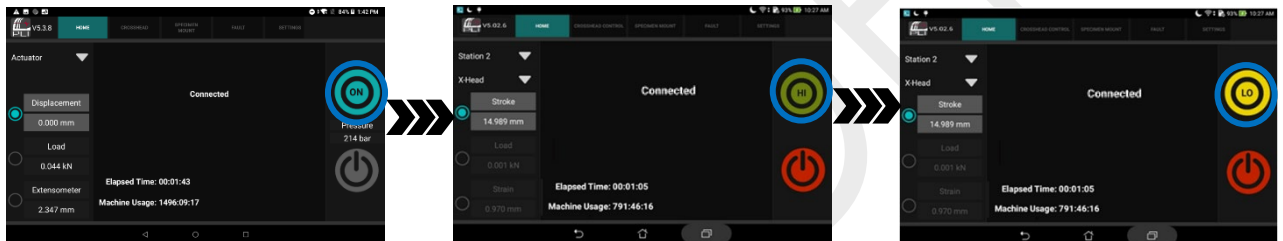
- Double click on the PLI icon  on the tab & click the connect button as shown below



- A page like this will appear once the communication is completed successfully



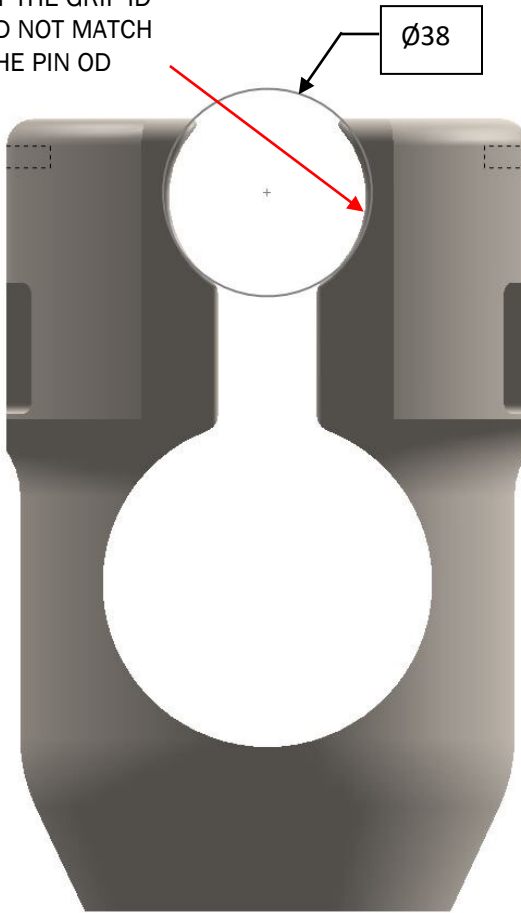
- Ensure system is in “Stroke” control mode before switching “ON”
- Click the “ON” button to turn ON the system (system switches ON in low pressure), wait for 5 s & click the “HI” button again (switches to High pressure). Once the system is in high pressure, the “HI” button will change to “LO”



❖ Insert Suitable Wedges & Mount Specimen

1. Apply 30N-m torque to M12X55 Allen bolt in a clockwise direction in order to open the grip

IDEALLY THE GRIP ID
SHOULD NOT MATCH
WITH THE PIN OD



M12 ALLEN BOLT

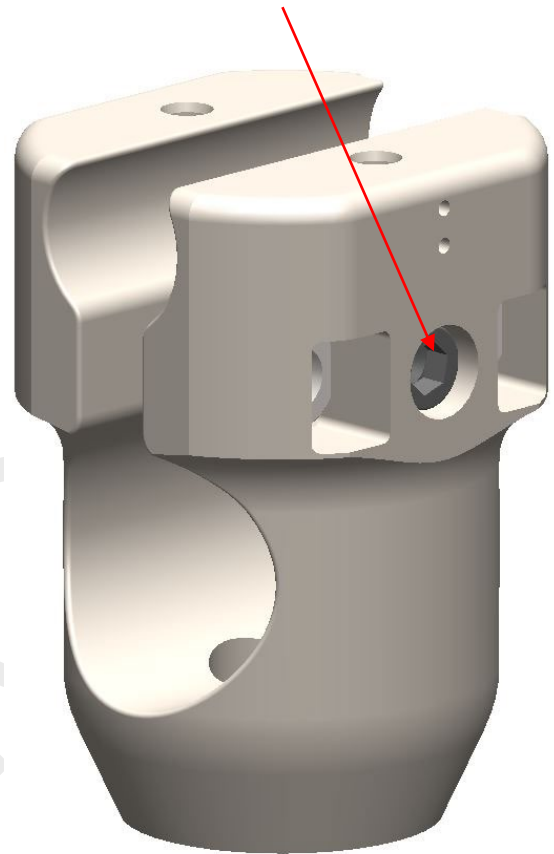
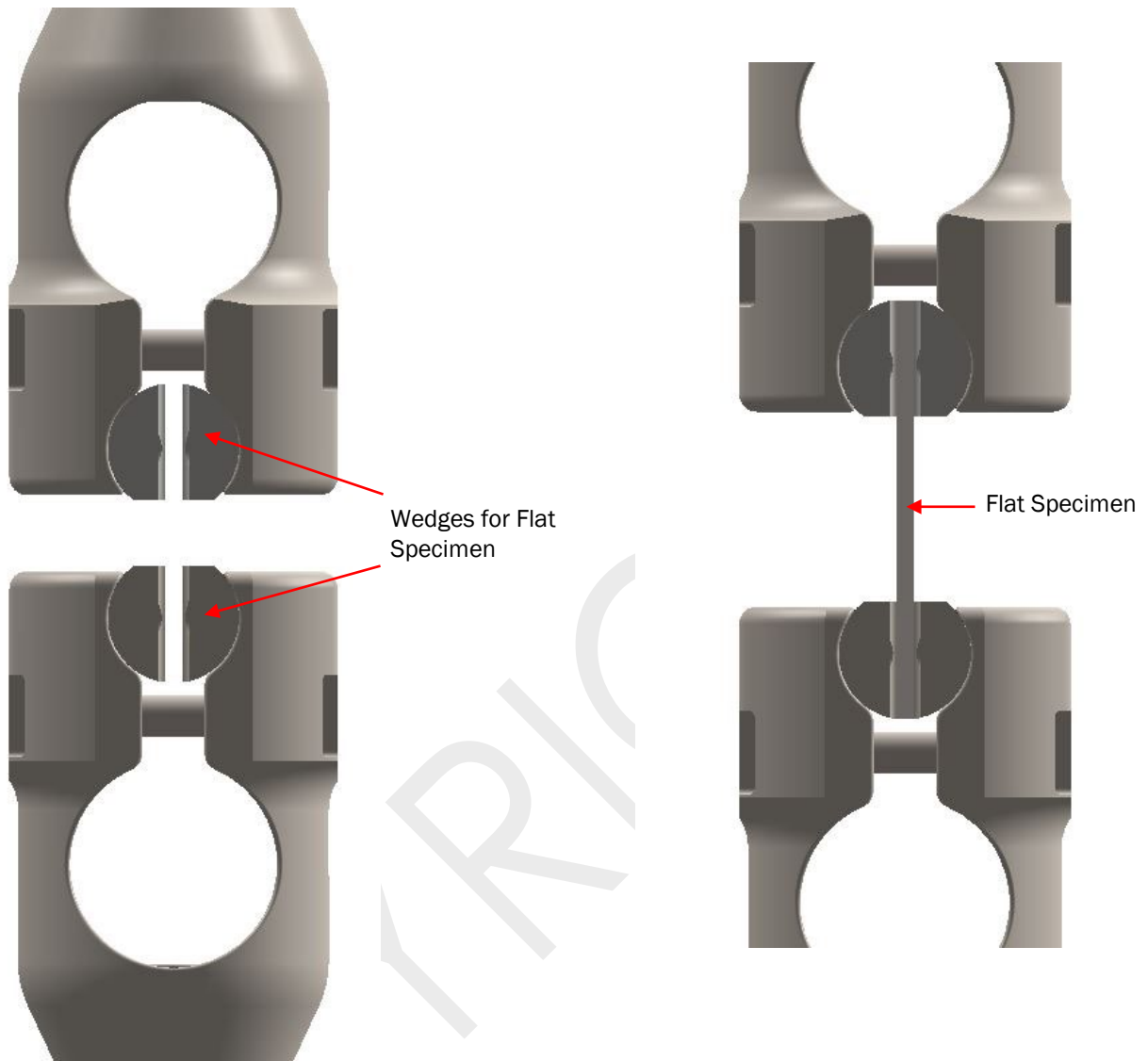


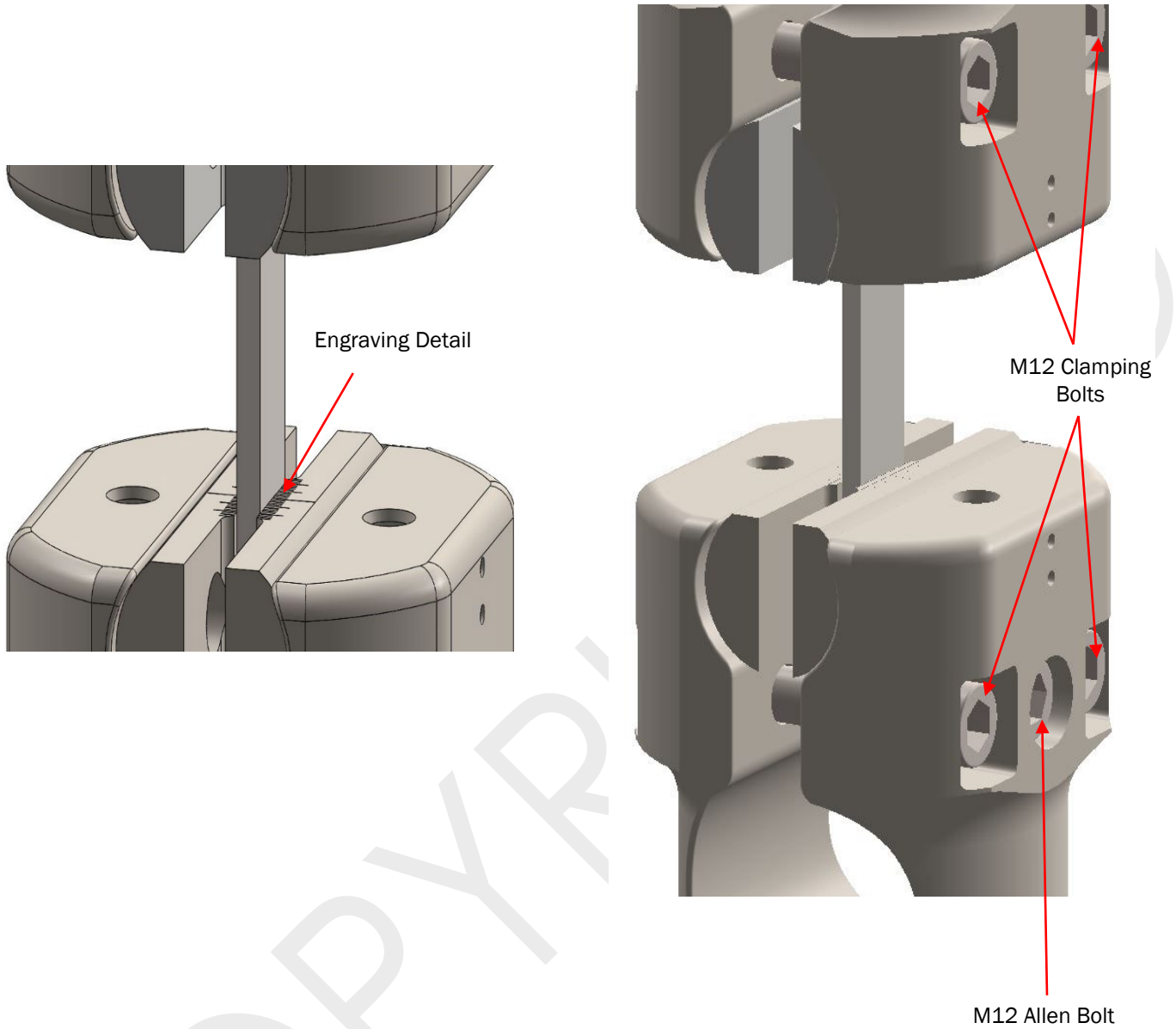
Table 1: Bolt Details for grip opening

SI No.	BOLT SIZE	QUANTITY	RECOMMENDED TORQUE
1	M12X55	1 No	30 Nm

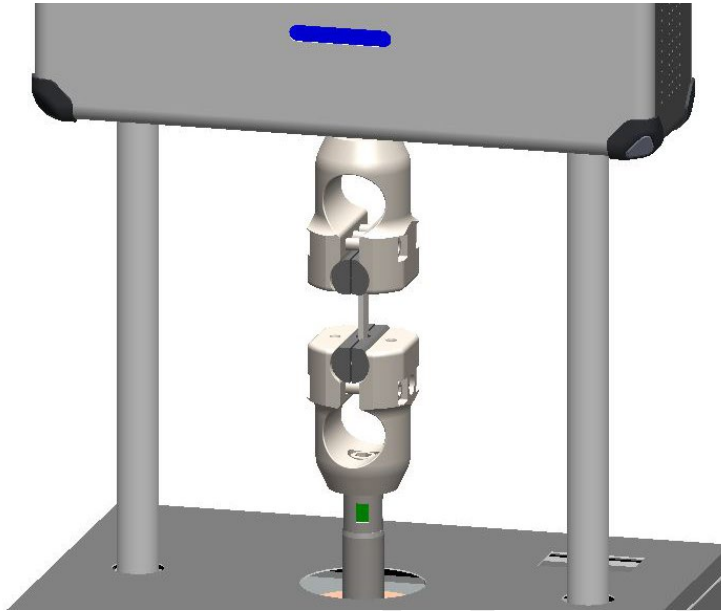
2. Insert the appropriate wedges suitable for specified specimen thickness. Then insert the flat specimen



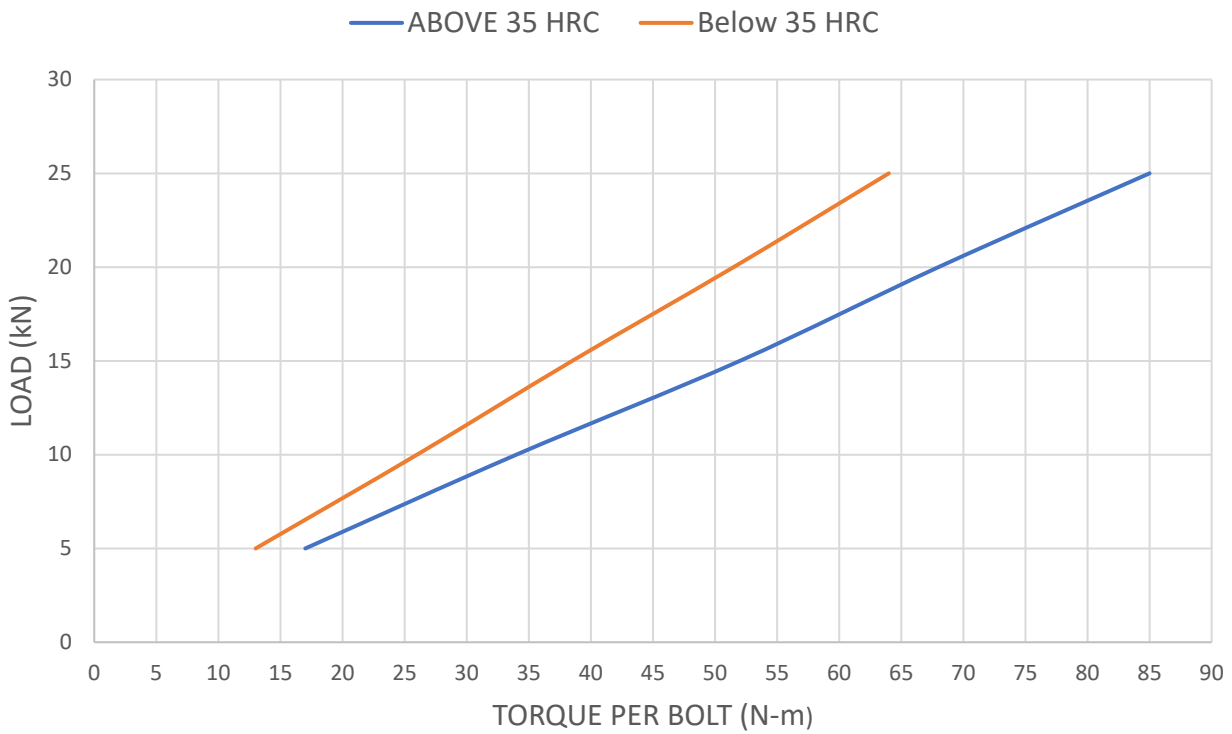
3. Align the flat specimen to the centre of the grip using the engraving provided on the wedges. Then loosen the M12 Allen bolt on both the top & bottom grips to clamp the flat specimen.



4. Switch to load control mode and set the load to zero before proceeding to next step.
5. Finally, apply torque (as mentioned in the torque chart) to each clamping bolt.
6. The Grip is ready for testing.



TORQUE CHART FOR 25kN UNIVERSAL GRIP

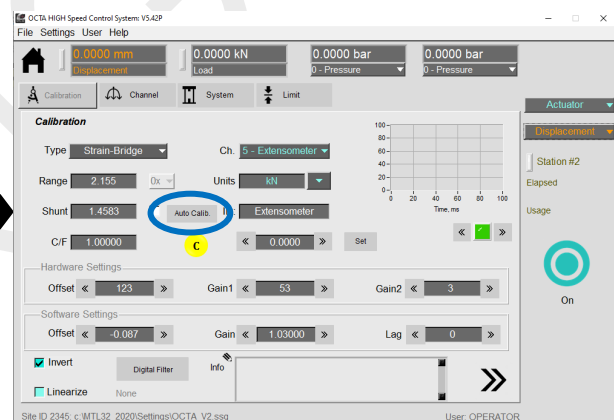
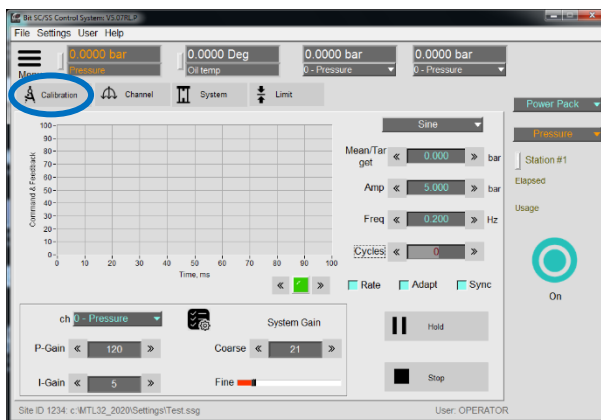


❖ Extensometer Mounting

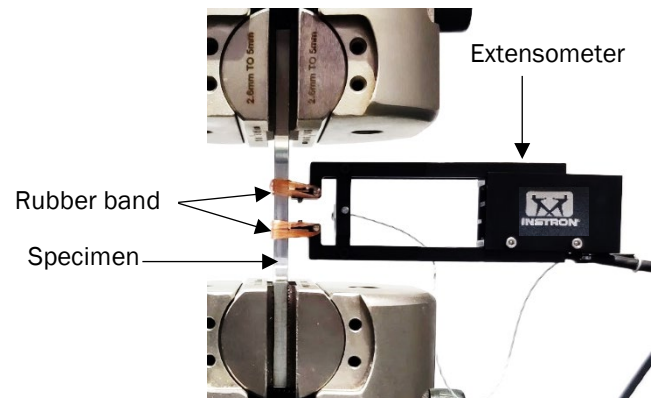
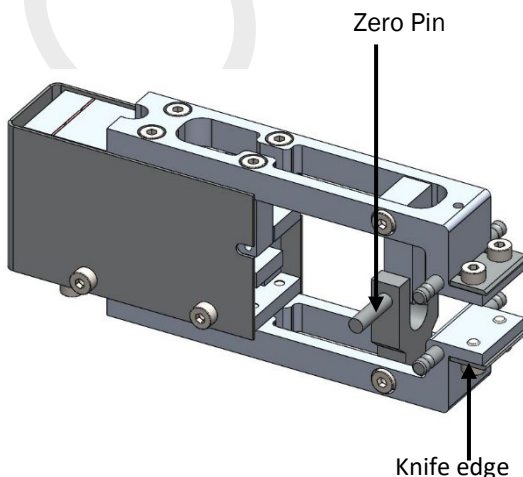
- Locate strain 1 channel (on load frame) & connect extensometer to it as shown below



- Before mounting the extensometer, **ensure that the control mode is not in Extensometer (select stroke)** & it is recommended to auto calibrate it.
- To auto calibrate, go to MTL32 “Calibration” page & click on “Auto calibrate” button & wait for 30-45s for auto calibration to complete



- Without removing the zero pin mount the extensometer on the specimen by rubber band, so that the blades coincide with the lines marked on specimen. Mount rubber band gently, do not over tight



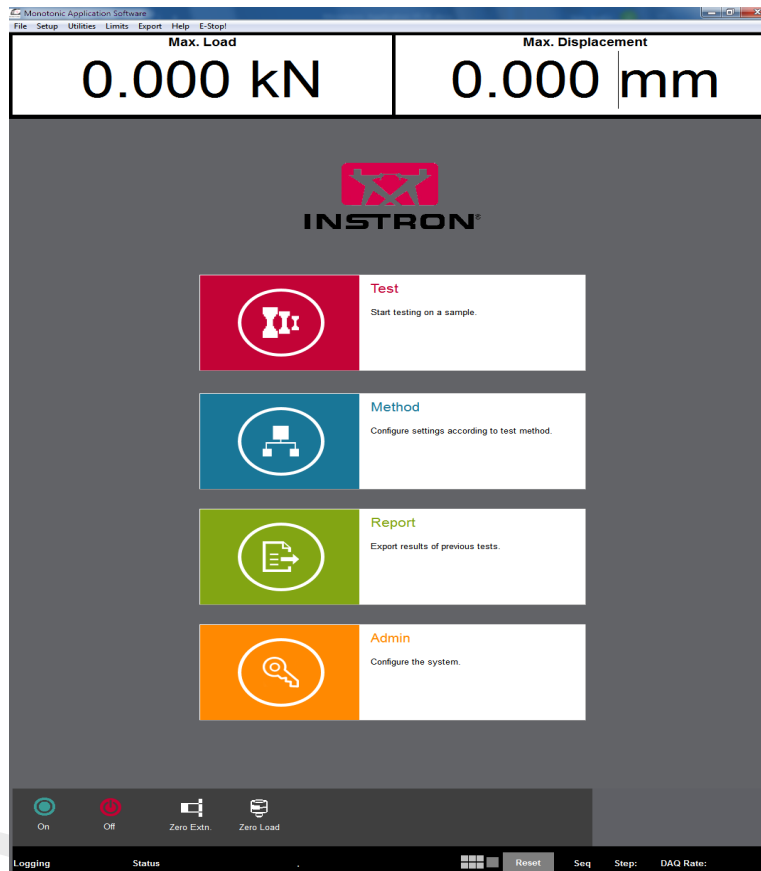
- Auto calibrate again & remove the zero pin slowly so that the blades position is not disturbed. Ensure that the extensometer read out on system is close to zero, if not remove the extensometer and mount it again
- Set the Extensometer limits in the limit panel



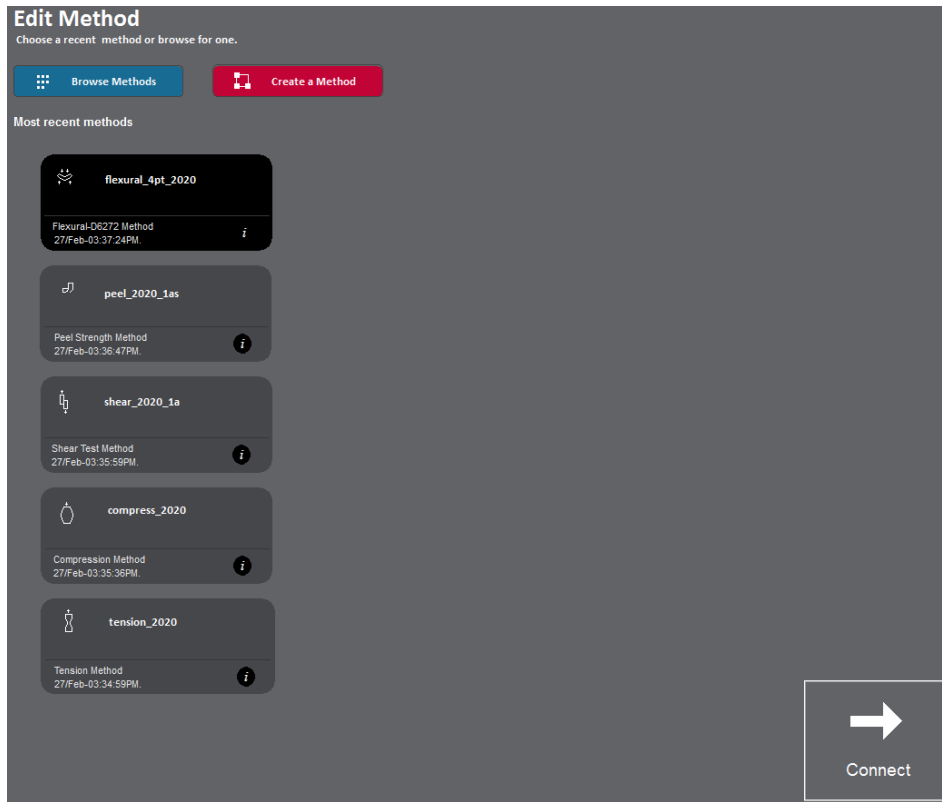
*Zero pin should be inserted while mounting/removing the extensometer to/from the specimen
Zero pin should be removed only when conducting a test.*

9.1.4 Run a test

- Double click on the Bluehill2ST logo & select “Test” tab



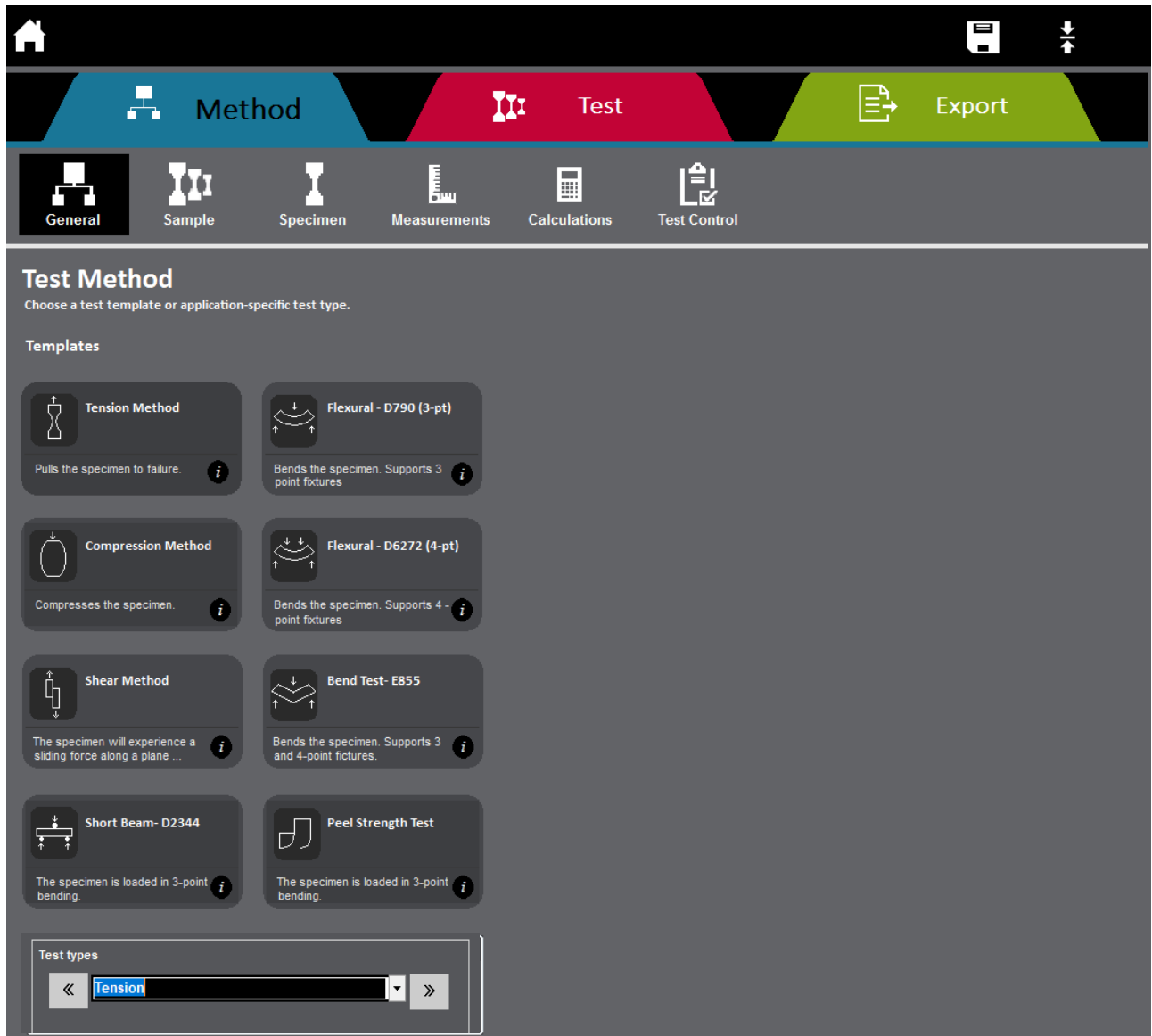
- The last loaded Test Settings file will be used automatically



- To load a different settings file, click on Browse Method button.
- To load previous file settings the user can click on the any of previous settings files from Most recent methods as shown below.
- To create a new settings file, click on Create a Method.
- To generate reports for completed tests, click on Report button.
- Change Screen Layout if required: either Horizontal or Vertical, depending on the screen being long or wide.
- Click on Connect button to go online.
- To get information on usage of software, click on Help >> Manual.

Settings

- Use the outer arrows to navigate between main pages.



- Use the inner arrows to navigate between sub-pages.

Test Type

- Test Type: Select required type of test. Settings will change according to test type.
- Extensometer/Strain Gauge/Deflectometer mounted: This option does not come into picture for Peel Strength Test, Shear Fasteners Test, Dent Resistance Test and Quick Test, Grab test and Trapezoidal tear test.
- Enable this when Extensometer/Strain Gauge/Deflectometer is mounted on the specimen.
- For Tension and Compression tests, enable Extensometer Mounted or Strain Gauge Mounted option depending on what is mounted on the Specimen.
- Extensometer removal settings: Active range of extensometer will be shown. Enter required range to remove extensometer. Stress and Strain % can also be specified for removal by enabling the checkbox next to it.
- Use extensometer removal option to avoid damage to extensometer.
- Remove Extensometer: After reaching specified range during Test, window shown below will appear. Confirm machine control mode is not in extensometer. Confirm all limits on extensometer are disabled. Then remove extensometer and click on “OK” button to continue test.

- >> Confirm control mode is Stroke or Load.
- >> Confirm all limits on the Extensometer are disabled.
- >> Then remove extensometer.
- >> After removing extensometer click OK to continue test.



- Auto Continue Test: Enable it, to continue test automatically after extensometer removal condition is met. The message shown below will appear for user information. User should take care of removing extensometer without disturbing test and extensometer.

Extensometer Removal condition hit!

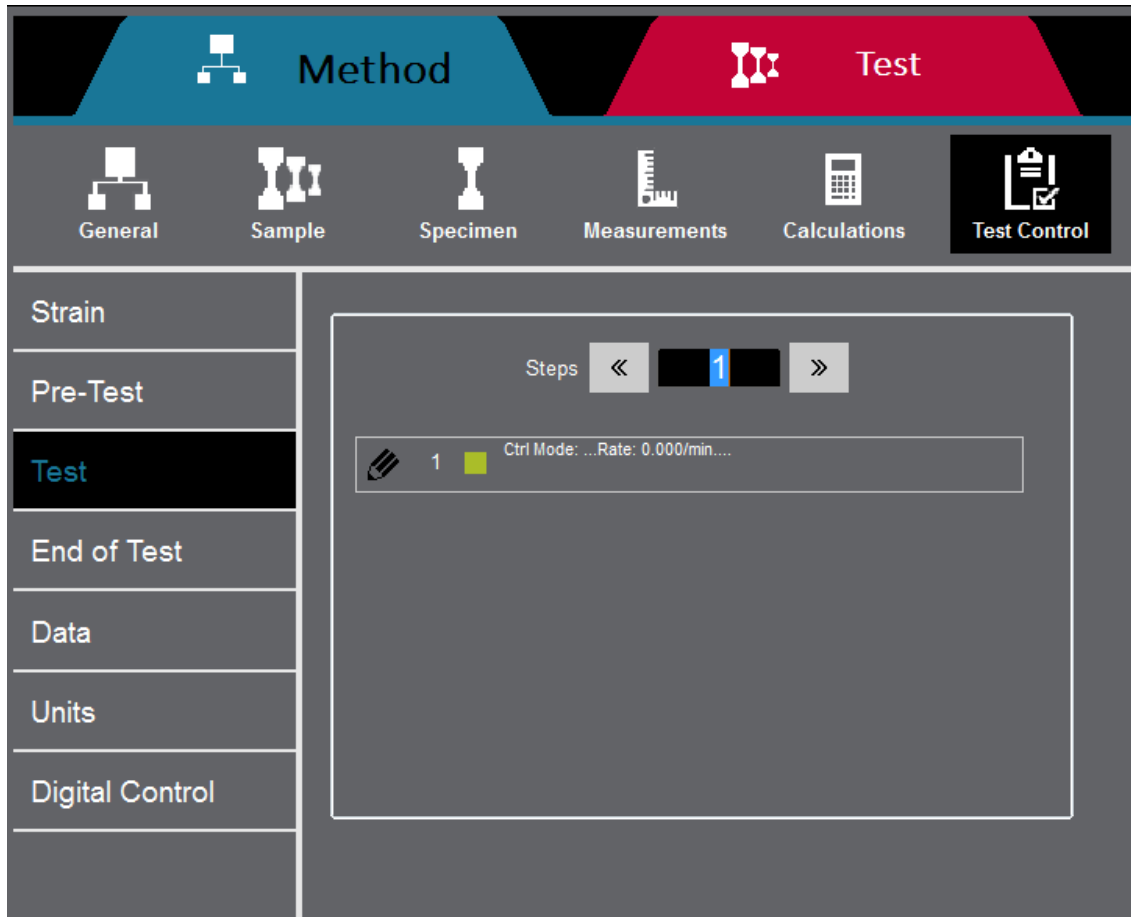
Remove Extensometer to avoid damage..



- For Tensile, Compression and Shear Test for Composites test types, if a 2nd extensometer is mounted, then enable Extensometer 2 Mounted option. If this is enabled, then Gauge Length for this extensometer must be entered. Also, Strain 2 Channel must be selected in DAQ Channel Settings panel.
- Strain Gauge Mounted option: Select Strain unit and Strain channel correctly in the System Settings section. If another strain gauge is mounted, then also enable Strain Gauge 2 Mounted and select Strain 2 channel correctly.
- Flexural D790 and Flexural D6272 Test: The option Stop at 5% Flexural Strain can be enabled. If enabled, Test will be terminated when this condition is met.

Test Settings

- Step Mode: Step by Step.



- Steps are loaded one after the other.
- Enter the number of Steps required. Maximum being 10 and minimum being 1.
- Click on  button provided for every step to edit parameters in the Edit Step window.

Edit Step:

The screenshot shows the 'Edit Step 1' window with the following settings:

- Control Mode:** Stroke
- Rate, /min:** 2.000
- Hold After End of Step:** ☒
- Hold Control Mode:** Stroke
- Hold Time Unit:** sec
- Hold Time, s:** 0.000
- End of Step:**
 - Option 1:** ☒
 - End of Step:** Displacement
 - End Value:** 0.000
 - Option 2:** ☐

- Select required Control Mode for the step. For tests not involving extensometer and Shear Test for Composites, do not select the Strain control mode.
- Enter the step rate either control channel unit/minute or /second. Or select Time in seconds and enter time required for the step. Also, if extensometer is mounted, there is the 3rd rate option: strain/s.
- End of Step: There are two options which can be used to end a step. Only if the step is the last one, none of them need be used.
- But, if it isn't, enable at least one option. In case of extensometer removal being used as an option, it is required to enable even the other one.
- The Displacement option under End of Step tracks the displacement. If Extensometer Mounted is enabled, then displacement is calculated from the extensometer channel. If not, then the stroke channel.
- Hold after End of Step: To dwell at the end of step, enable this option. Select Control Mode during the hold. Ensure that Strain control is not selected when Extensometer is not used.
- Select Time unit as seconds or minutes. Enter the required Hold Time.
- Use Left and Right arrows to load previous and next step respectively.
- Ensure that all parameters are correctly entered.
- Use Save button to save entries and selection.

The screenshot shows the 'Test Termination' configuration window. The top navigation bar includes 'Method' and 'Test' tabs. Below it, a row of icons represents different test parameters: General, Sample, Specimen, Measurements, Calculations, and Test Control. The left sidebar lists various test stages: Strain, Pre-Test, Test, End of Test (which is currently selected and highlighted in blue), Data, Units, and Digital Control. The main content area is titled 'Test Termination' and contains four adjustable parameters, each with a numerical display and a checkbox to enable or disable the condition:

- Drop in Load, %: Set to 0.000, checkbox is checked.
- Load Threshold, kN: Set to 0.000, checkbox is checked.
- Max. Load, kN: Set to 0.000, checkbox is checked.
- Max. Displacement, : Set to 0.000, checkbox is checked.

- Drop in Load, %: Test will stop automatically, if load drops by specified %. Load Threshold: Enter the Load value up to which drop in load is not considered.
- Max Load: If current load value is greater than or equal to this value, test stops.
- Max Displacement: If displacement is greater than or equal to this value, test stops.
- Strain %: If Strain % during test exceeds this value, it is terminated. For Bend Test - E855, this will be displayed as Permanent Strain %. This is very important for Bend Proof Strength calculation as desired Permanent set value comes into picture. It is essential that loading-unloading sequence be maintained without which the Permanent Strain % cannot be calculated.
- For all the above conditions, enable their respective checkboxes to activate them. Without doing this, they will not work as terminating conditions!

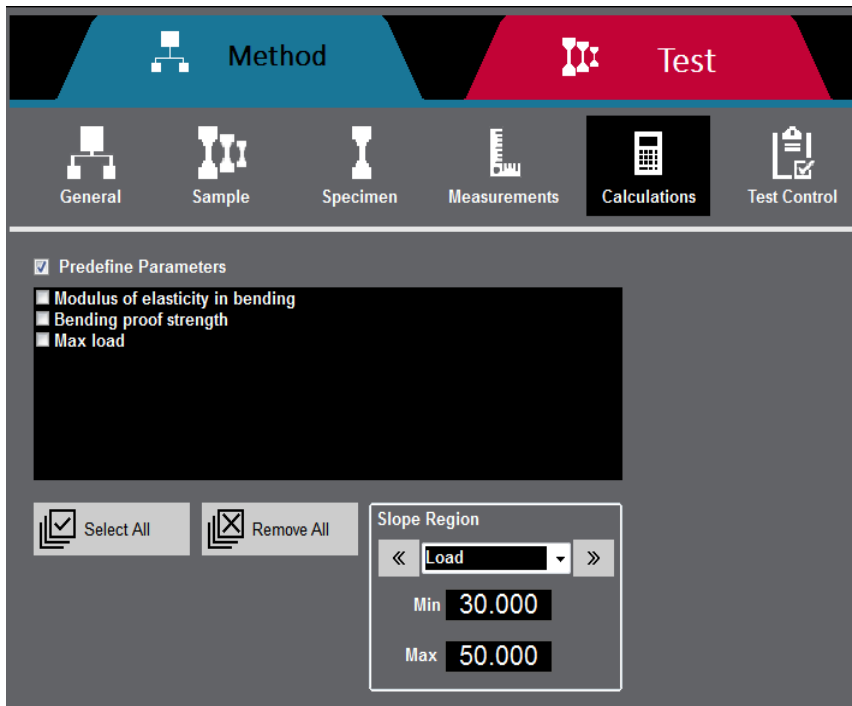
Pre-Test

- Home Position: Enable this option to use it to return to specified position at the specified rate. This feature is most useful before and after a test.
- Preload Settings: Enable this option to preload in required control mode to required Set Point at required rate. Also, Alternate mode can be selected. This can be used just before a test.
- Failure Mode Info: Use this to enter information or observation regarding failure of specimen. A popup window will open after test completion. Write in the given space as brief as possible.

The screenshot displays the 'Pre-Test' configuration window. At the top, there are two main tabs: 'Method' (blue) and 'Test' (red). Below these is a horizontal navigation bar with icons and labels for 'General', 'Sample', 'Specimen', 'Measurements', 'Calculations', and 'Test Control'. On the left side, a vertical sidebar contains menu items: 'Strain', 'Pre-Test' (highlighted in black), 'Test', 'End of Test', 'Data', 'Units', and 'Digital Control'. The main content area is divided into sections. The 'Home Position' section includes a checkbox, a 'Home Position' field with a value of 0.000, and a 'Rate, /s' field with a value of 0.000. The 'Pre-Load Settings' section includes a checkbox, a 'Control Mode' dropdown set to 'Stroke', an 'Alternate Mode' dropdown set to 'Same Mode', a 'Set Point' field with a value of 0.000, and a 'Rate, /s' field with a value of 0.000. At the bottom, there is a checkbox for 'Failure Mode Info'.

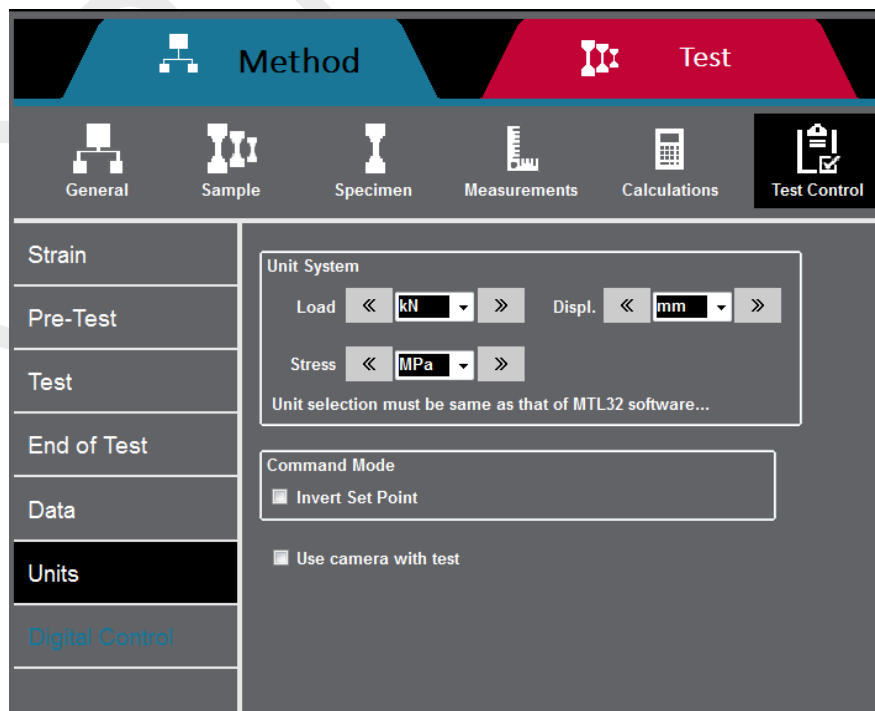
Predefine Parameters:

- User can predefine the output parameters before conducting test. These parameters will be automatically selected during report generation to appear in reports.
- E.g., select the parameters for first sample of the test and then continue doing tests. Selected parameters apply for all the specimens tested until user changes them or the test type.
- These parameters will change depending on the type of test.
- Slope Region can also be predefined for Report. Enter appropriate values of Max and Min depending on the parameter selected (% of Load, Load, Stress or Strain %).

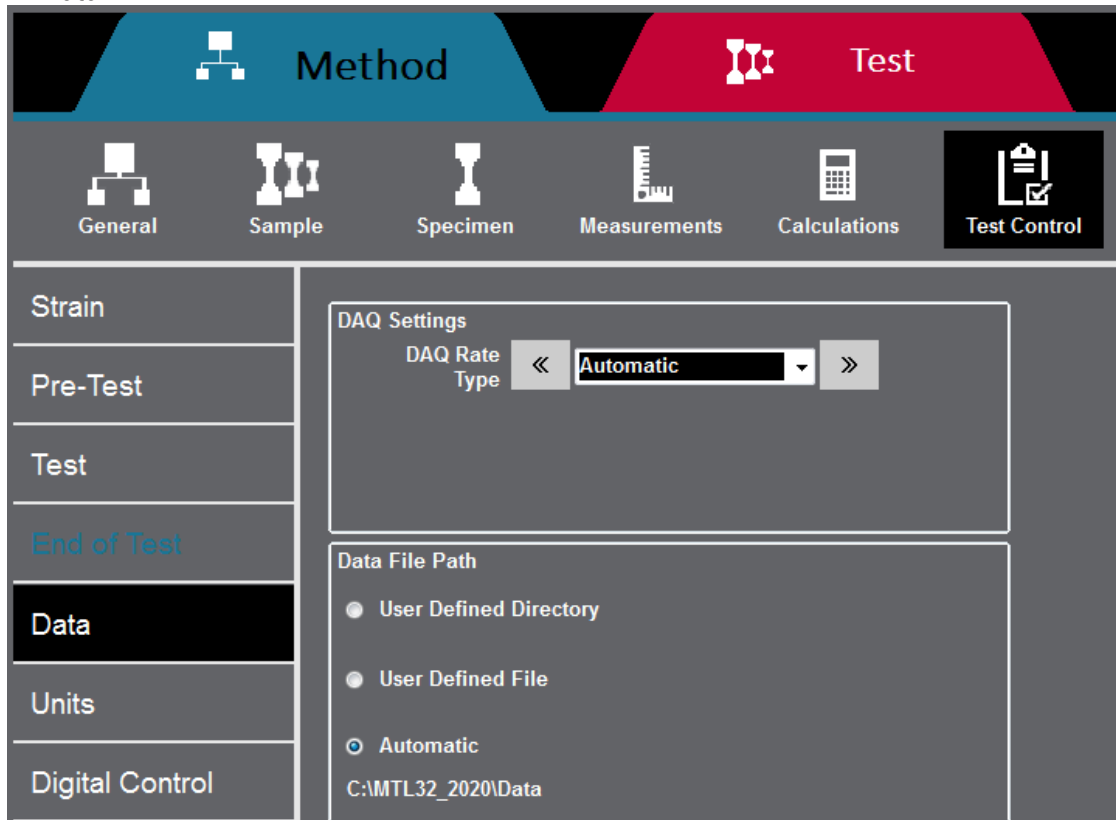


System Settings

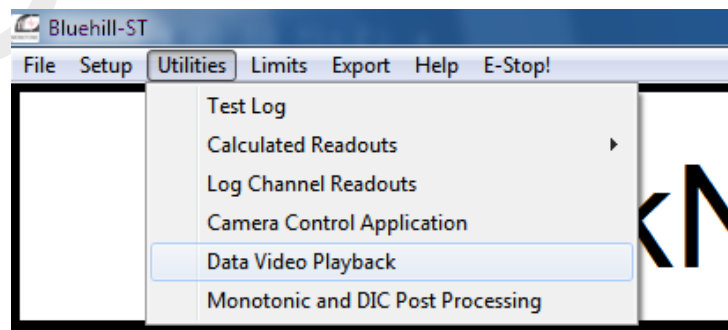
- **Unit System:** Select the load and displacement unit as calibrated in MTL32_2020. In case of Strain Gauge mounted option being enabled for Tension or Compression or Shear D5379 tests, select the Strain unit by referring the MTL32_2020 software. Select required unit for Stress value display.



- **Data File path:** If “Automatic” is selected, data files automatically get saved in “C:\MTL32_2020\Data”. If “User Defined Directory” is selected, click on the **Select/Create Directory** to create your own directory. To create a data file in any location and with any name, use the “User Defined File” option.
- **DAQ Settings:** Can be automatic or manual or manual-dual rate. *Automatic* will set a DAQ Rate depending on the Test speed. To set a DAQ Rate of choice, select *Manual* option and enter it. To set two DAQ rates select the third option. Enter the value in Time or Stress that triggers change in DAQ Rate.



- **DAQ Channel Settings:** Match the log channels according to what is displayed. It is important that Stroke, Load and Strain channels are selected correctly.
- To record video with test or to trigger time periodic clicking of photos, check the *Use camera with test* option. This works only when Video-With Bluehill ST Test or Time Periodic-With Bluehill ST Test option is selected in the Camera Control Application.
- To open Camera Control Application, Data, and Video Playback Application and Bluehill ST and DIC Post Processing application, go to Utilities list of the main menu bar.

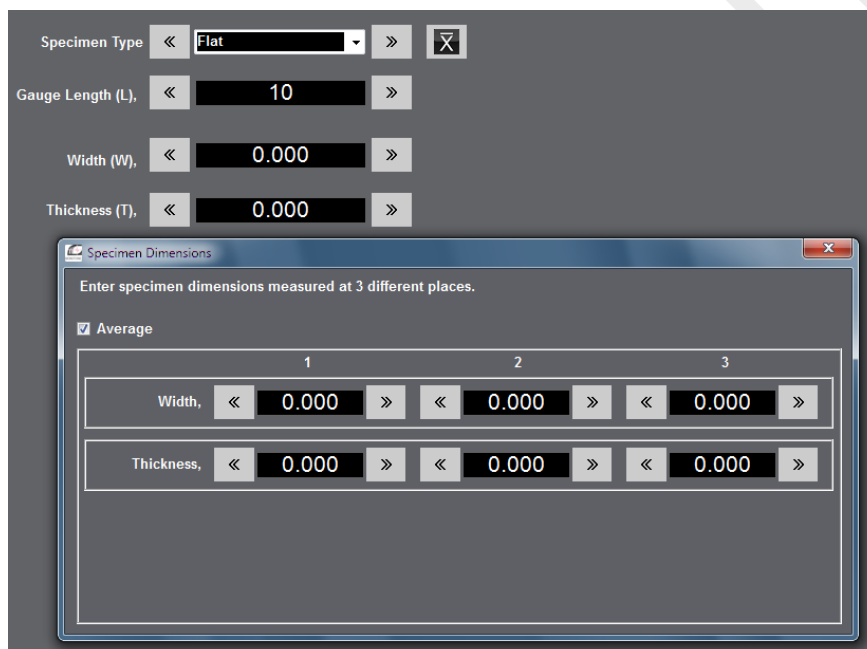


Specimen

- Select required specimen type. Enter required dimensions of specimen. These parameters change according to the Specimen Type.
- In case of Extensometer 2 Mounted option being enabled, then an extra parameter Gauge Length 2 must be entered correctly.

Average value

- In case of Specimen dimensions being measured at different places and to get their averaged value, click on Dimensions  button.
- Enable Average option and enter dimensions. The averaged value is reflected in the main Specimen panel as shown below.
- Ensure that the parameters are correctly entered without which Test cannot be started!



Specimen Type << Flat >> 

Gauge Length (L), << 10 >>

Width (W), << 0.000 >>

Thickness (T), << 0.000 >>

Specimen Dimensions

Enter specimen dimensions measured at 3 different places.

☒ Average

	1	2	3
Width,	<< 0.000 >>	<< 0.000 >>	<< 0.000 >>
Thickness,	<< 0.000 >>	<< 0.000 >>	<< 0.000 >>

- Description: User can describe the Specimen ID, Operator and Batch etc. in their respective fields.
- Also, any other test related details can be entered under the Title and Description section.

Description

- Click on Method-> Sample
- Enable the checkbox of respective row of Title and Description. Enter here. The rows appear in Report only if enabled.
- Click Save button to save details entered so far.

Method

Test

Export

General

Sample

Specimen

Measurements

Calculations

Test Control

Test Description

	Title	Description
☒	COMPANY	BISS
☒	TEST	TENSILE
☐		
☐		
☐		
☐		
☐		
☐		
☐		
☐		
☐		

Batch

File Setup Utilities Limits Export Help E-Stop

Max. Load

0.000 kN

Max. Displacement

0.000 mm

Method

Test

Export

General

Sample

Specimen

Measurements

Calculations

Test Control

Properties

Inputs

Specimen Type

Flat

<<

>>

✕

Gauge Length (L)

10

<<

>>

Width (W)

3

<<

>>

Thickness (T)

1.90

<<

>>

<<

<

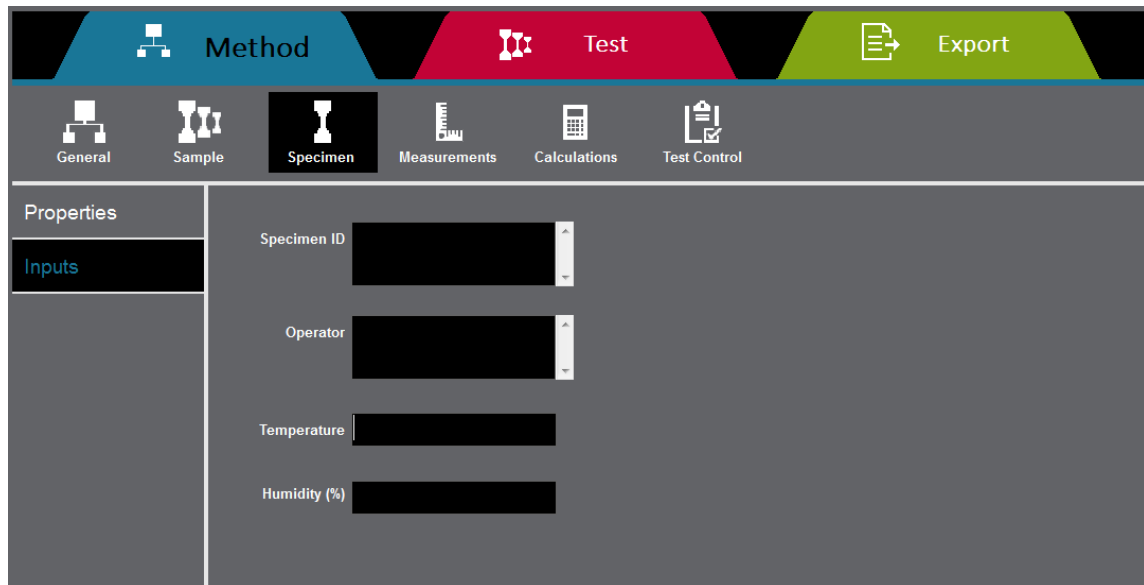
>

>>

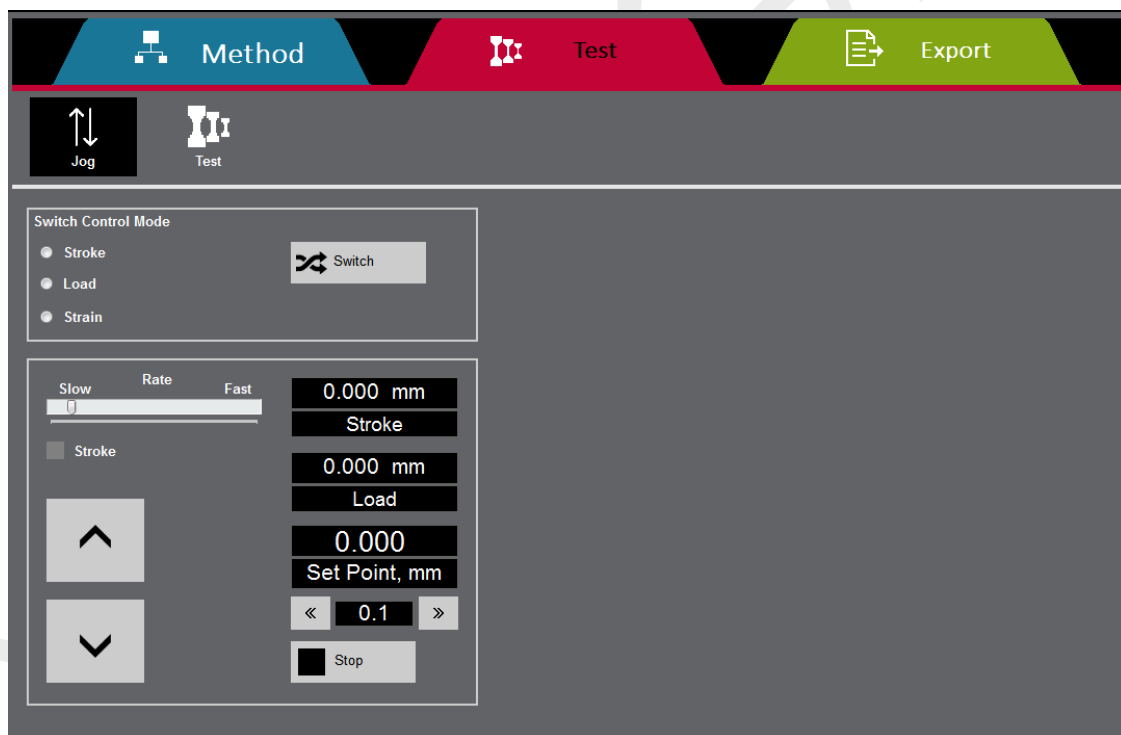
W

L

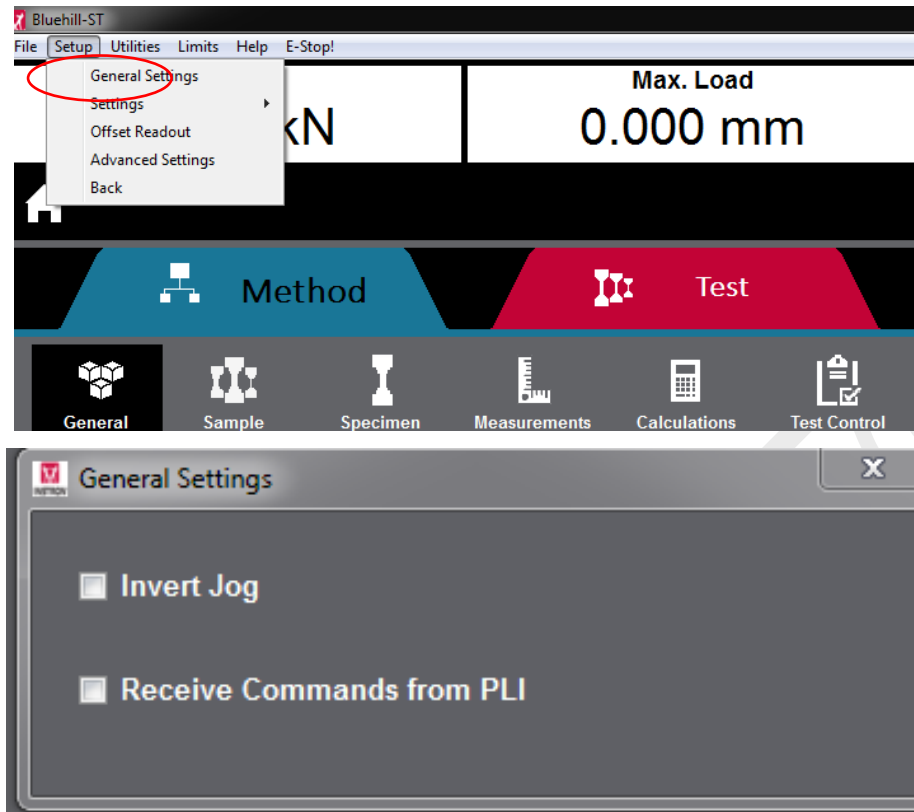
T



Jog

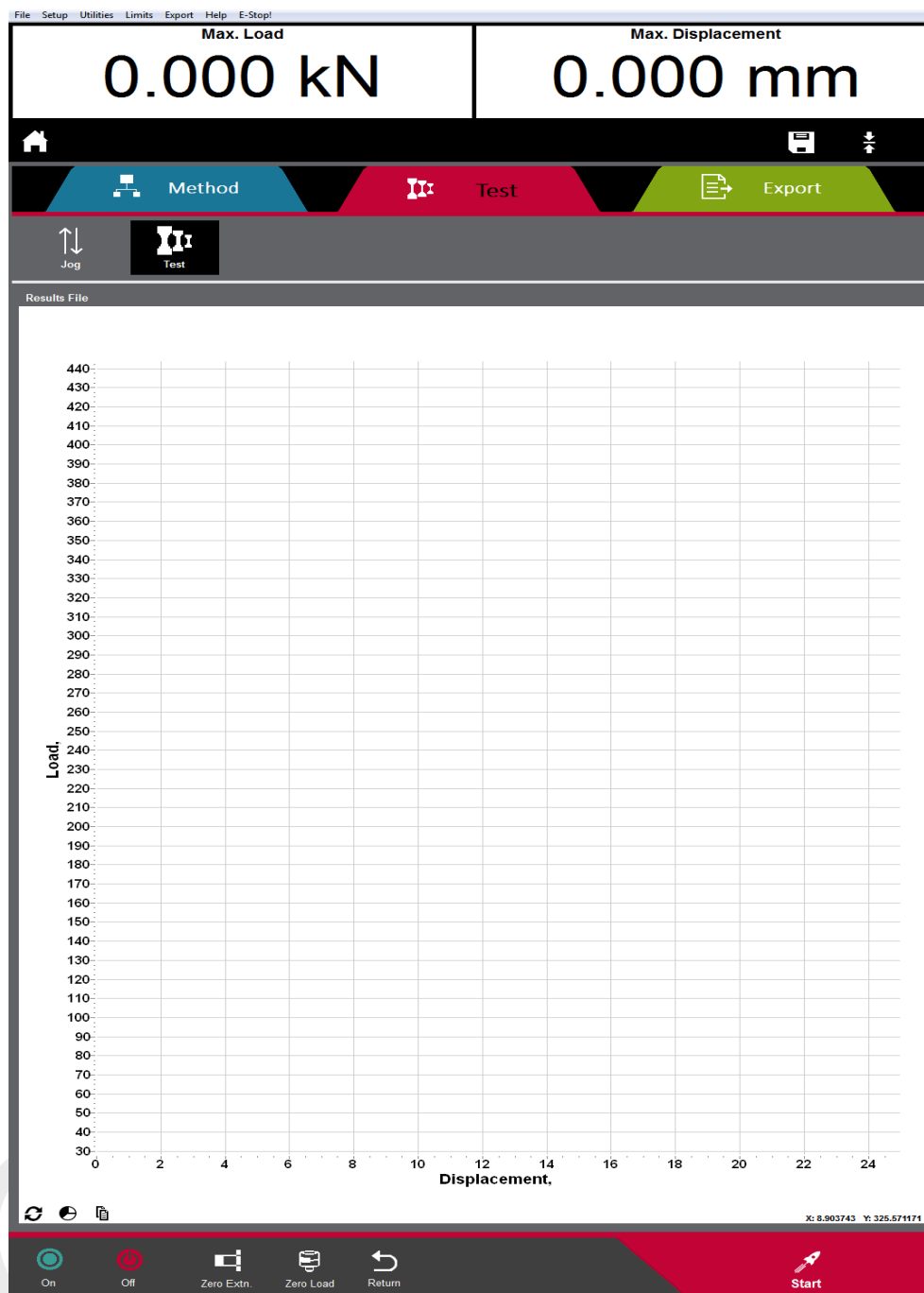


- Switch Control Mode: To change the current control mode, select required control channel. Then, click Switch button.
- Jog section:
- Ensure power is on before using this section. Also, that set point is not on hold!
- To change the current position, use the Top and Bottom arrow buttons. Click and hold either of them depending on the required change in position. The rate of Jog can be adjusted by moving the slider. By default, its rate will be 60 units/min.
- If the Jog Top and Bottom arrow buttons are moving the actuator in reverse directions, i.e., Top moving it down and Bottom moving it up, then use the Invert Jog option by opening the General Settings. From menu bar, click on Setup->General settings->Invert jog

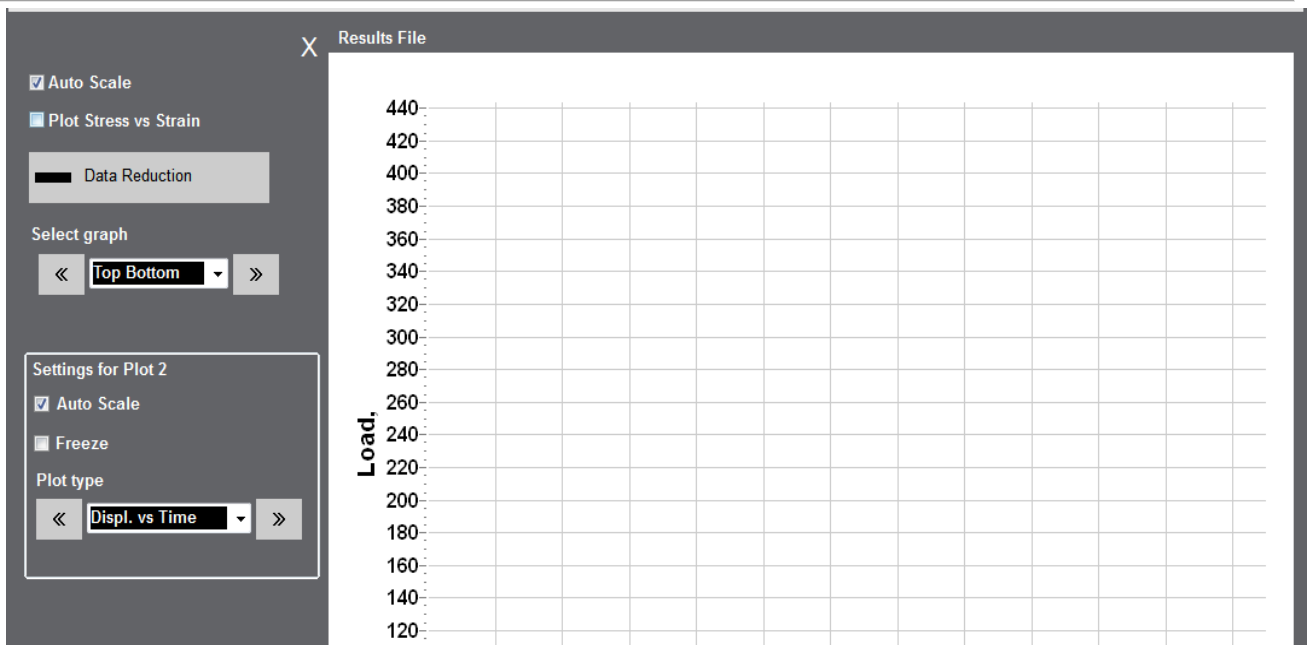


- Also, required position can be achieved by entering Set Point.
- Another form of jogging is using the << and >> buttons. Enter the value relative to current value by which it is decremented or incremented respectively.
- To stop the jogging, click the Stop button.
- Use any of the options based on convenience.
- This is mainly used while mounting specimen.

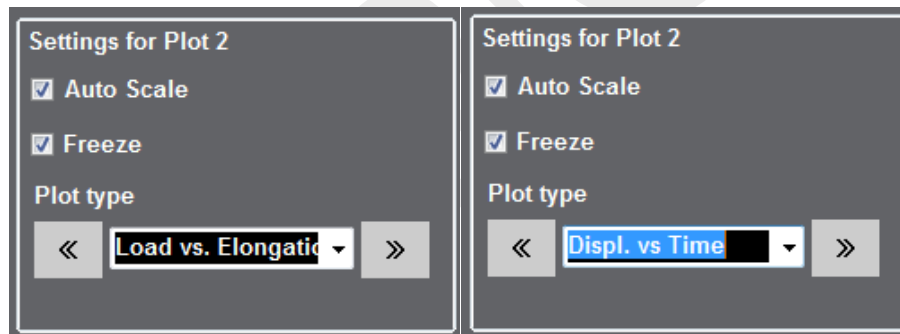
Run test



- User can save the readout meters, change the graph settings, set the extensometer readouts to zero when test is idle.
- Graph Settings: Click on the Pie Chart on the Bottom Left corner of the chart to view Graph Settings Panel as shown below.
- Use Refresh button to clear graphs when it's needed.
- Auto Scaling is checked by default.
- To get Stress vs. Strain plot instead of Load vs. Displacement plot, check Plot Stress vs. Strain. This option is available only for relevant tests. For Shear Test for Composites, Shear Strain % vs. Shear Stress plot is shown.
- To get Displacement vs. Time graph, select required orientation: either Top Bottom or Left Right, as shown below. Strain/Extensometer channel is selected as the displacement channel in case of Extensometer/Deflectometer, etc. being mounted. Else, position channel is selected.



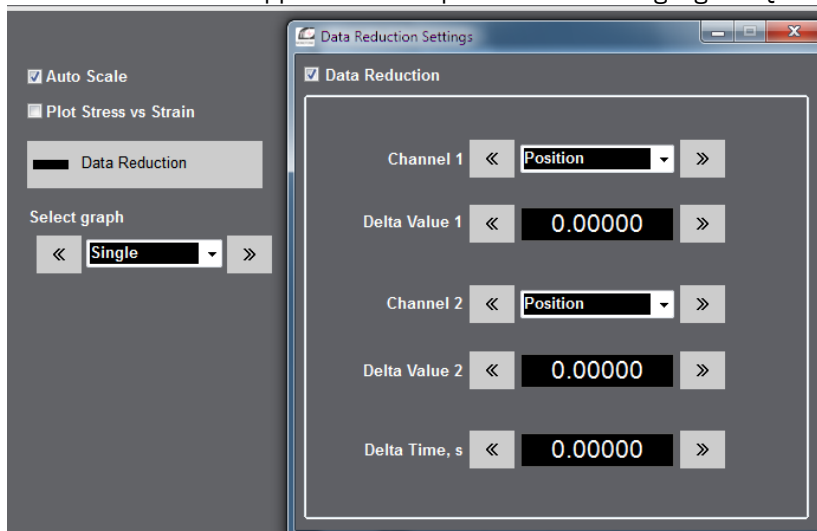
- Chart 2 has its own settings. Auto-scale is enabled by default.
- Use *Freeze* option to pause plotting. Uncheck it to resume plotting from that time onwards.
- In case of elongation parameter being calculated for the selected test type, Load vs Elongation plot option is available. Just select it under the Plot type option.



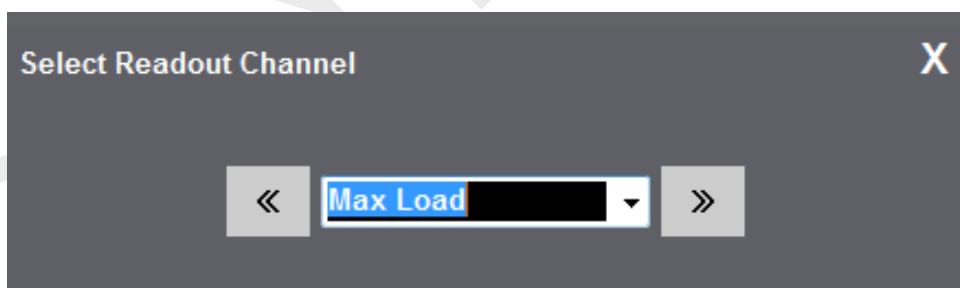
- Use « » buttons to decrease and increase respectively, the maximum value of Time axis.

Data Reduction

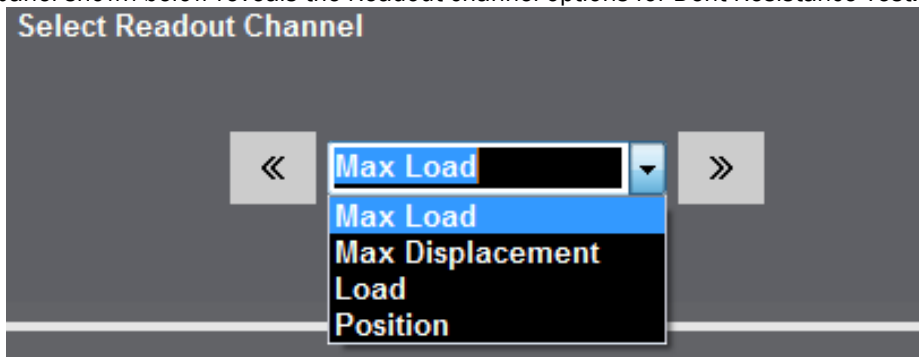
- This feature is used so that the Application is responsive even during high DAQ rates.



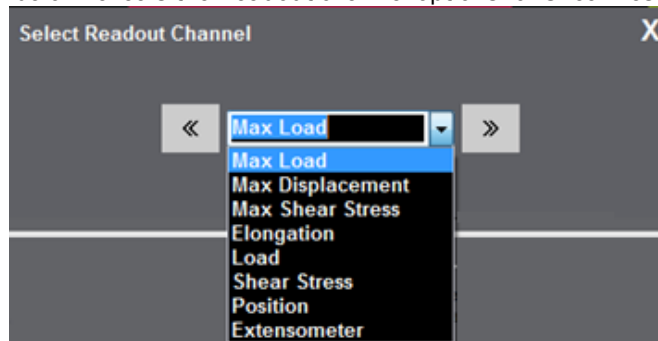
- Enable Data Reduction to activate this feature.
- Select any feedback or set point channel and enter delta value. This value is to be between 0 and 1. Entering zero indicates that the channel will not be used for data reduction.
- Enter Delta Time in seconds also between 0 and 1.
- Data will be plotted in graph only when there is change in channel or time by respective delta values.
- If this feature is not required, then uncheck Data Reduction.
- Readouts:** Double Click on Readout to change its channel.
- The panel shown below opens. Select required channel. The panel shown below reveals channels according to the Test Type.



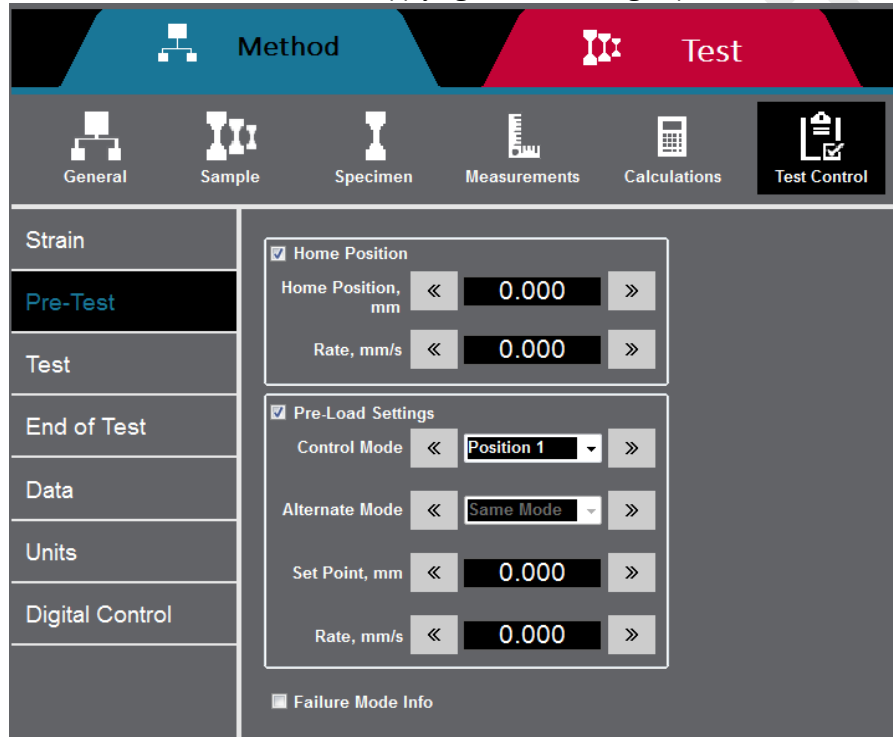
- The panel shown below reveals the Readout channel options for Dent Resistance Test.



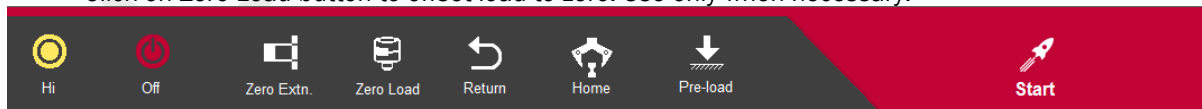
- The panel shown below reveals the Readout channel options for Shear Test for Composites.



- Use the Preload or Home buttons for applying load or moving to positions as defined in test settings.

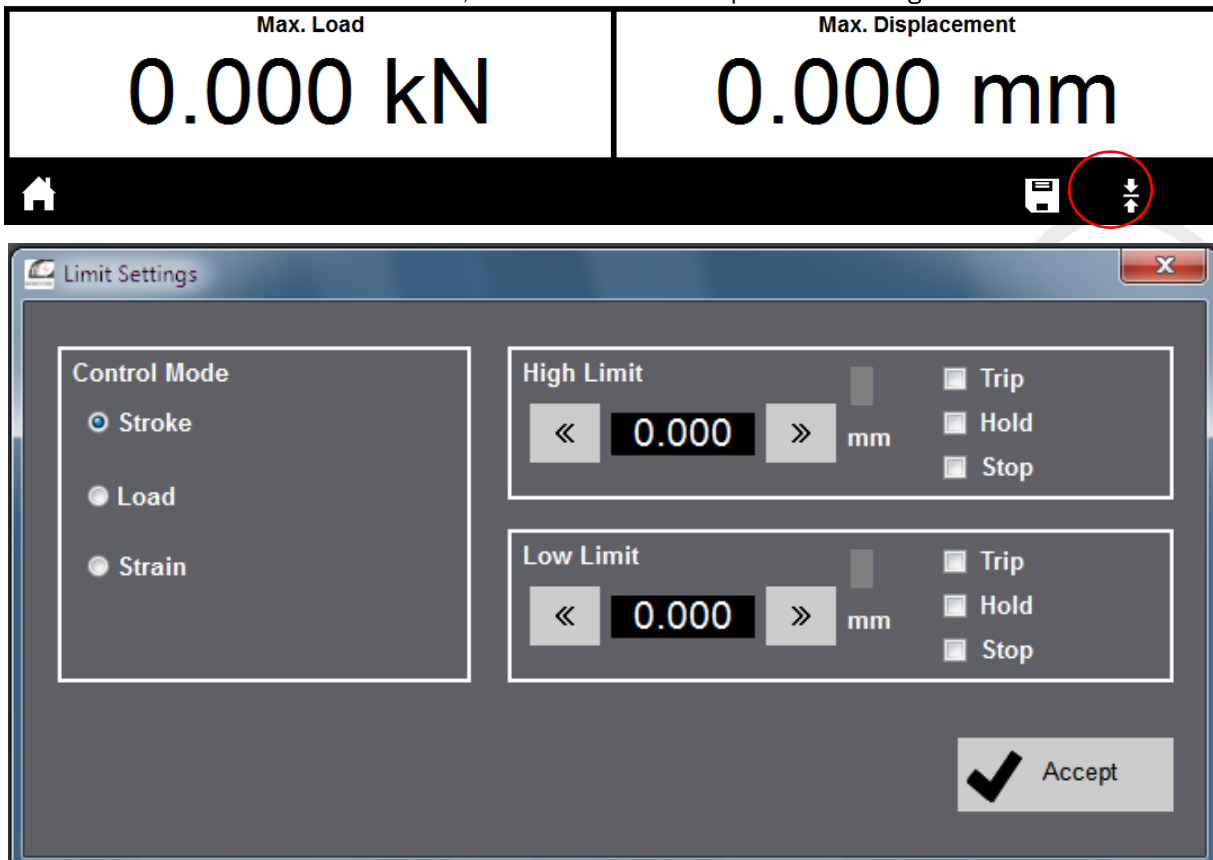


- To return to the start position after a test, simply use the Return button.
- Click on 'Zero Extn.' button to offset Strain/Extensometer channel(s) to zero. Ensure that Strain channel is not in control while doing this.
- Click on Zero Load button to offset load to zero. Use only when necessary.

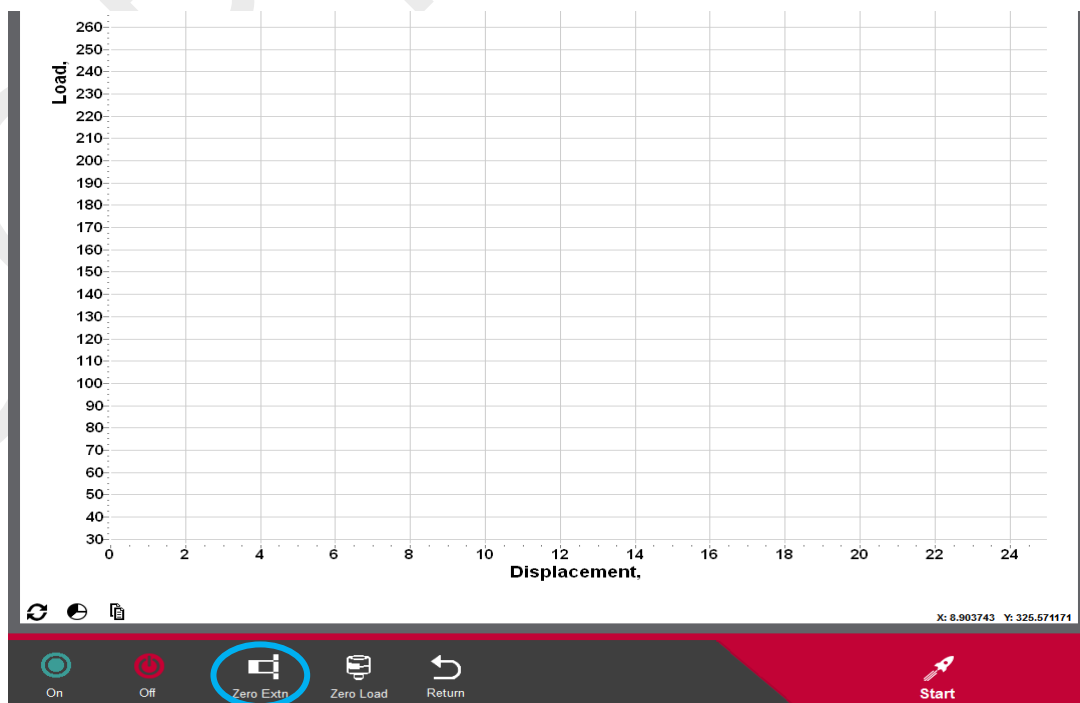


Limits

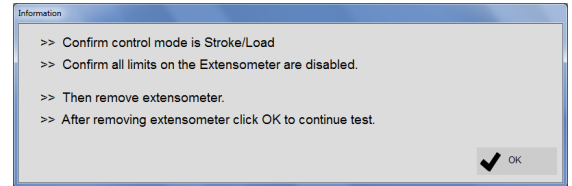
- To set Limits for control channels, click *Limits* button to open Limit Settings.



- Select Stroke, Load or Strain. Set high limit for this channel. Tick the action to be made when limit is hit. Similarly set low limit and its action.
- Click Accept button to set the limits.
- Now system is ready to run a test before that offset extensometer readout to zero by clicking “Zero Extn” as shown below



- Go to the “Run” page & click on “Start” button to start the test
- After the test as reached a specified range over a time period, a window will appear requesting for Extensometer removal (as shown below)



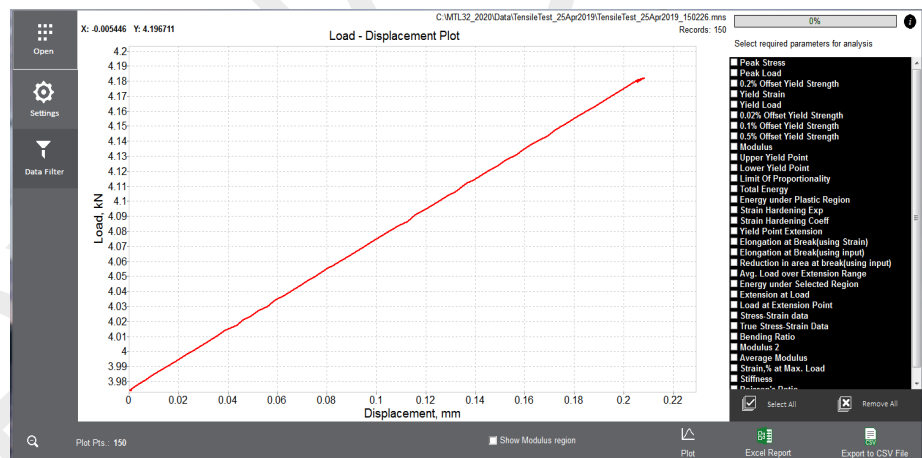
- Ensure that the Extensometer/Strain limits are removed before removing the extensometer (Use a blade/knife to cut the rubber band/O-ring)

Note: Take extra care while removing the extensometer.

- A graph would be drawn on the application showing load vs strain relationship
- Observe the specimen and the graph simultaneously and you will notice the specimen break after a certain period.
- The test will come to a halt as the per the test parameter specifications.

9.1.5 Exporting Data

- To export result to MS Excel sheet, follow steps mentioned as below
- Click on “Report” in the home page (Monotonic/Bluehill)



- a. Post Processing a window will open with the latest data file by default,
- b. Here the user can view “Modulus” by enabling “Show Modulus Region” check box
- c. Select the predefined parameters from “Select Required Parameters for analysis” window. These parameters will be analysed & will subsequently appear in the Excel report
- d. Then click on “Excel Report” for result to appear in Excel sheet

Note: Refer to the Monotonic/Bluehill application manual for more details on data processing in single files or Multiple files/Batch processing.

Note: If no other test is being conducted, its recommended to Turn OFF/Shut down the following devices

- a. System hydraulic pressure (In application)
- b. Monotonic/Bluehill application
- c. MTL32 basic software
- d. Controller
- e. Computer
- f. UPS
- g. UPS input
- h. Selector Switch
- i. Lastly HPS Mains.

9.2 Flexural Testing (3 & 4 Point)

TITLE Standard test method for Flexural properties of Unreinforced and Reinforced plastics and Electrical Insulating materials

TEST REFERENCE ASTM D790 (3-Point bend) & ASTM D6272 (4-Point Bend)

Requirements

- Grips/Fixtures: 3- & 4-Point Bend

Displacement/strain-controlled test procedure

- Powering up the system
- Test setup
 - Mount grips
 - Insert suitable wedges & mount specimen
 - Mount extensometer
- Run the test
- Export data

9.2.1 Powering up the system

- Please ensure to switch ON
 - System Mains
 - Host Computer/Laptop
- Double click on MTL 32  icon on the desktop
- Click "Go-Online" button & wait for 15-30 seconds

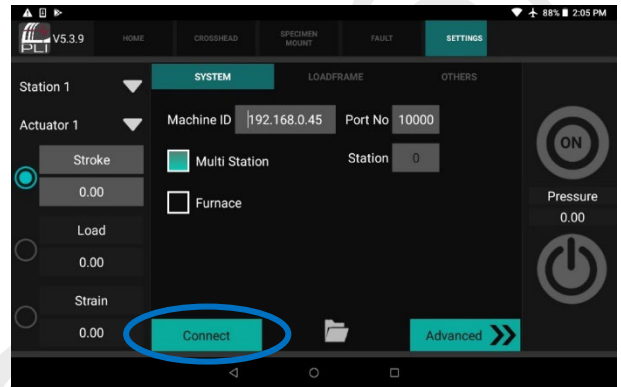


9.2.2 Test Setup

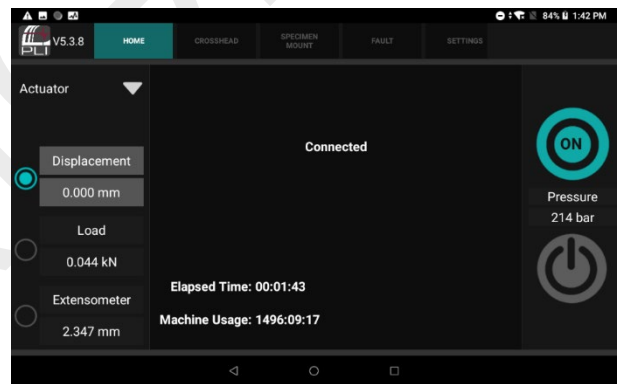
❖ Mount grips

Top grip on the actuator rod end and bottom grip to the load cell will already be mounted. Only accessories pin for respective tests needs to be mounted (refer to the Machine & Maintenance manual>> Universal Grips manual for more details on universal grips.) but before that ensure that the PLI application on tab is open.

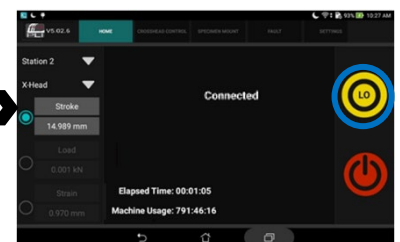
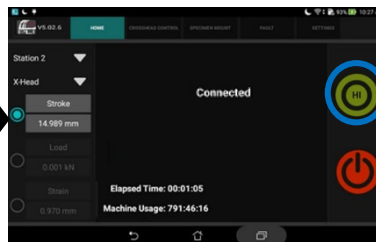
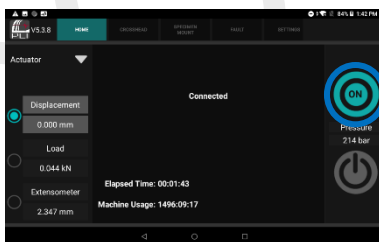
- Double click on the PLI icon  on the tab & click the connect button as shown below



- A page like this will appear once the communication is completed successfully



- Ensure system is in "Stroke" control mode before switching "ON"
- Click the "ON" button to turn ON the system (system switches ON in low pressure), wait for 5 s & click the "HI" button again (switches to High pressure). Once the system is in high pressure, the "HI" button will change to "LO"



Mounting instruction and alignment

- Place a top cup on the bottom plate and connect both using M10 x 27mm length stud.

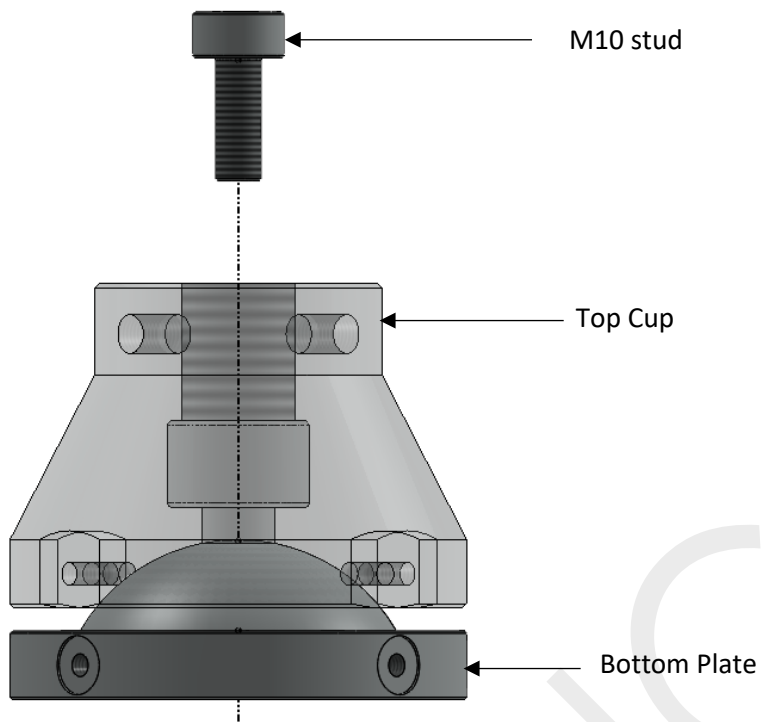


Figure: Top cup mounting

- After connecting the Top cup and bottom plate, use M27 x 60mm adaptor to mount the assembly to the piston rod.

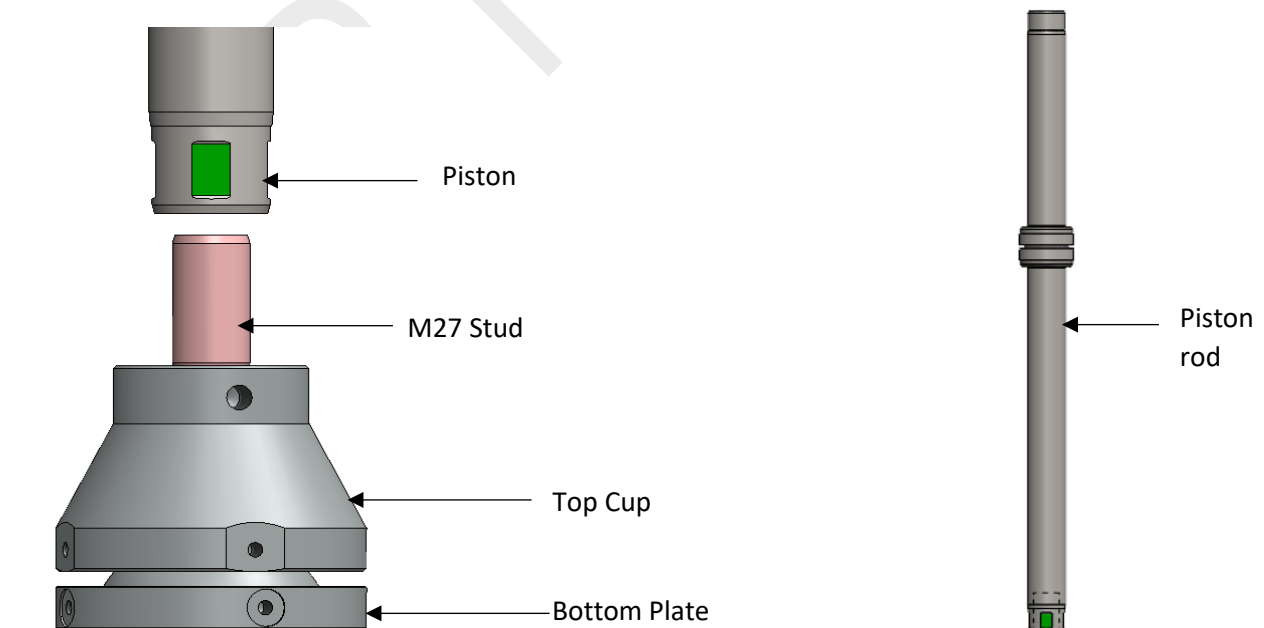


Figure: Assembly mounting to piston

- Assemble the top support block with the bottom plate using M8 x 20mm bolt.

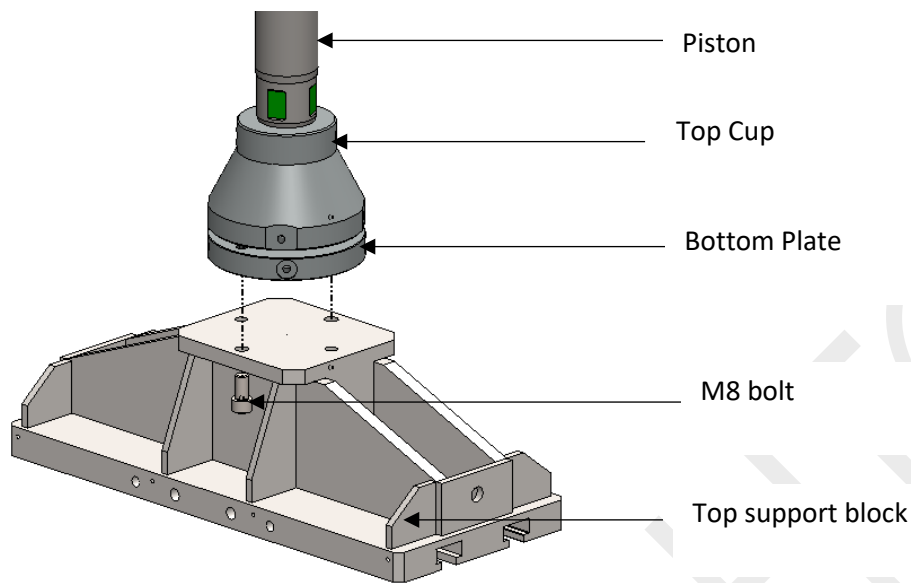


Figure: Top support block assembly

- Attach an anvil scale with the top support block using M3 x 8mm socket head cap screw.

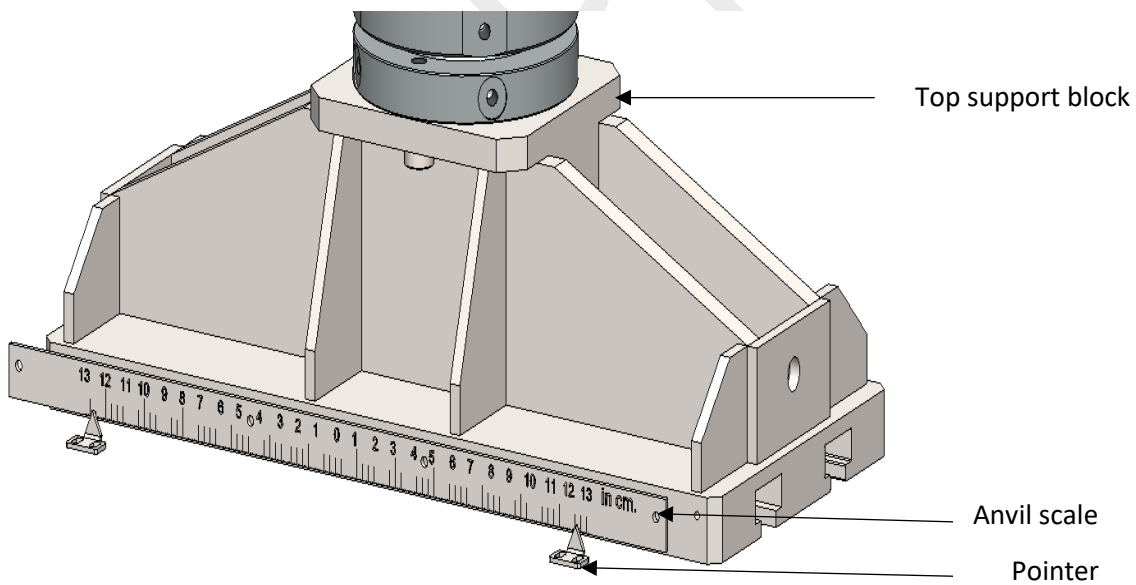


Figure: Scale mounting

- Mount the Top roller mount in the T slot of top support block using the M10 Nut and M10 x 25mm bolt.
- Attach pointer with the roller mount using M3x 10mm socket head cap screw.

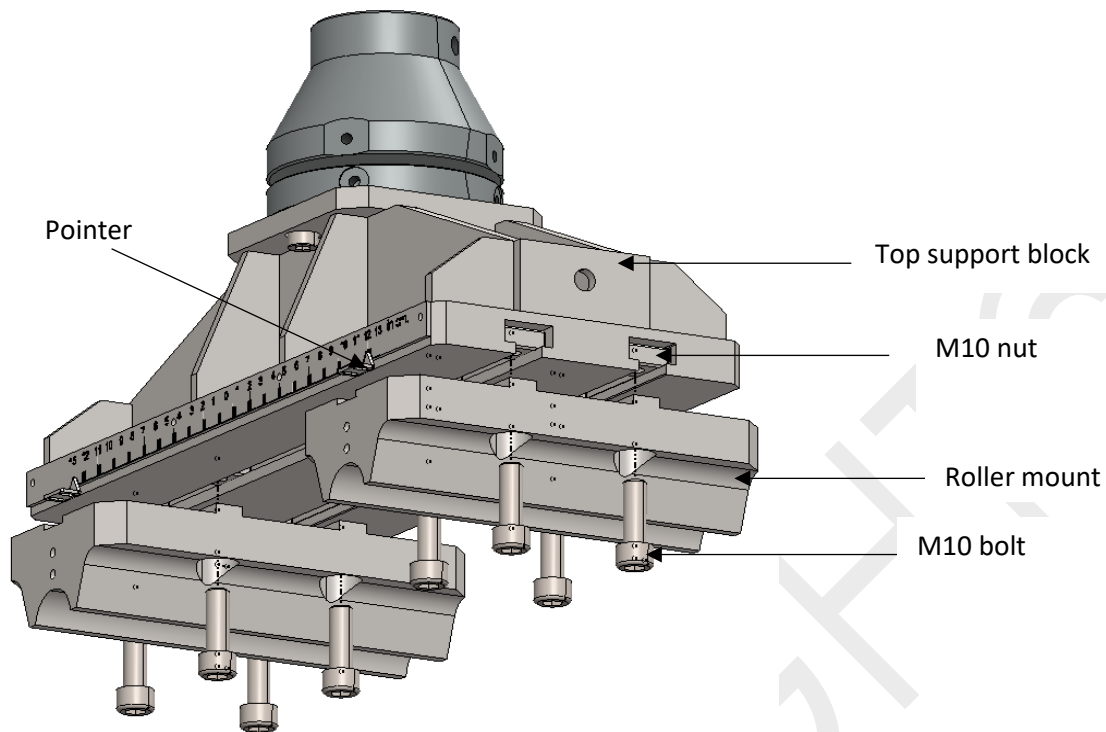


Figure: Roller mount assembly

- Attach a roller with roller mount using roller retainer clamp using M5 x 16mm bolt.

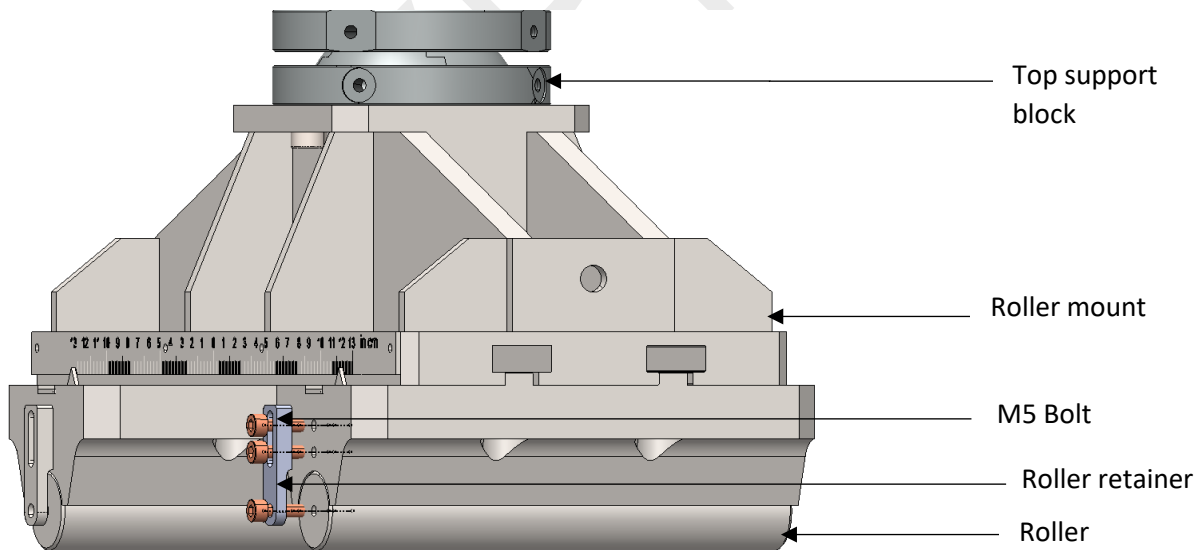


Figure: Roller assembly

- Place bottom support block on load cell.
- Place the loadcell spacer in between the bottom support block and load cell.
- Insert M27 X M16 conversion adaptor into the loadcell.
- After connecting the bottom support block with load cell using M27 to M16 x 28mm connector adapter, fasten partially the M16 x 140mm Allen bolt (Do not tighten completely at this point).
- Place the 3PBT or 4PBT fixture in diagonal position, such that it should not touch the load frame.
- After the setting diagonal position tighten the M16 x 140mm bolt.

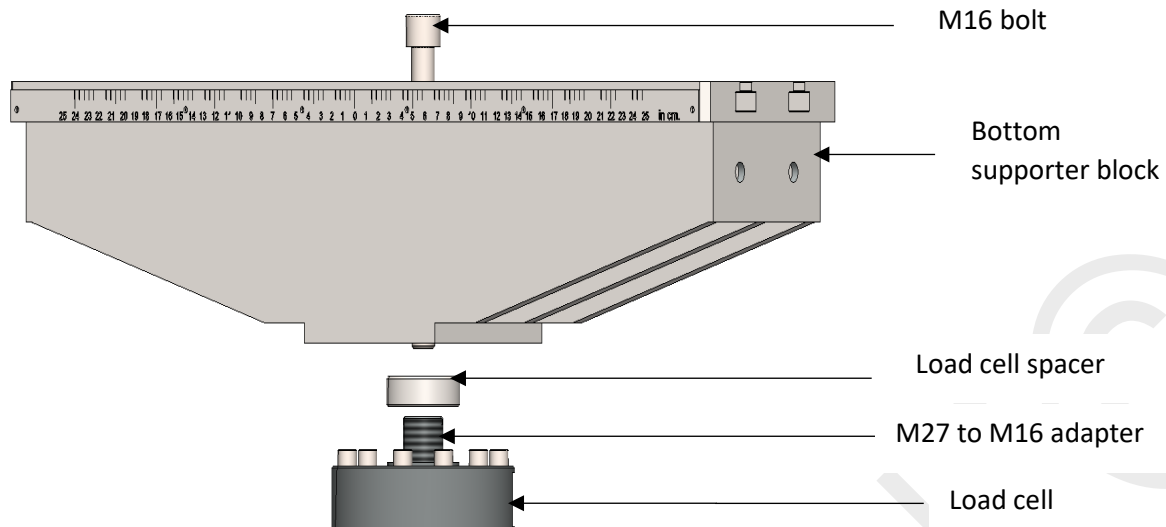


Figure: Bottom support block assembly

- Attach roller mount with the t slot bottom support block using M10 nut and M10 x 25mm bolt.
- After attaching roller mount place a roller on top of the roller mount.
- Connect the roller and roller mount using roller retainer and M5 x 10mm bolt.

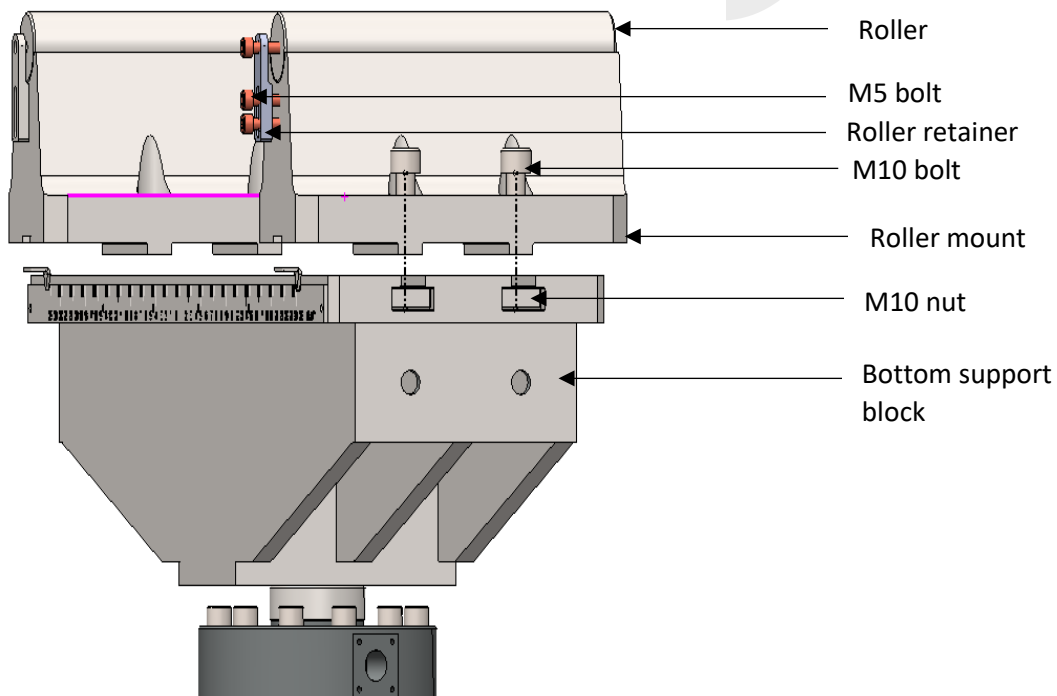


Figure: Roller mount & roller assembly

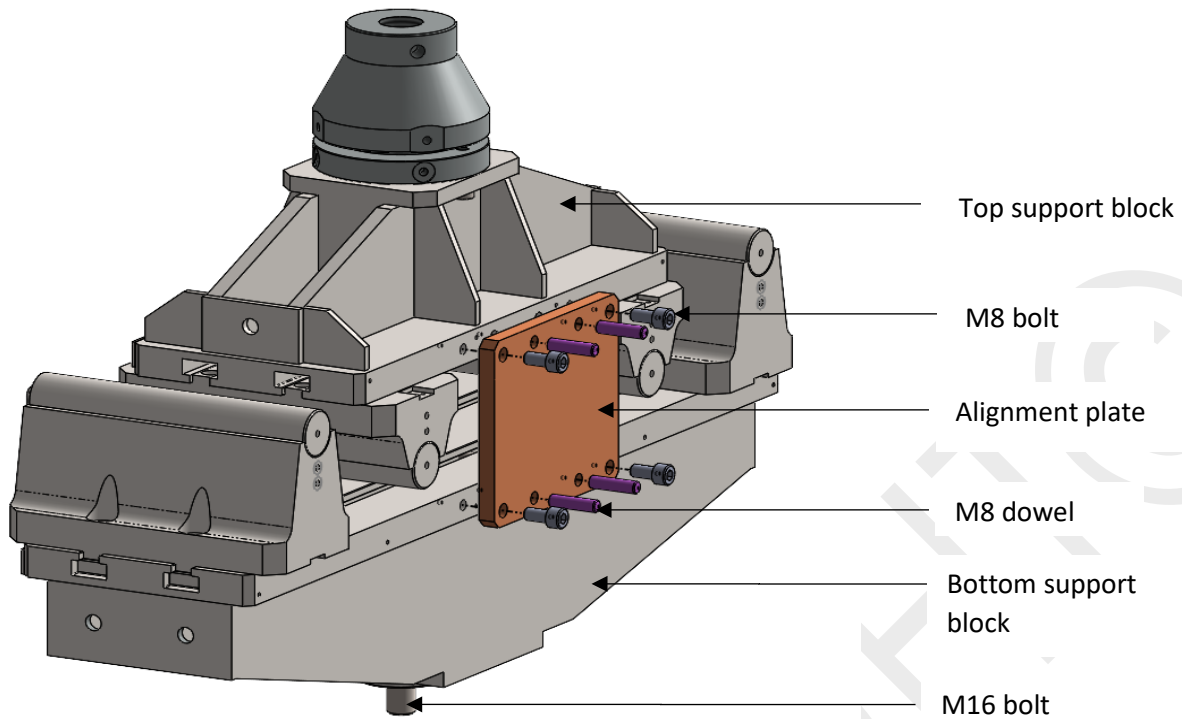
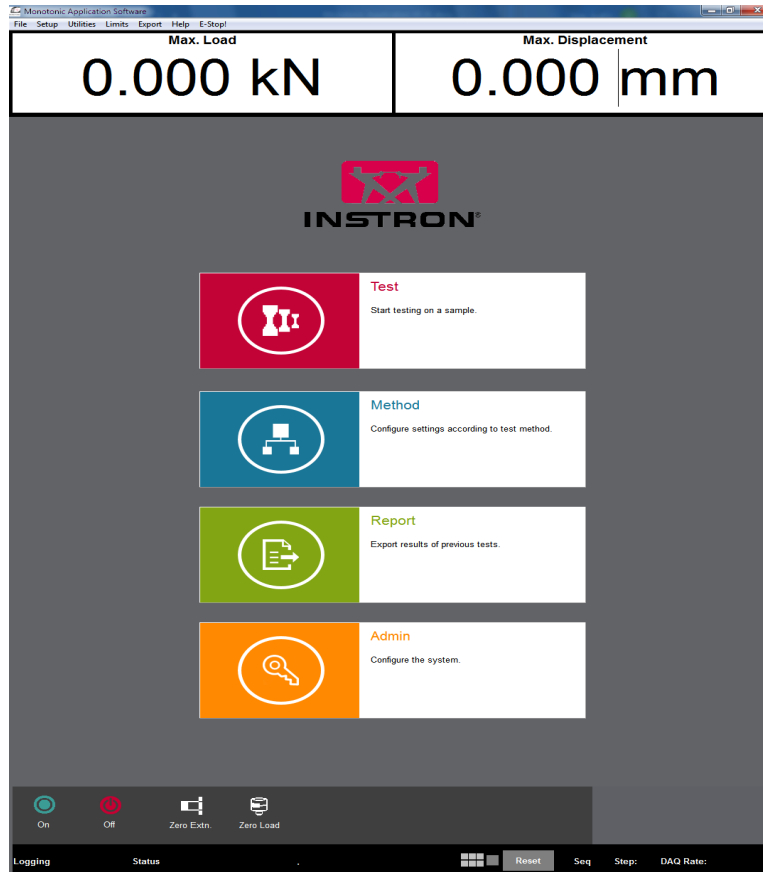


Figure: Fixtures alignment

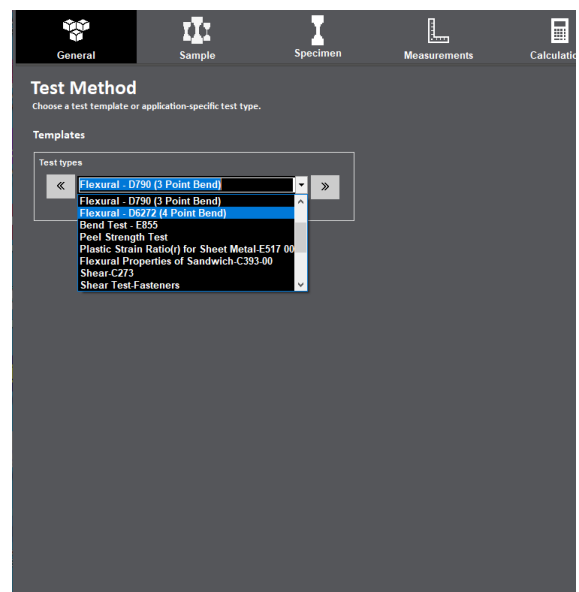
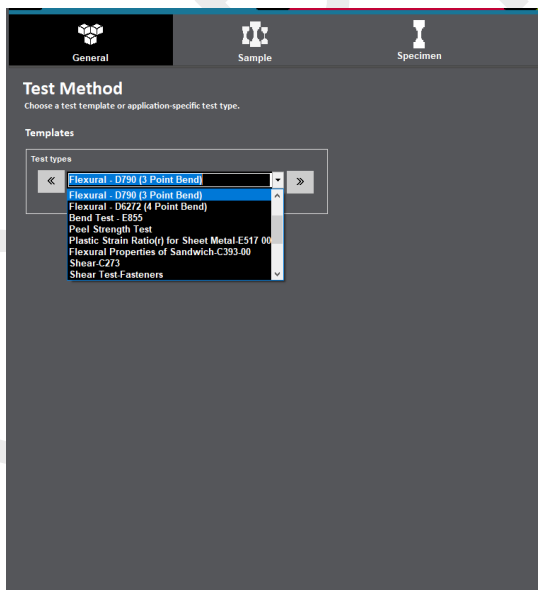
- Set top & bottom fixtures in the same direction using the alignment plate.
- Alignment plate will mount on the both the fixtures using M8 x 20mm dowel and M8 x 20mm bolt.
- Place the 3PBT & 4PBT in diagonal position, such that it should not touch the load frame.
- After the setting diagonal position tight the M16 x 140mm bolt.

9.2.3 Run a test

- Double click on the BluehillST logo & select “Test” tab



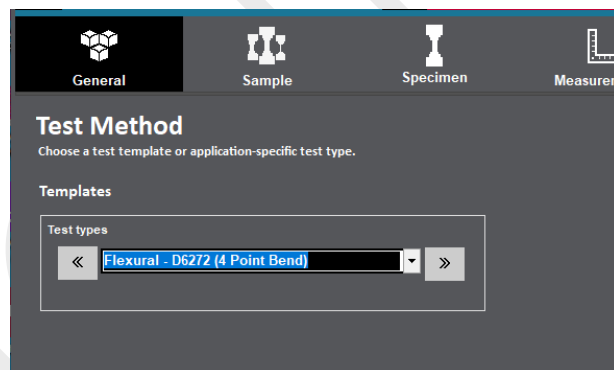
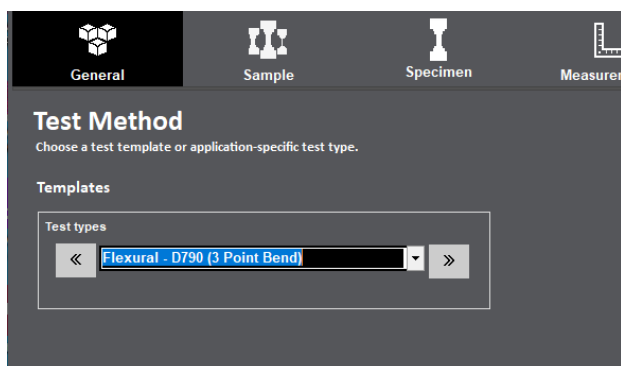
- The last loaded Test Settings file will be used automatically



- To load a different settings file, click on Browse Method button.
- To load previous file settings the user can click on the any of previous settings files from Most recent methods as shown below.
- To create a new settings file, click on Create a Method.
- To generate reports for completed tests, click on Report button.
- Change Screen Layout if required: either Horizontal or Vertical, depending on the screen being long or wide.
- Click on Connect button to go online.
- To get information on usage of software, click on Help >> Manual.

Settings

- Use the outer arrows to navigate between main pages.



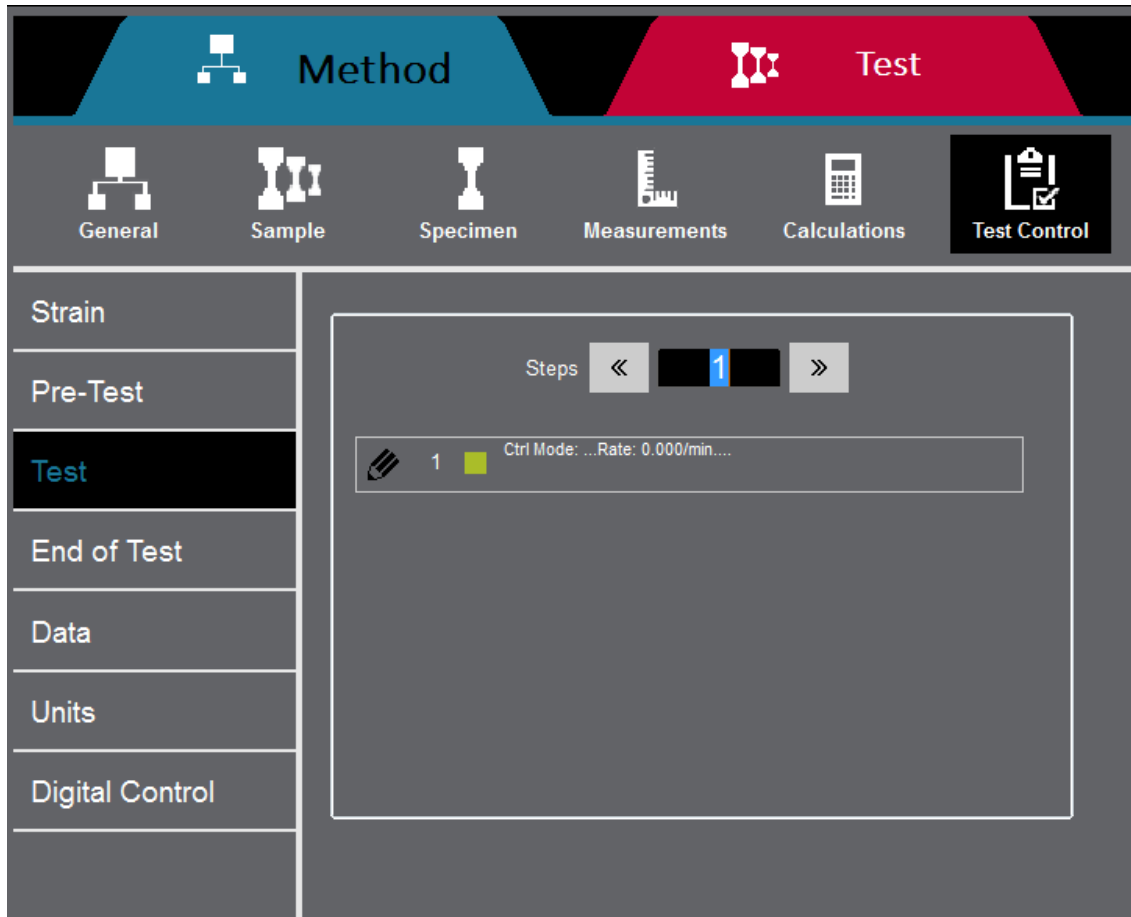
- Use the inner arrows to navigate between sub-pages.

Test Type

- Test Type: Select required type of test. Settings will change according to test type.
- Flexural D790 and Flexural D6272 Test: The option Stop at 5% Flexural Strain can be enabled. If enabled, Test will be terminated when this condition is met.

Test Settings

- Step Mode: Step by Step.



- Steps are loaded one after the other.
- Enter the number of Steps required. Maximum being 10 and minimum being 1.
- Click on  button provided for every step to edit parameters in the Edit Step window.

Edit Step:

- Select required Control Mode for the step. For tests not involving extensometer and Shear Test for Composites, do not select the Strain control mode.
- Enter the step rate either control channel unit/minute or /second. Or select Time in seconds and enter time required for the step. Also, if extensometer is mounted, there is the 3rd rate option: strain/s.
- End of Step: There are two options which can be used to end a step. Only if the step is the last one, none of them need be used.
- But, if it isn't, enable at least one option. In case of extensometer removal being used as an option, it is required to enable even the other one.
- The Displacement option under End of Step tracks the displacement. If Extensometer Mounted is enabled, then displacement is calculated from the extensometer channel. If not, then the stroke channel.
- Hold after End of Step: To dwell at the end of step, enable this option. Select Control Mode during the hold. Ensure that Strain control is not selected when Extensometer is not used.
- Select Time unit as seconds or minutes. Enter the required Hold Time.
- Use Left and Right arrows to load previous and next step respectively.
- Ensure that all parameters are correctly entered.
- Use Save button to save entries and selection.

The screenshot shows the 'Test Termination' configuration window. The top bar has 'Method' and 'Test' tabs. The navigation bar includes 'General', 'Sample', 'Specimen', 'Measurements', 'Calculations', and 'Test Control'. The left sidebar lists 'Strain', 'Pre-Test', 'Test', 'End of Test' (selected), 'Data', 'Units', and 'Digital Control'. The main panel is titled 'Test Termination' and contains four settings, each with a numeric input field and a checkbox:

- Drop in Load, %: 0.000
- Load Threshold, kN: 0.000
- Max. Load, kN: 0.000
- Max. Displacement, : 0.000

- Drop in Load, %: Test will stop automatically, if load drops by specified %. Load Threshold: Enter the Load value up to which drop in load is not considered.
- Max Load: If current load value is greater than or equal to this value, test stops.
- Max Displacement: If displacement is greater than or equal to this value, test stops.
- Strain %: If Strain % during test exceeds this value, it is terminated. For Bend Test - E855, this will be displayed as Permanent Strain %. This is very important for Bend Proof Strength calculation as desired Permanent set value comes into picture. It is essential that loading-unloading sequence be maintained without which the Permanent Strain % cannot be calculated.
- For all the above conditions, enable their respective checkboxes to activate them. Without doing this, they will not work as terminating conditions!

Pre-Test

- Home Position: Enable this option to use it to return to specified position at the specified rate. This feature is most useful before and after a test.
- Preload Settings: Enable this option to preload in required control mode to required Set Point at required rate. Also, Alternate mode can be selected. This can be used just before a test.
- Failure Mode Info: Use this to enter information or observation regarding failure of specimen. A popup window will open after test completion. Write in the given space as brief as possible.

The screenshot displays the 'Pre-Test' configuration window. At the top, there are two main tabs: 'Method' (blue) and 'Test' (red). Below these is a horizontal navigation bar with icons and labels for 'General', 'Sample', 'Specimen', 'Measurements', 'Calculations', and 'Test Control'. On the left side, a vertical sidebar lists various test stages: 'Strain', 'Pre-Test' (highlighted in black), 'Test', 'End of Test', 'Data', 'Units', and 'Digital Control'. The main content area is divided into sections. The 'Home Position' section includes a checkbox, a 'Home Position' field with a value of 0.000, and a 'Rate, /s' field with a value of 0.000. The 'Pre-Load Settings' section includes a checkbox, a 'Control Mode' dropdown set to 'Stroke', an 'Alternate Mode' dropdown set to 'Same Mode', a 'Set Point' field with a value of 0.000, and a 'Rate, /s' field with a value of 0.000. At the bottom, there is a checkbox for 'Failure Mode Info'.

Predefine Parameters:

- User can predefine the output parameters before conducting test. These parameters will be automatically selected during report generation to appear in reports.
- E.g., select the parameters for first sample of the test and then continue doing tests. Selected parameters apply for all the specimens tested until user changes them or the test type.
- These parameters will change depending on the type of test.
- Slope Region can also be predefined for Report. Enter appropriate values of Max and Min depending on the parameter selected (% of Load, Load, Stress or Strain %).

Method **Test**

General Sample Specimen Measurements **Calculations** Test Control

☒ Predefine Parameters

- ☐ Modulus of elasticity in bending
- ☐ Bending proof strength
- ☐ Max load

☒ Select All ☐ Remove All

Slope Region

<< Load >>

Min 30.000

Max 50.000

System Settings

- **Unit System:** Select the load and displacement unit as calibrated in MTL32_2020. In case of Strain Gauge mounted option being enabled for Tension or Compression or Shear D5379 tests, select the Strain unit by referring the MTL32_2020 software. Select required unit for Stress value display.

Method **Test**

General Sample Specimen Measurements Calculations **Test Control**

Strain

Pre-Test

Test

End of Test

Data

Units

Digital Control

Unit System

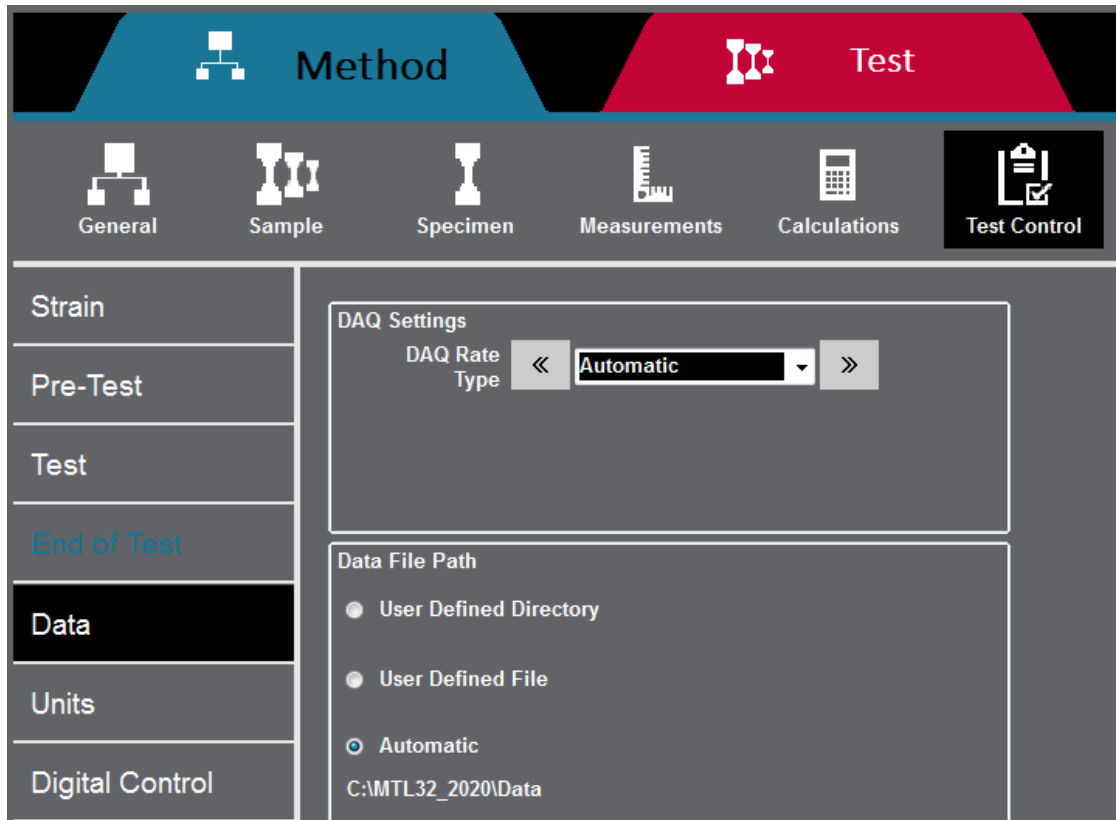
Load << kN >> Displ. << mm >>

Stress << MPa >>

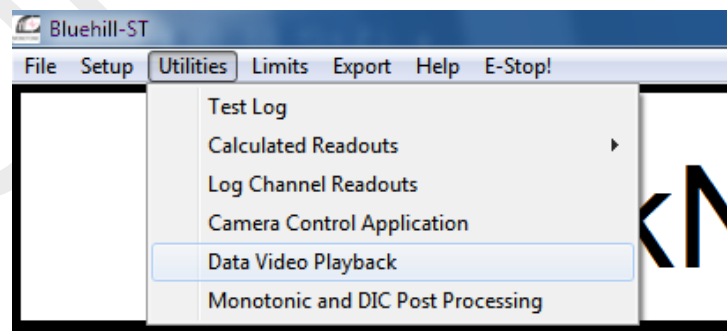
Unit selection must be same as that of MTL32 software...

- **Data File path:** If "Automatic" is selected, data files automatically get saved in "C:\MTL32_2020\Data". If "User Defined Directory" is selected, click on the **Select/Create Directory** to create your own directory. To create a data file in any location and with any name, use the "User Defined File" option.

- **DAQ Settings:** Can be automatic or manual or manual-dual rate. *Automatic* will set a DAQ Rate depending on the Test speed. To set a DAQ Rate of choice, select *Manual* option and enter it. To set two DAQ rates select the third option. Enter the value in Time or Stress that triggers change in DAQ Rate.



- **DAQ Channel Settings:** Match the log channels according to what is displayed. It is important that Stroke, Load and Strain channels are selected correctly.
- To record video with test or to trigger time periodic clicking of photos, check the *Use camera with test* option. This works only when Video-With Bluehill ST Test or Time Periodic-With Bluehill ST Test option is selected in the Camera Control Application.
- To open Camera Control Application, Data, and Video Playback Application and Bluehill ST and DIC Post Processing application, go to Utilities list of the main menu bar.



Specimen

- Select required specimen type. Enter required dimensions of specimen. These parameters change according to the Specimen Type.

Average value

- In case of Specimen dimensions being measured at different places and to get their averaged value, click on Dimensions  button.
- Enable Average option and enter dimensions. The averaged value is reflected in the main Specimen panel as shown below.
- Ensure that the parameters are correctly entered without which Test cannot be started!

Enter specimen dimensions measured at 3 different places.

☒ Average

	1	2	3
Width,	0.000	0.000	0.000
Thickness,	0.000	0.000	0.000

- Description: User can describe the Specimen ID, Operator and Batch etc. in their respective fields.
- Also, any other test related details can be entered under the Title and Description section.

Description

- Click on Method-> Sample
- Enable the checkbox of respective row of Title and Description. Enter here. The rows appear in Report only if enabled.
- Click Save button to save details entered so far.

Method Test Export

General Sample Specimen Measurements Calculations Test Control

Test Description

	Title	Description
☒	COMPANY	BISS
☒	TEST	TENSILE
☐		
☐		
☐		
☐		
☐		
☐		
☐		
☐		
☐		

Batch

Method Test Export

General Sample Specimen Measurements Calculations Test Control

Properties

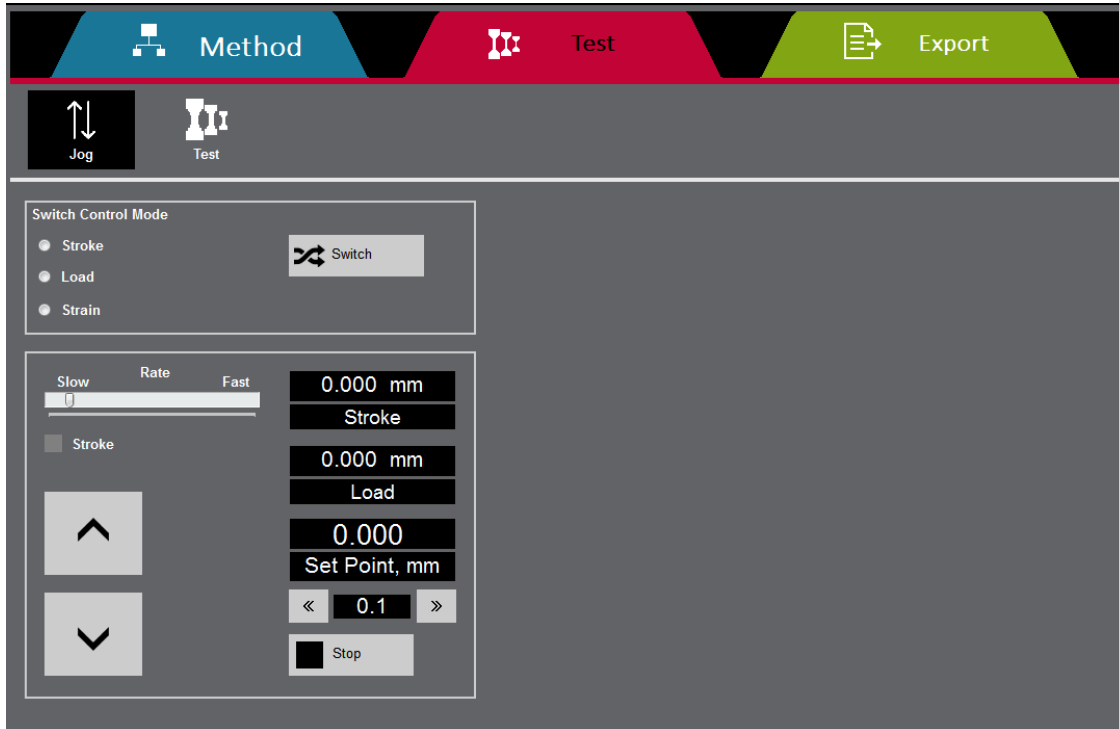
Inputs

Specimen ID

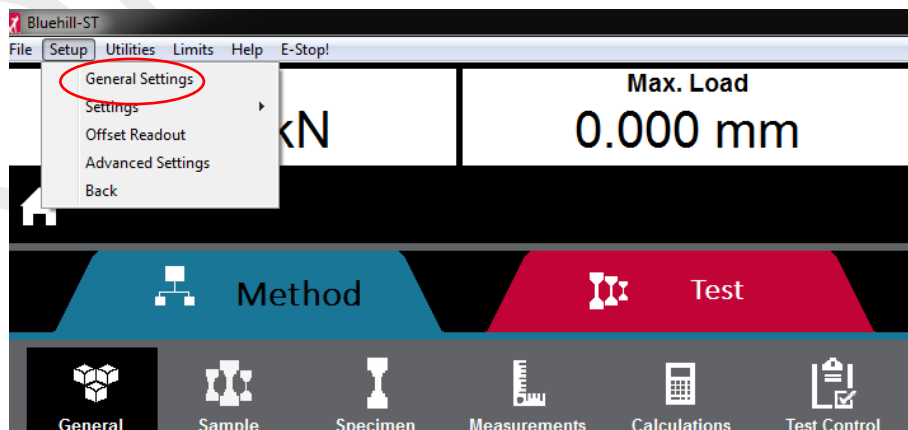
Operator

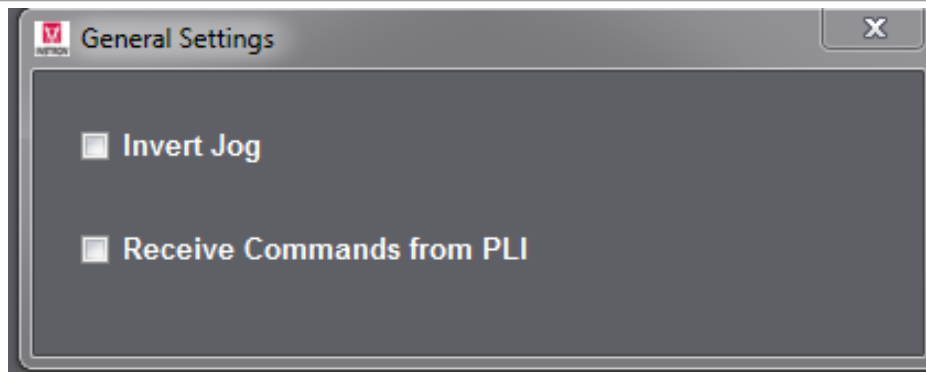
Temperature

Humidity (%)



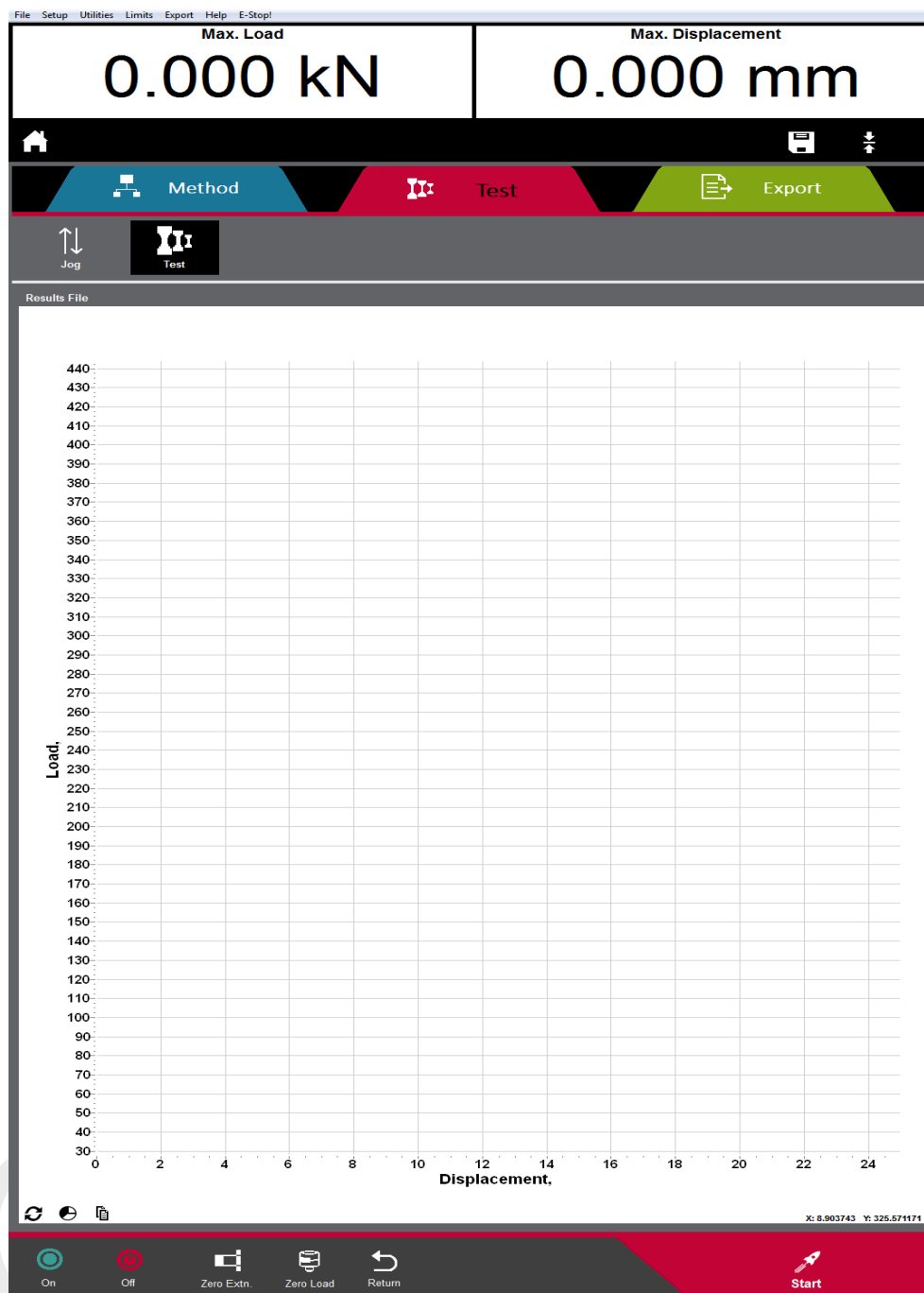
- Switch Control Mode: To change the current control mode, select required control channel. Then, click Switch button.
- Jog section:
- Ensure power is on before using this section. Also, that set point is not on hold!
- To change the current position, use the Top and Bottom arrow buttons. Click and hold either of them depending on the required change in position. The rate of Jog can be adjusted by moving the slider. By default, its rate will be 60 units/min.
- If the Jog Top and Bottom arrow buttons are moving the actuator in reverse directions, i.e., Top moving it down and Bottom moving it up, then use the Invert Jog option by opening the General Settings. From menu bar, click on Setup->General settings->Invert jog



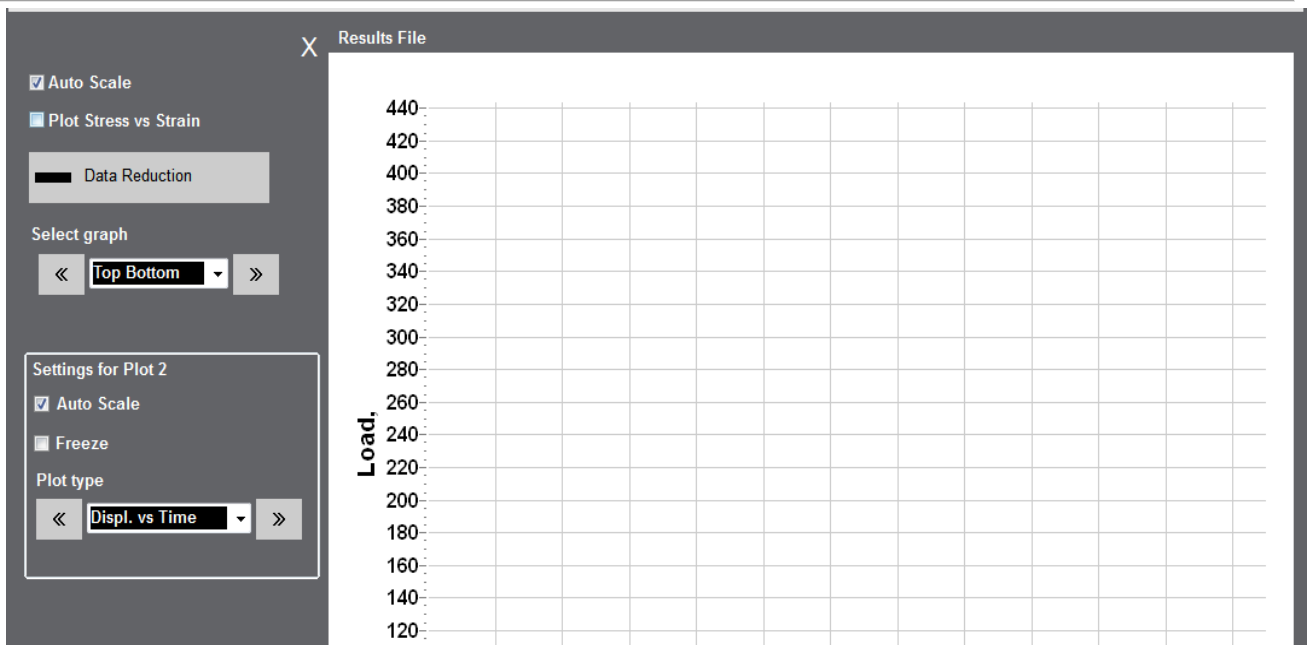


- Also, required position can be achieved by entering Set Point.
- Another form of jogging is using the << and >> buttons. Enter the value relative to current value by which it is decremented or incremented respectively.
- To stop the jogging, click the Stop button.
- Use any of the options based on convenience.
- This is mainly used while mounting specimen.

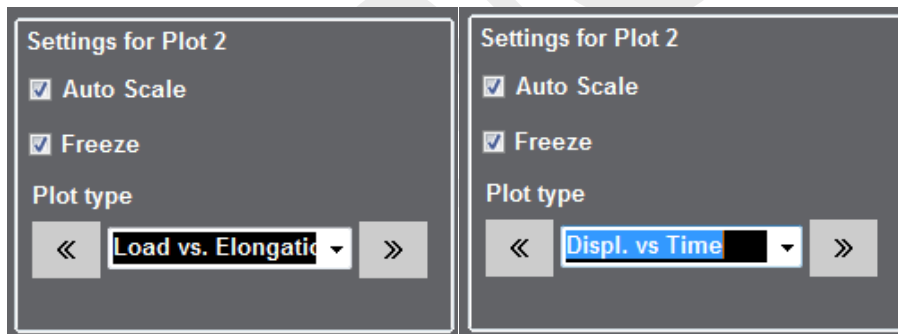
Run test



- User can save the readout meters, change the graph settings, set the extensometer readouts to zero when test is idle.
- Graph Settings: Click on the Pie Chart on the Bottom Left corner of the chart to view Graph Settings Panel as shown below.
- Use Refresh button to clear graphs when it's needed.
- Auto Scaling is checked by default.
- To get Stress vs. Strain plot instead of Load vs. Displacement plot, check Plot Stress vs. Strain. This option is available only for relevant tests. For Shear Test for Composites, Shear Strain % vs. Shear Stress plot is shown.
- To get Displacement vs. Time graph, select required orientation: either Top Bottom or Left Right, as shown below. Strain/Extensometer channel is selected as the displacement channel in case of Extensometer/Deflectometer, etc. being mounted. Else, position channel is selected.



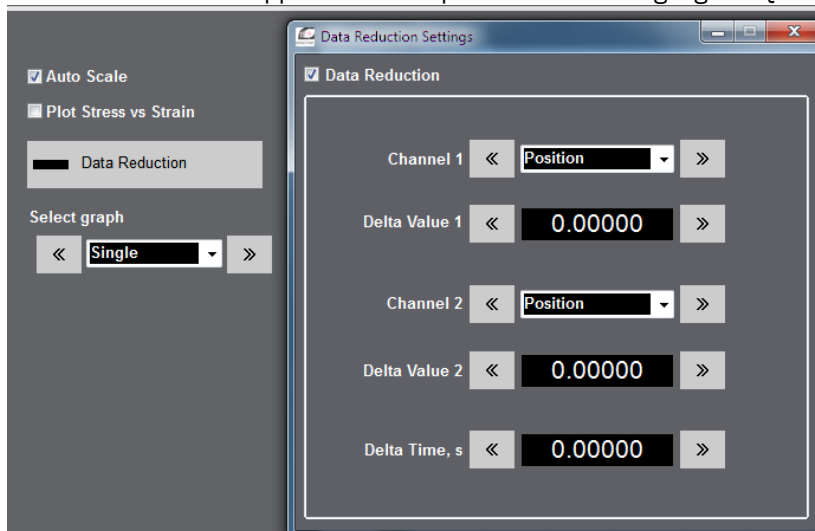
- Chart 2 has its own settings. Auto-scale is enabled by default.
- Use *Freeze* option to pause plotting. Uncheck it to resume plotting from that time onwards.
- In case of elongation parameter being calculated for the selected test type, Load vs Elongation plot option is available. Just select it under the Plot type option.



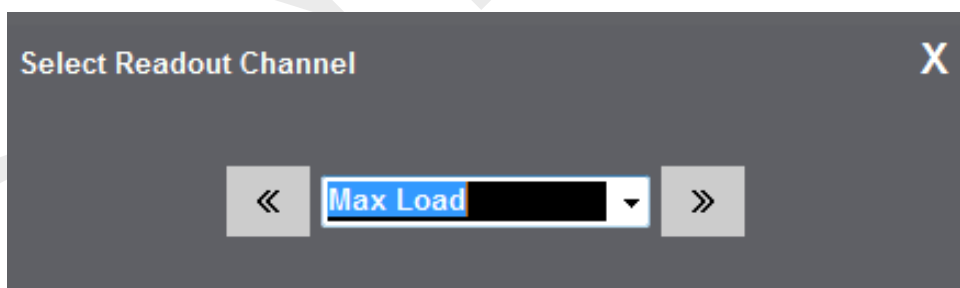
- Use « » buttons to decrease and increase respectively, the maximum value of Time axis.

Data Reduction

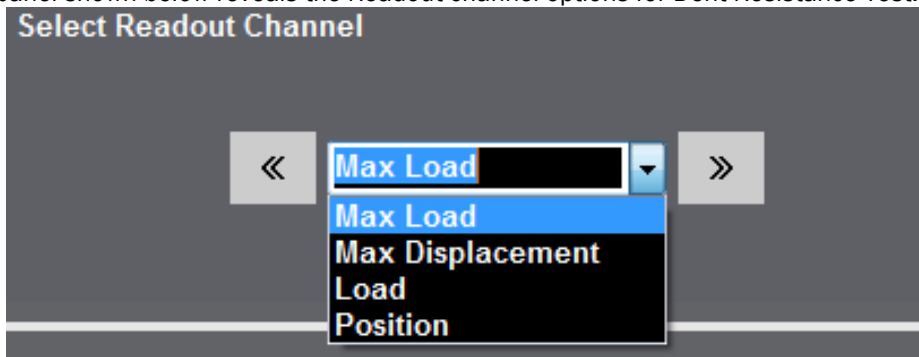
- This feature is used so that the Application is responsive even during high DAQ rates.



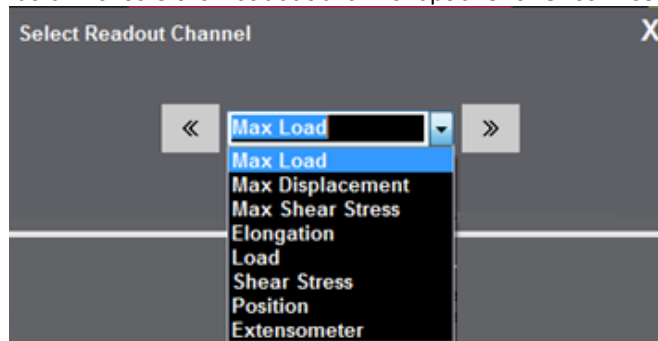
- Enable Data Reduction to activate this feature.
- Select any feedback or set point channel and enter delta value. This value is to be between 0 and 1. Entering zero indicates that the channel will not be used for data reduction.
- Enter Delta Time in seconds also between 0 and 1.
- Data will be plotted in graph only when there is change in channel or time by respective delta values.
- If this feature is not required, then uncheck Data Reduction.
- Readouts:** Double Click on Readout to change its channel.
- The panel shown below opens. Select required channel. The panel shown below reveals channels according to the Test Type.



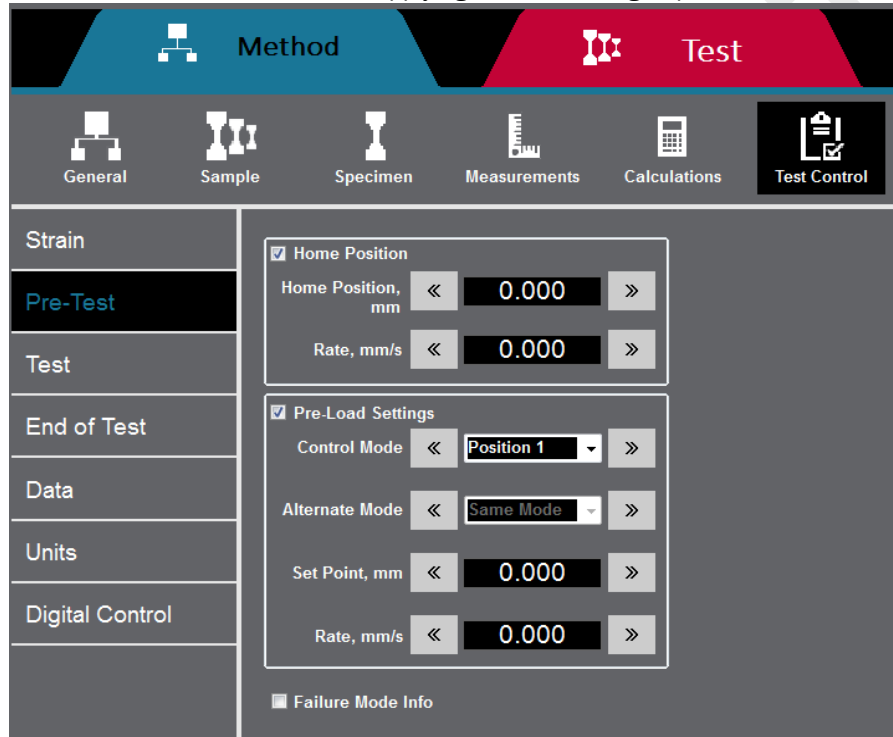
- The panel shown below reveals the Readout channel options for Dent Resistance Test.



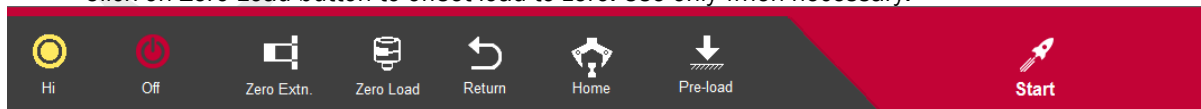
- The panel shown below reveals the Readout channel options for Shear Test for Composites.



- Use the Preload or Home buttons for applying load or moving to positions as defined in test settings.

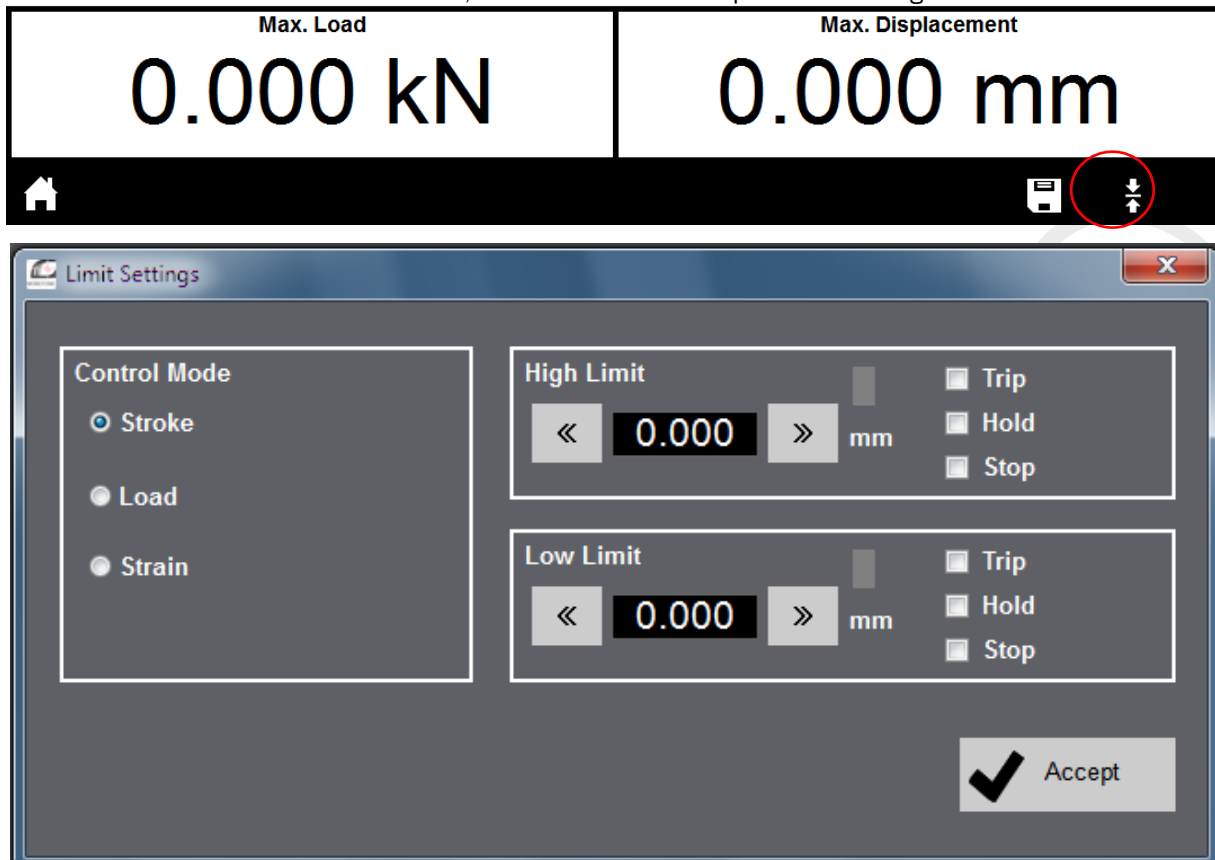


- To return to the start position after a test, simply use the Return button.
- Click on 'Zero Extn.' button to offset Strain/Extensometer channel(s) to zero. Ensure that Strain channel is not in control while doing this.
- Click on Zero Load button to offset load to zero. Use only when necessary.

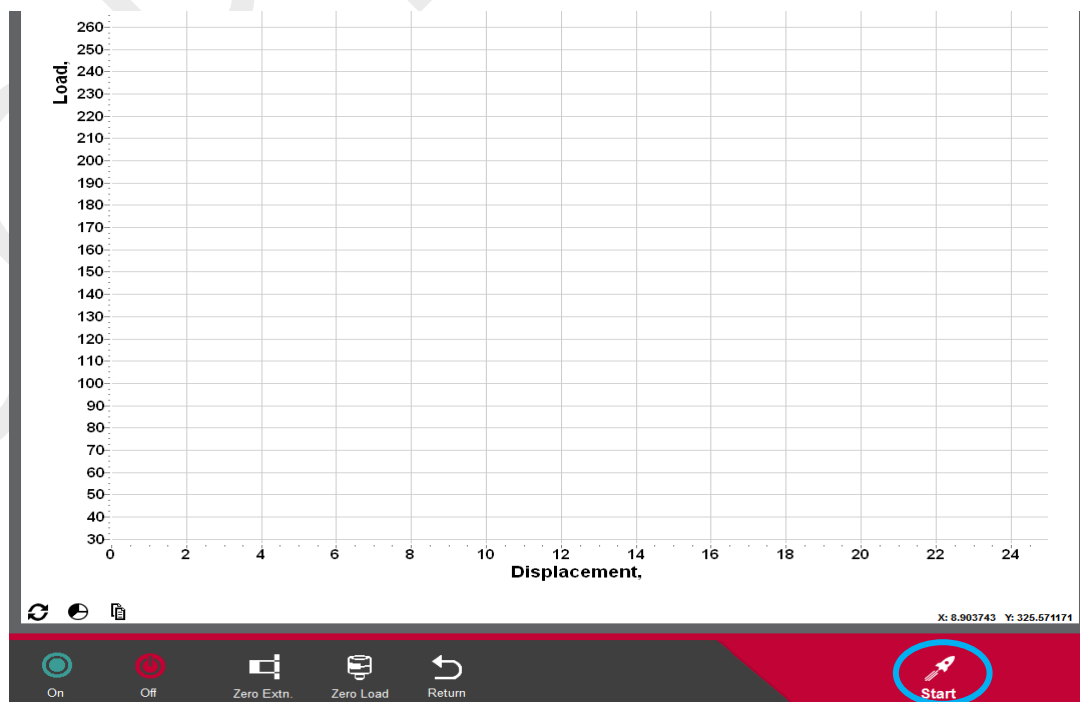


Limits

- To set Limits for control channels, click *Limits* button to open Limit Settings.



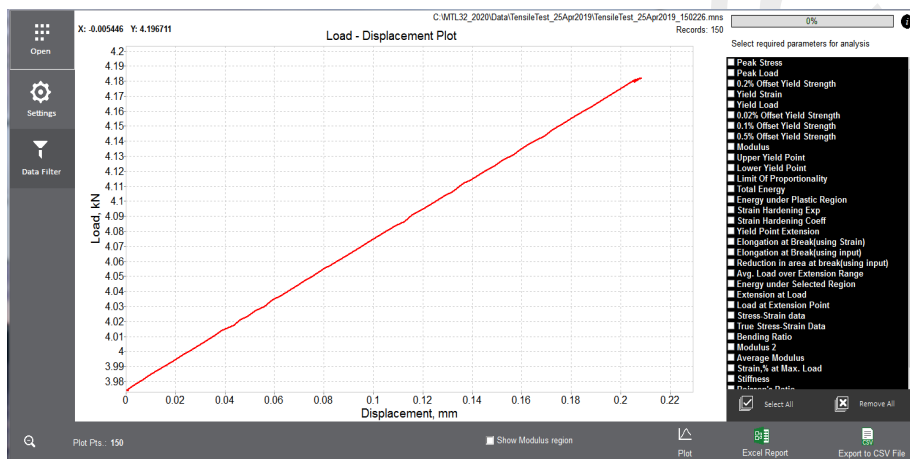
- Select Stroke, Load or Strain. Set high limit for this channel. Tick the action to be made when limit is hit. Similarly set low limit and its action.
- Click Accept button to set the limits.
- Now system is ready to run a test



- Go to the “Run” page & click on “Start” button to start the test
- A graph would be drawn on the application showing load vs strain relationship
- Observe the specimen and the graph simultaneously and you will notice the specimen break after a certain period.
- The test will come to a halt as the per the test parameter specifications.

9.2.4 Exporting Data

- To export result to MS Excel sheet, follow steps mentioned as below
- Click on “Report” in the home page (Monotonic/Bluehill)



- Post Processing a window will open with the latest data file by default,
- Here the user can view “Modulus” by enabling “Show Modulus Region” check box
- Select the predefined parameters from “Select Required Parameters for analysis” window. These parameters will be analysed & will subsequently appear in the Excel report
- Then click on “Excel Report” for result to appear in Excel sheet

Note: Refer to the Monotonic/Bluehill application manual for more details on data processing in single files or Multiple files/Batch processing.

Note: If no other test is being conducted, its recommended to Turn OFF/Shut down the following devices

- System hydraulic pressure (In application)
- Monotonic/Bluehill application
- MTL32 basic software
- Controller
- Computer
- UPS
- UPS input
- Selector Switch
- Lastly HPS Mains.

9.3 Fatigue Crack Propagation Testing

TITLE	Standard test method for Fatigue Crack Propagation Testing
TEST REFERENCE	ASTM E 647

Requirements

- Grips/Fixtures: C(T) Cups/3Point Bend fixture
- Wedges :

Sl No	Available Cups (Thickness)	Specimen	
		Width	Thickness
1	6.35	25.4	6.35

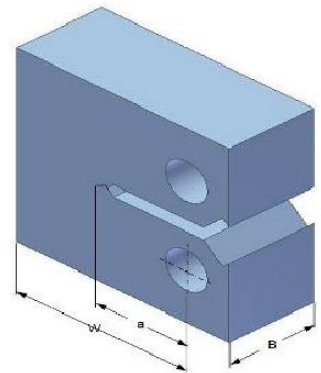
- COD: Recommended 5 mm gage length extensometer.
- Torque Wrench (M10 socket), M5 Allen key, Wooden blocks

Displacement/strain-controlled test procedure

- Specimen geometry
- Powering up the system
- Test setup
 - Mount grips
 - Insert suitable cups & mount specimen
 - Mount extensometer
- Run the test
- Export data

9.3.1 Specimen Geometry

- a. Measure specimen width, thickness & crack size
- Instrument required - Vernier Calliper
- b. Take the average value of 3 readings



Refer ASTM E 647 for more details on specimen

9.3.2 Powering up the system

- a. Please ensure to switch ON
 - System Mains
 - Host Computer/Laptop
- b. Double click on MTL 32  icon on the desktop
- c. Click “Go-Online” button & wait for 15-30 seconds



9.3.3 Test Setup

❖ Mount grips & insert suitable wedges

Top grip on the actuator rod end and bottom grip to the load cell will be already mounted only accessories pin for respective tests must be mounted (refer to the Machine & Maintenance manual>> Universal Grips manual for more details on universal grips. Before that ensure PLI application on tab is open.

Refer to Tension testing (Mount grips) for details on cross head adjustment

❖ Insert suitable cups & Mount specimen

1. Release the torque on the M12 bolt, insert the accessories pin and butt it to stopper in top and bottom grip. Do not apply clamping force to the clamping bolts.

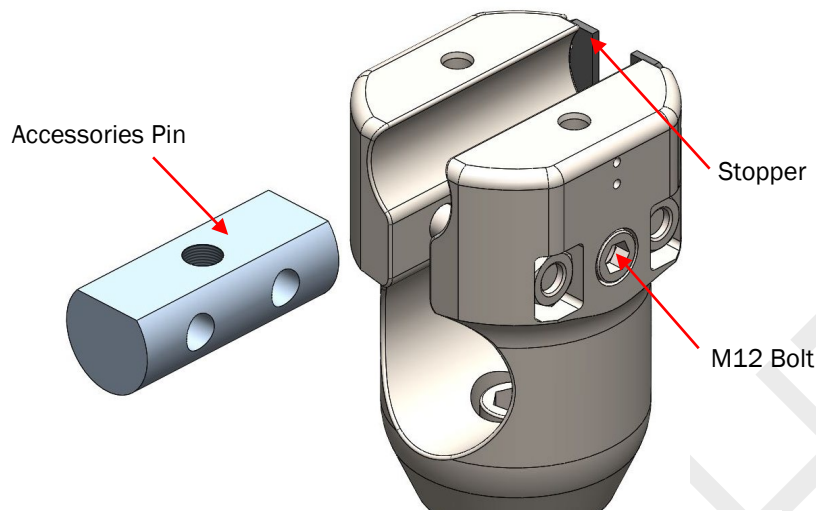


Figure: Accessories Pin insertion.

2. Fasten the M16 adaptor to accessories pin tightly (30 Nm) and mount M36 male adaptor and M36 lock nut to top and bottom grips.

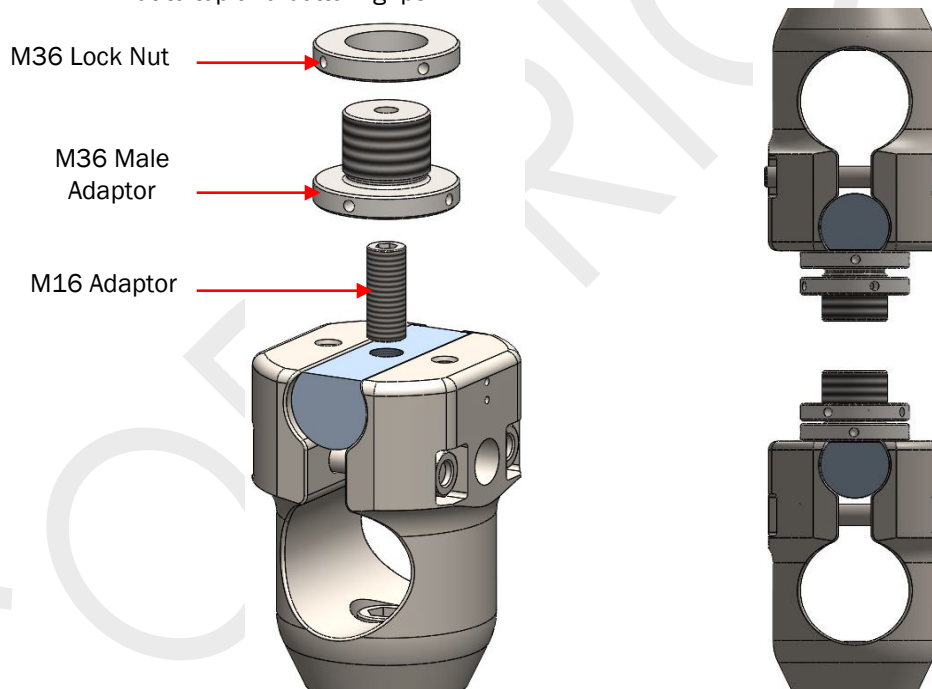


Figure: Accessories Mounting.

Note: This assembly remains common for 3-PB, 4-PB, compression platen & clevis grips.

3. Tighten the M36 male adaptor to grip face using the lock nut pin as shown in figure 17.

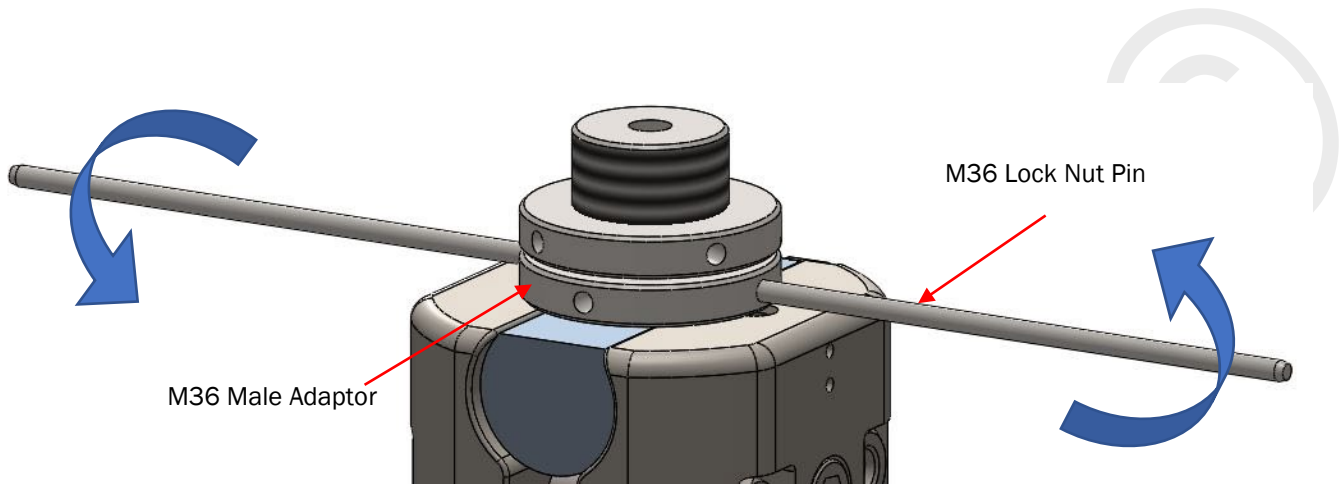


Figure: Tightening of M36 Male adaptor.

4. Ensure to apply a torque equivalent to 30 Nm to each clamping bolt.

Fracture Mechanics Test Setup (using Clevis Grip)

1. Mount the clevis grips to the top and bottom M36 male adaptor.
2. Place the preloading specimen and Insert the clevis pin on both the grips
3. Apply a tensile load of 5kN to the align top and bottom clevis grips.
4. Tighten the M36 lock nut towards clevis grips using a lock nut pin.

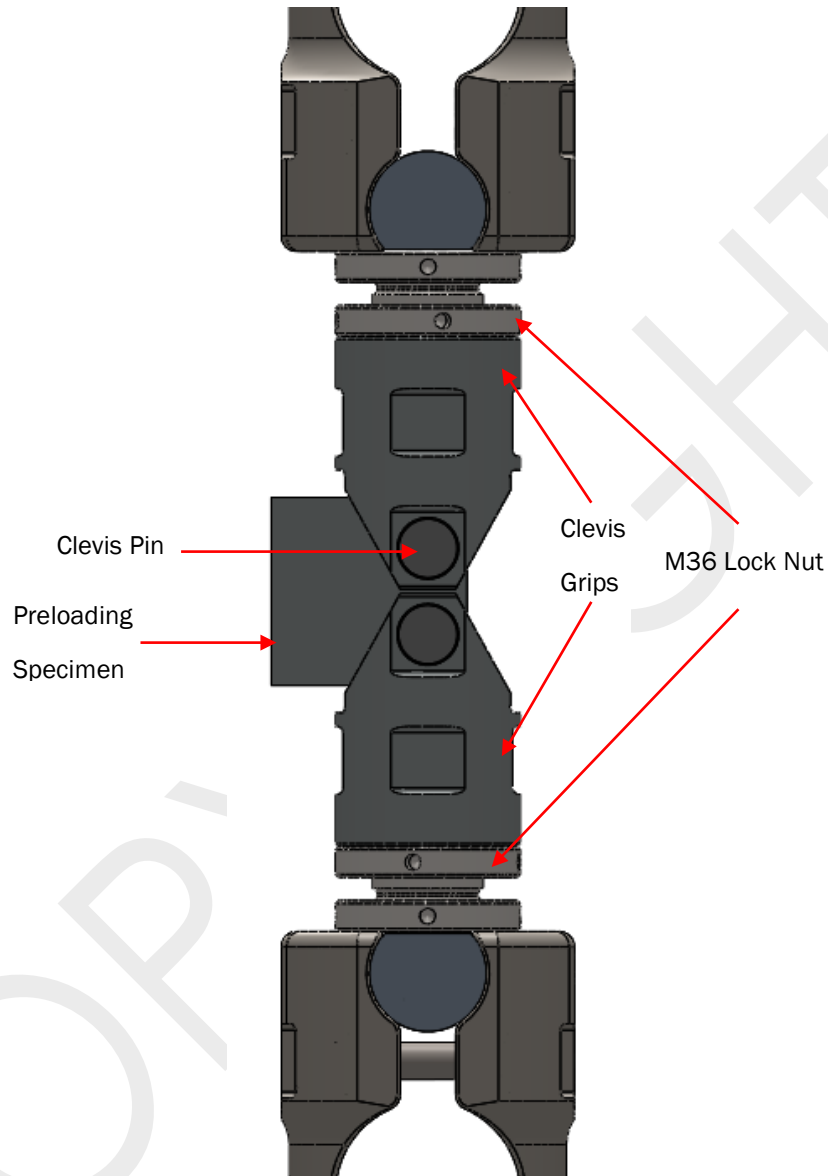


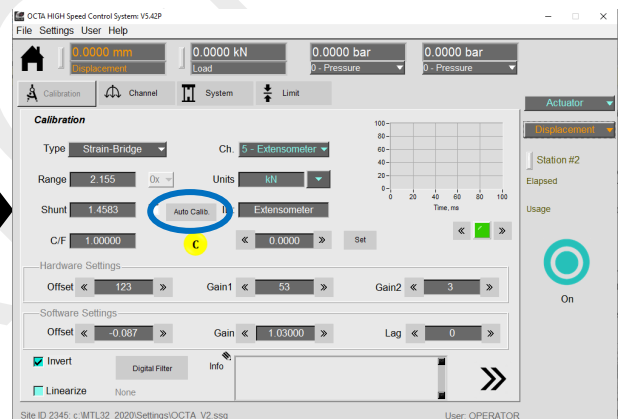
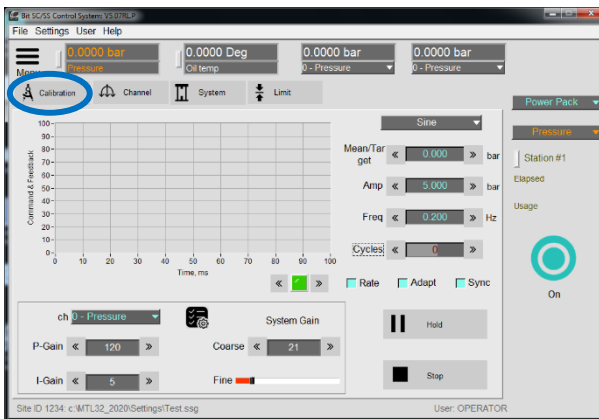
Figure: Fracture Mechanics Test Assembly.

❖ COD Mounting

- Locate strain 1 channel (on load frame) & connect COD to it as shown below



- Before mounting the extensometer, **ensure that the control mode is not in COD (select stroke)** & it is recommended to auto calibrate it.
- To auto calibrate, go to the MTL32 “Calibration” page & click on “Auto calibrate” button & wait for 30-45s for auto calibration to complete

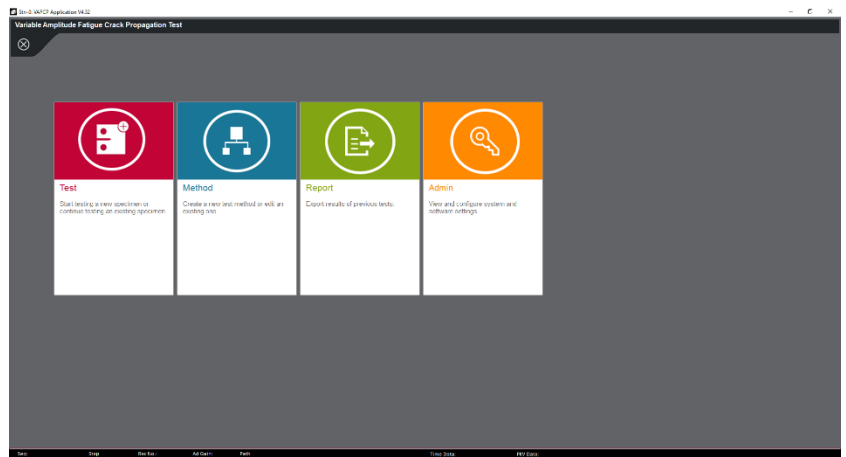


- Compress the cod blade by 2-3 mm and place it on specimen groove

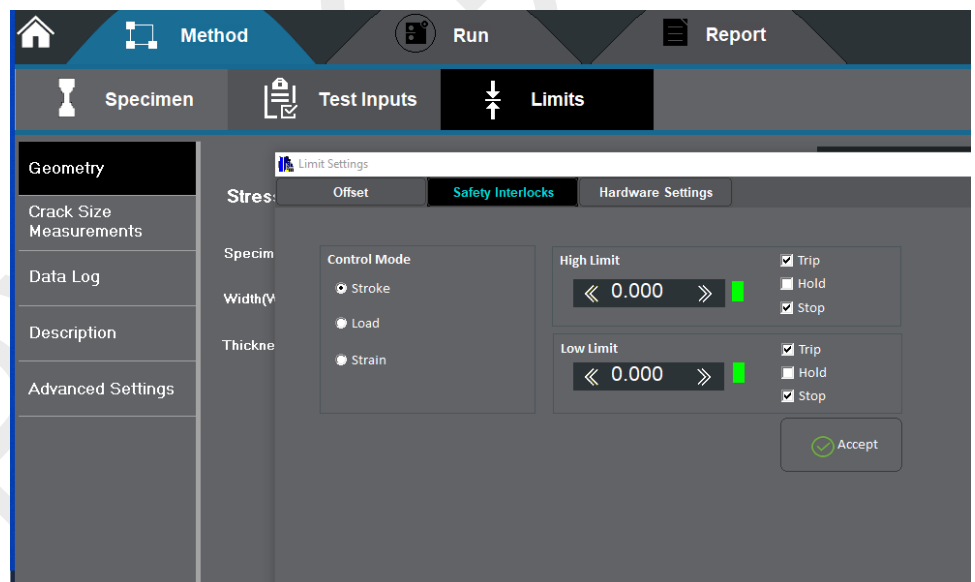


- Auto calibrate again, ensure that the COD read out on system is close to zero
- Set COD, Stroke & Load limits in the limit panel

- Double click on the Application software **VAFCP** icon

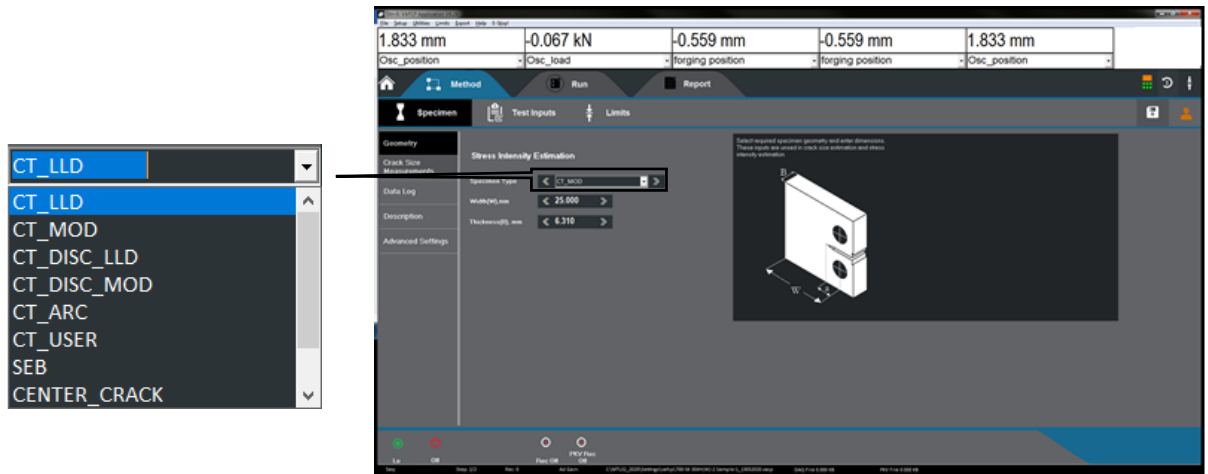


- Click “Method” to create a new test method or edit an existing one
- Click “Create a method” button to create a new Settings file (Enter file name to be saved) or choose an existing file from “Most recent methods”
- Ensure Site ID & station number match those entered in MTL32_2020 software & click next to go to “Method Page”
- Go to “Limits” page & set “High” & “Low” limits to Stroke & COD controls. Enable check box from the given options, and find out what action is required when the limit is hit

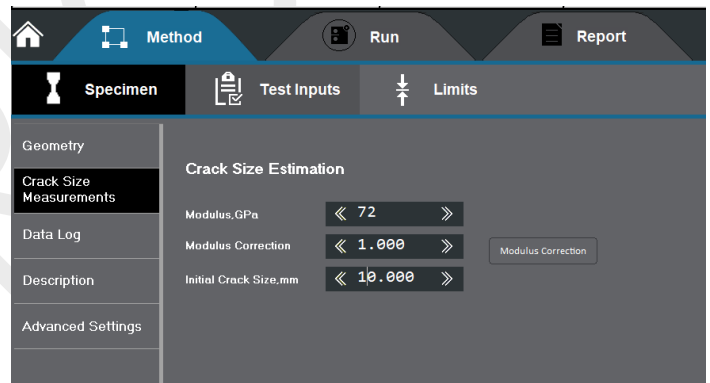


9.3.4 Run a test

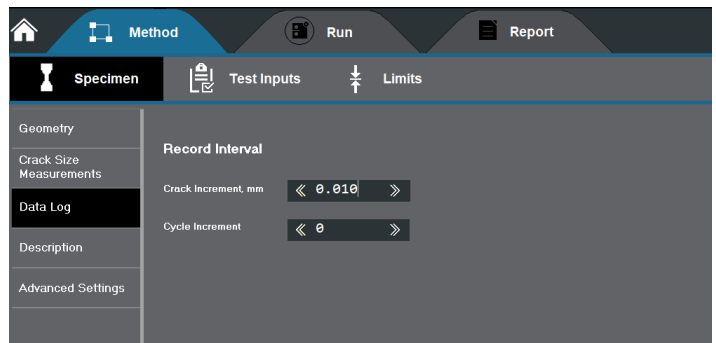
- Go to “Specimen page”, select specimen type from the drop-down menu & enter specimen width & thickness, specimen example pictures for respective specimen types will be available to aid the user



- Under “Crack size measurement”, enter Young’s modulus & initial crack size of the specimen. Modulus Correction should be default 1. If there is any deviation in crack size measurement, modulus correction can be done. Modulus correction should be within 5%. Click the Modulus Correction button to correct the entered value



- Record Interval: Enter the Crack Increment in mm. For every increment of crack size by a specified value, data will be recorded into the results file. Data will be logged to assignment file (.vacp). This Data logging function is automatic
- Cycle Increment: For every increment of cycle count by specified value, data will be logged into the assignment file (.vacp)



- Under “Description” page, enter the details which will appear in the test report

Geometry	Specimen ID 700-5K 304H(W)-2 Sample-1	Manufacturer Brigade
Crack Size Measurements	Batch 700-5K 304H(W)-2 Sample-1	Customer IIT
Data Log	Operator Vishwas	Specimen Description 6.25mm RT test
Description	Company and Department BISS	Others
Advanced Settings		

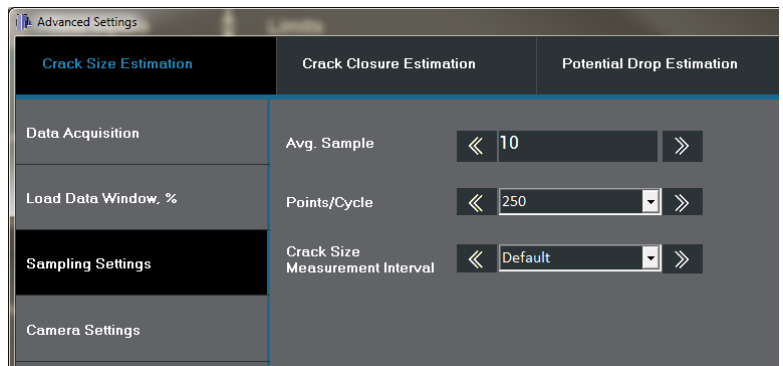
- Under “Advanced setting” >> “Crack Size Estimation” >> “Data Acquisition”, select the respective channels

Advanced Settings	
Crack Size Estimation	Crack Closure Estimation
Data Acquisition	Load Channel << Load >>
Load Data Window, %	COD Channel << COD >>
Sampling Settings	SP Channel << SP-Act >>
Camera Settings	Cycle Channel << CY-Act >>

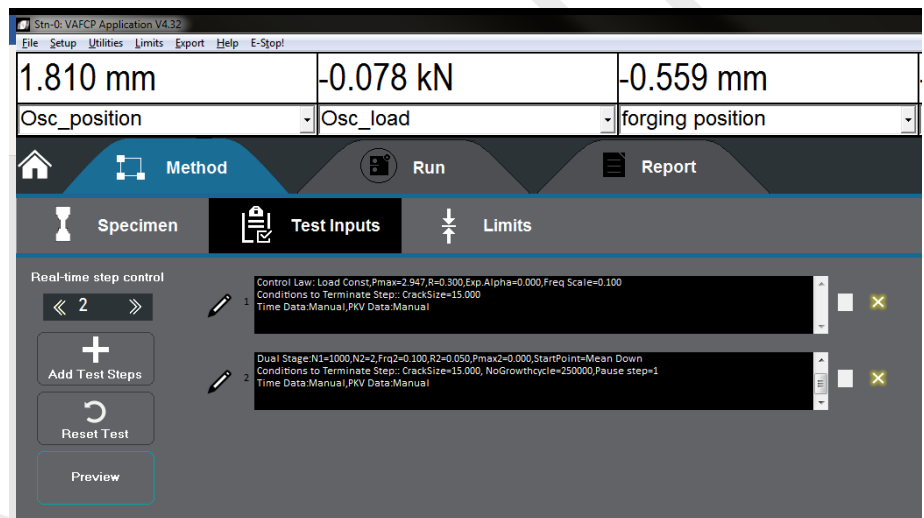
- Load Data Window limits: Enter “From” and “To” fields. This window is used to calculate compliance. “From” and “To” inputs are in % of Pmax
- In Compliance Segment- Three options available Unloading, Loading and Both. Default selection is Unloading. Other options can be selected based on requirement

Advanced Settings	
Crack Size Estimation	Crack Closure Estimation
Data Acquisition	From << 95.000 >>
Load Data Window, %	To << 40.000 >>
Sampling Settings	Compliance Segment << Unloading >>
Camera Settings	

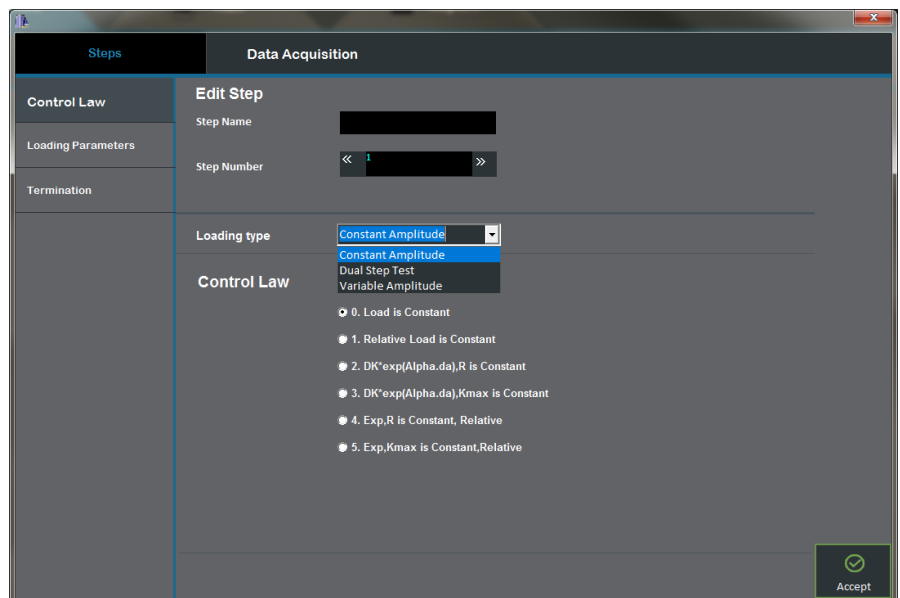
- Average Sample: This is moving average sample size for crack size
- Points/Cycle: Number of data points per cycle
- Crack size measurement interval: “Default” selection will measure crack size once in a second if frequency is more than one Hz. “All cycles” selection will measure crack size for all cycles if frequency less than or equal to 10Hz



- Go to “Test Inputs”, click “Add Test Steps” to add steps, here the user can add a maximum of 20 steps with different loading condition.
- Click edit logo  to edit respective steps, cancel  logo to cancel respective step



- After clicking edit logo, the User can select “loading type” from the drop-down menu & “Control Law” from the available options as per the requirement



- Under “Loading Parameters” the user needs to enter crack size and Kmax. With these inputs, load will be calculated, considering the specimen geometry and instrumentation used. This can be used to calculate required Pmax while starting the test

Steps	Data Acquisition
Control Law	Edit Step
Loading Parameters	Step Name
	Step Number << 1 >>
Termination	Loading type Constant Amplitude
	Loading Parameters Pmax, kN << 0.000 >> R << 0.000 >> Exponential Alpha << 0.000 >> Frequency, Hz << 0.000 >>



Fracture Mechanics Calculator	
P, kN << 1.572 >>	☐ K-Control
K, Mpa/m << 15.000 >>	K0, Mpa/mm << 0.000 >>
a/W << 0.500 >>	a0, mm << 0.000 >>
a, mm << 12.700 >>	Alpha << 0.000 >>
Compliance mm/kN << 0.119808 >>	Close

- After determining Pmax from “Fracture Mechanics Calculator”, enter Pmax, R value & Frequency in loading parameters
- Under “Termination”, enter specific condition to stop test/go to next the step

Steps | **Data Acquisition**

Control Law

Loading Parameters

Termination

Edit Step

Step Name: []

Step Number: << 1 >>

Loading type: Constant Amplitude

Termination

Crack Size, mm: << 15.000 >>

Incremental Growth, mm: << 0.000 >>

No Growth Cycle: << 0 >>

Kmax, Mpa/m: << 0.000 >>

Cycle Increment: 0

☐ Pause at end-of-step

☐ Power off at end-of-step

Accept

- Enable “Pause at end-of-step” check box to hold condition to activate once current step hits the termination condition
- Enable “Power off at end of step” check box to turn off Power respectively
- Go to “Data Acquisition”, click “Create file” to create a new file with the desired file name & location

Create File

C:\MTL32_2020\Data\VAFCP Test 1.mtd

DAQ Type: Automatic

On Time, s: << 0.000 >>

On Cycles: << 0 >>

Cycle Channel: []

Off Time, s: << 0.000 >>

Off Cycles: << 0 >>

Manual

Manual

Automatic

Cycle Periodic

Time Periodic

None

- Select “DAQ Type” from the drop-down menu
- Go to “Limits” page & set “High” & “Low” limits to load controls (5 - 10% higher than Pmax). Enable check box from the given options, what action is required when the limit is hit

Limit Settings

Offset | **Safety Interlocks** | Hardware Settings

Control Mode

Stroke

Load

Strain

High Limit

<< 3.500 >>

☒ Trip

☒ Hold

☒ Stop

Low Limit

<< 0.000 >>

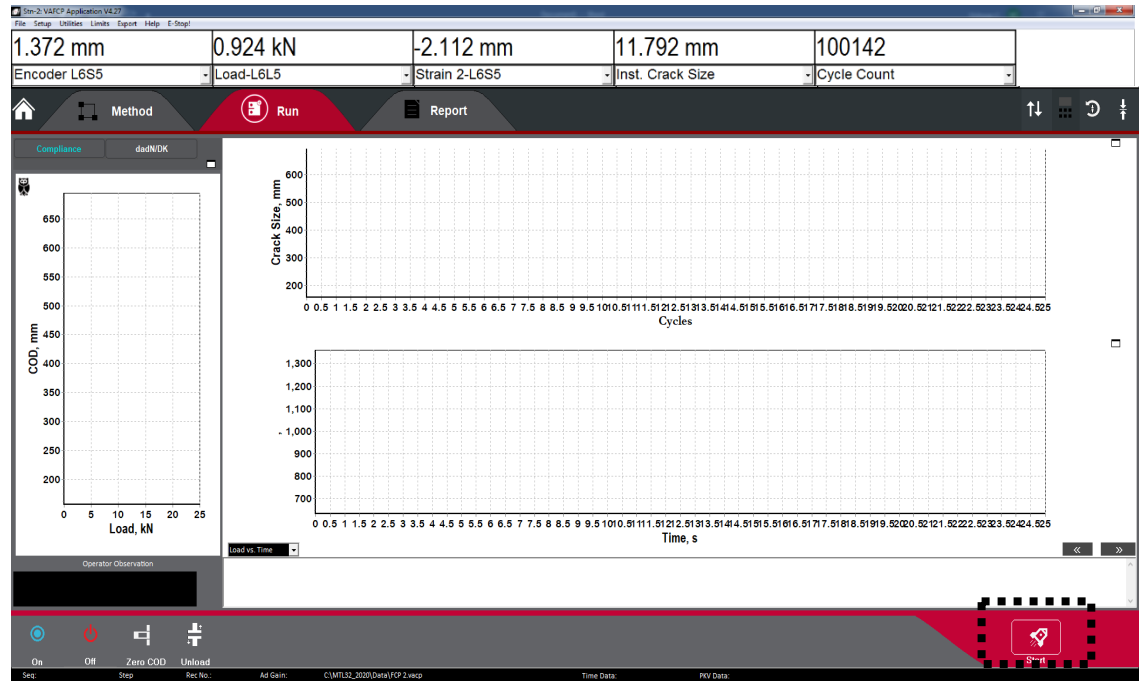
☒ Trip

☒ Hold

☒ Stop

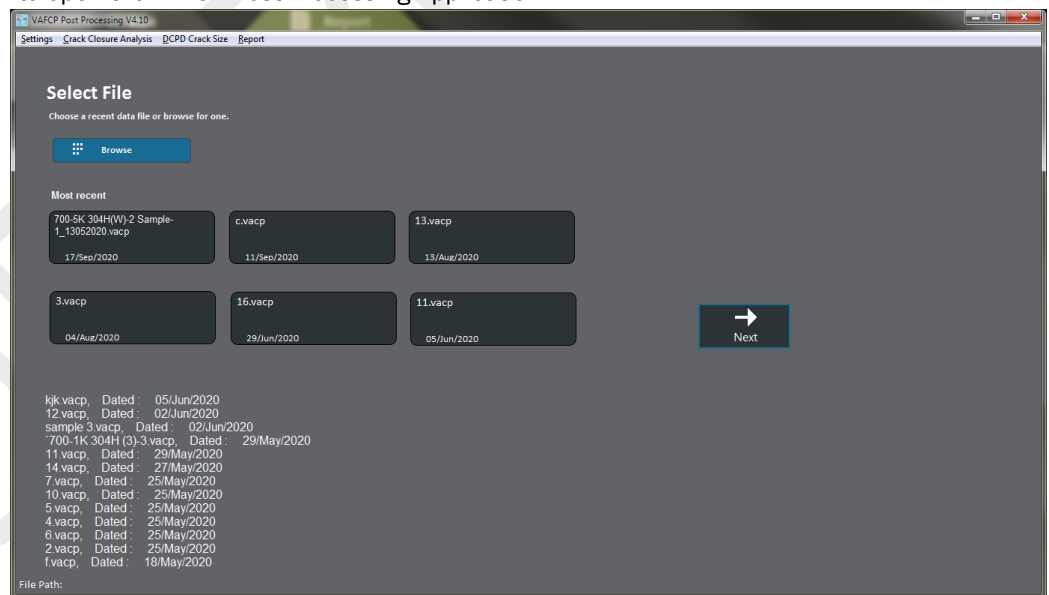
Accept

- Go to “Run” page & click “Start” button to start the test

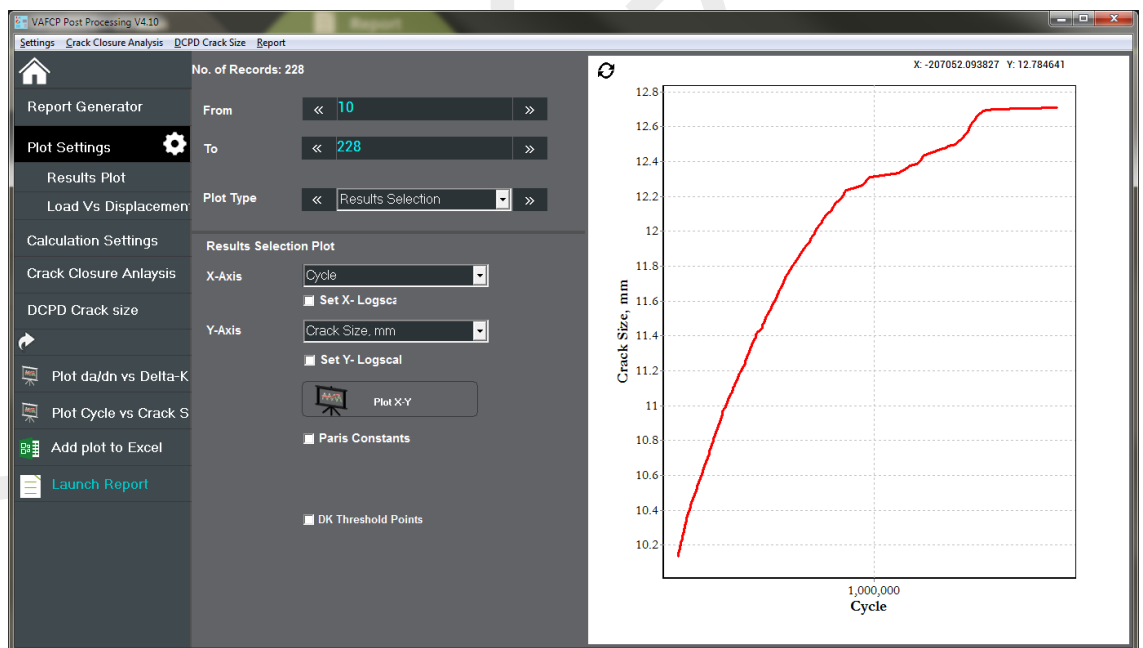
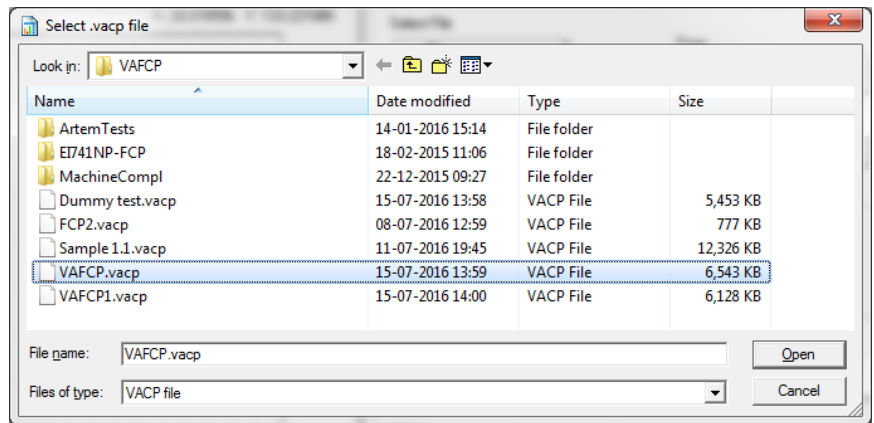


9.3.5 Exporting Data

- Go to “Report” to open the VAFCP Post Processing Application



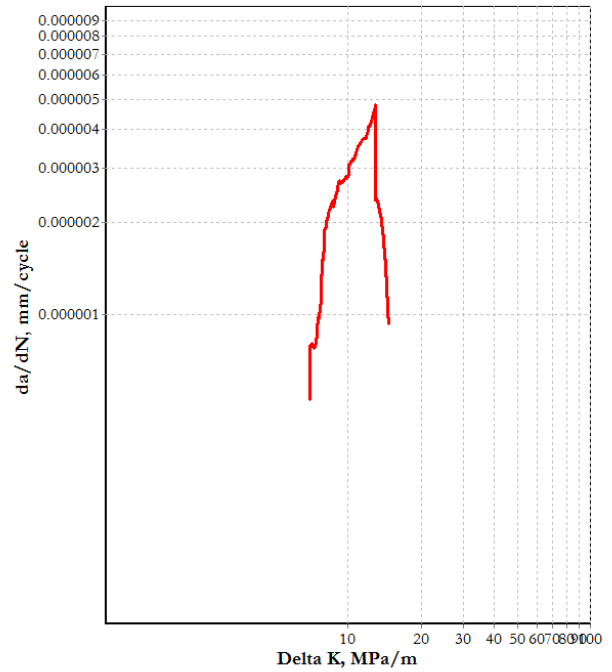
- Click “Browse” to select required data file (.vacp format only)



- Enter number of records to be plotted, enter “From” and “To” fields within the available number of records
- Plot type: Select “Results Selection” or “Load vs. Displacement”
- Under “Results Selection”, select required results as X-Axis and Y-Axis
- To get log scale for a particular axis, enable Log option. For normal scaling, disable the option
- After selection, click Plot X-Y to view the plot in the graph
- To get da/dN vs. Delta K with log scale, click Plot da/dN vs. Delta K button
- Click “Refresh” button to refresh the graph



X: 11.628913 Y: 0.000001



- Under “Load Vs Displacement”, enter any number as Rec. Point within the number of records. Or use the << and >> buttons to decrease/increase the record number, respectively

Plot Type << Load vs. Displacement >>

Load Vs Displacement

Rec. Point << 4 >>

First Cycle No.: 0

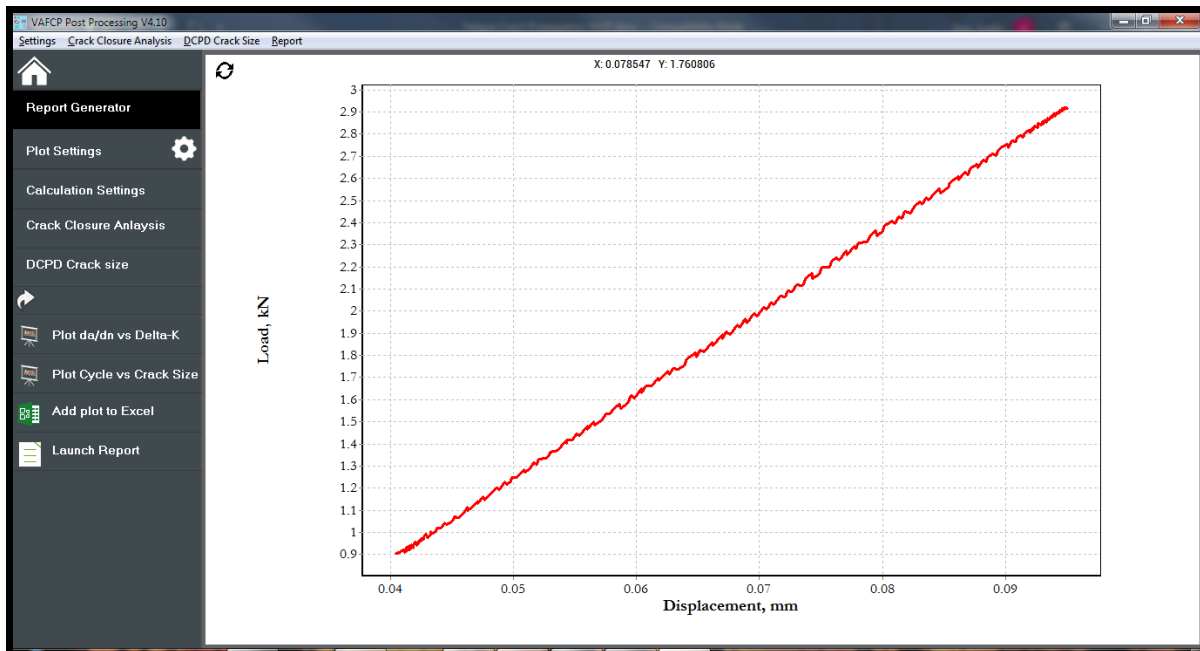
Last Cycle No.: 1813286

Crack Size, mm 0

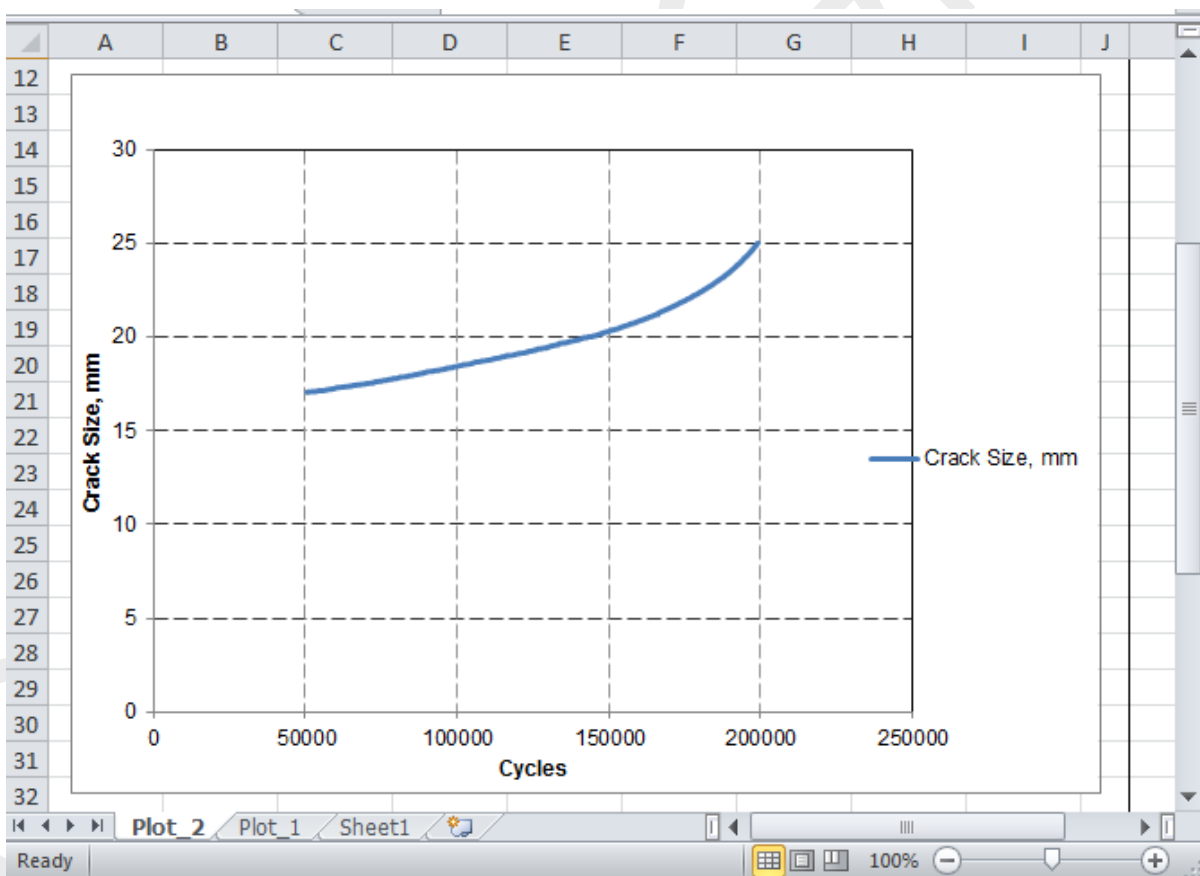
Cycle 0

☐ DK Threshold Points

- As the set record point, the points will be plotted, and Crack Size and Cycle number will be displayed



- To export “Results selection” or “Load vs. Displacement”, click Add Plot to Excel button.
- For the Results selection option, an excel plot will be available in the Report file



- In case of Load vs. Displacement, the Load and Displacement data points, Crack Size and Cycle number will be exported to an excel sheet
- For exporting a plot, a new sheet is created. But Load vs. Displacement data for any number of record points will be available in a single sheet

Modulus: 210.000 GPa					
Modulus Correction: 1.000					
Crack Size: 13.60000 mm			Crack Size: 13.64177 mm		
Cycle No.: 2002			Cycle No.: 2003		
Displacement, mm		Load, kN	Displacement, mm		Load, kN
	0.143	2.499		0.143	2.515
	0.143	2.503		0.143	2.519
	0.143	2.509		0.143	2.522
	0.143	2.513		0.143	2.53
	0.143	2.519		0.143	2.535
	0.143	2.53		0.143	2.54
	0.143	2.538		0.144	2.549
	0.144	2.548		0.144	2.561
	0.144	2.567		0.144	2.569
	0.144	2.576		0.144	2.581
	0.144	2.588		0.144	2.595
	0.144	2.603		0.144	2.606
	0.145	2.628		0.145	2.619
	0.145	2.638		0.145	2.639

- After exporting required data and plots, click on “Launch Report” button to view the Excel Report file. This marks the end of the Report file

Note: If no other test is being conducted, its recommended to Turn OFF/Shut down the following devices

- System hydraulic pressure (In application)
- VAFCP application
- MTL32 basic software
- Controller
- Computer
- UPS
- UPS input
- Selector Switch
- Lastly HPS

9.4 Fracture Toughness- J1c Testing

TITLE	Standard test method for Measurement of Fracture Toughness
TEST REFERENCE	ASTM E 1820-11 (J1c)

Requirements

- Grips/Fixtures: C(T) Cups/3Point Bend fixture
- Wedges :

Sl No	Available Cups (Thickness)	Specimen	
		Width	Thickness
1	6.35	25.4	6.35

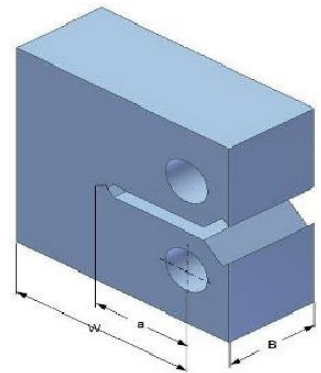
- COD: Recommended 5 mm gage length extensometer.
- Torque Wrench (M10 socket), M5 Allen key, Wooden blocks

Displacement/Strain-Controlled Test Procedure

- Specimen geometry
- Powering up the system
- Test setup
 - Mount grips
 - Insert suitable cups & mount specimen
 - Mount extensometer
- Run the test
- Export data

9.4.1 Specimen Geometry

- a. Measure specimen width, thickness & crack size
- Instrument required - Vernier Calliper
- b. Take the average value of 3 readings



Refer to ASTM E 1820-11 for more details on specimen

9.4.2 Powering up the system

- a. Please ensure to switch ON
 - System Mains



- Host Computer/Laptop



- b. Double click on MTL 32 icon on the desktop
- c. Click “Go-Online” button & wait for 15-30 seconds

9.4.3 Test Setup

❖ Mount grips & insert suitable wedges

Top grip on the actuator rod end and bottom grip to the load cell will be already mounted only accessories pin for respective tests must be mounted (refer to the Machine & Maintenance manual>> Universal Grips manual for more details on universal grips. Before that ensure PLI application on tab is open.

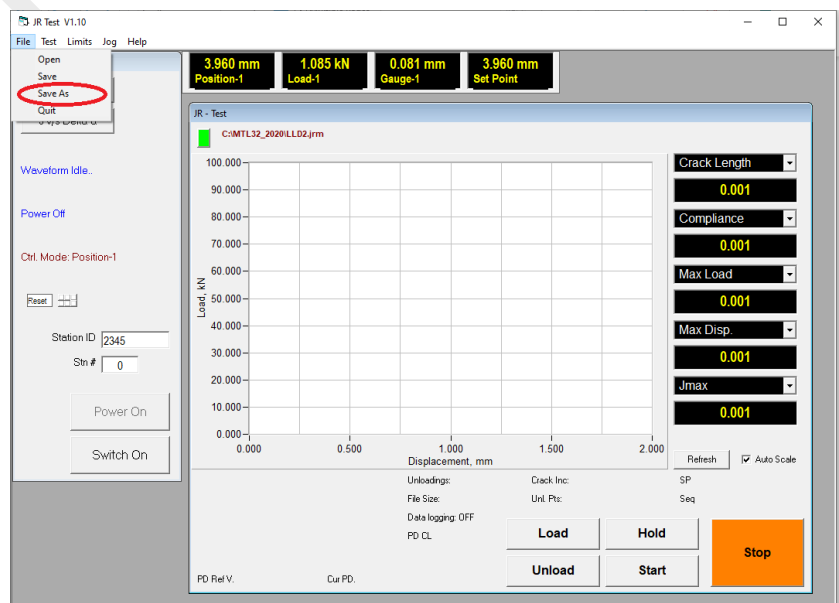
Refer to Tension testing (Mount grips) for details on cross head adjustment

❖ Insert suitable cups & mount specimen

Refer Fatigue crack propagation testing (test setup part) for J1c test setup & COD mounting which remains unchanged

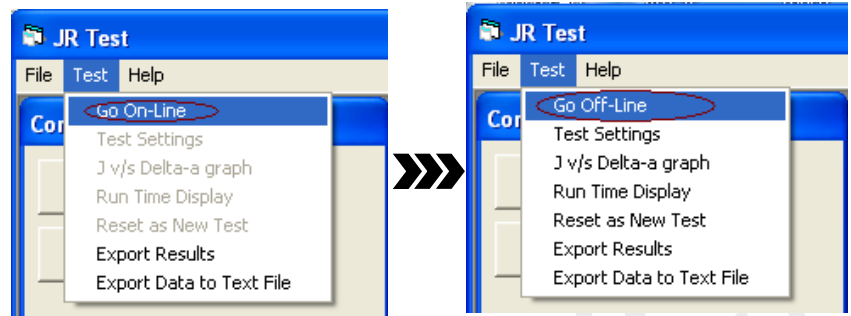
9.4.4 Run a test

- Double click on the Application software J1c icon

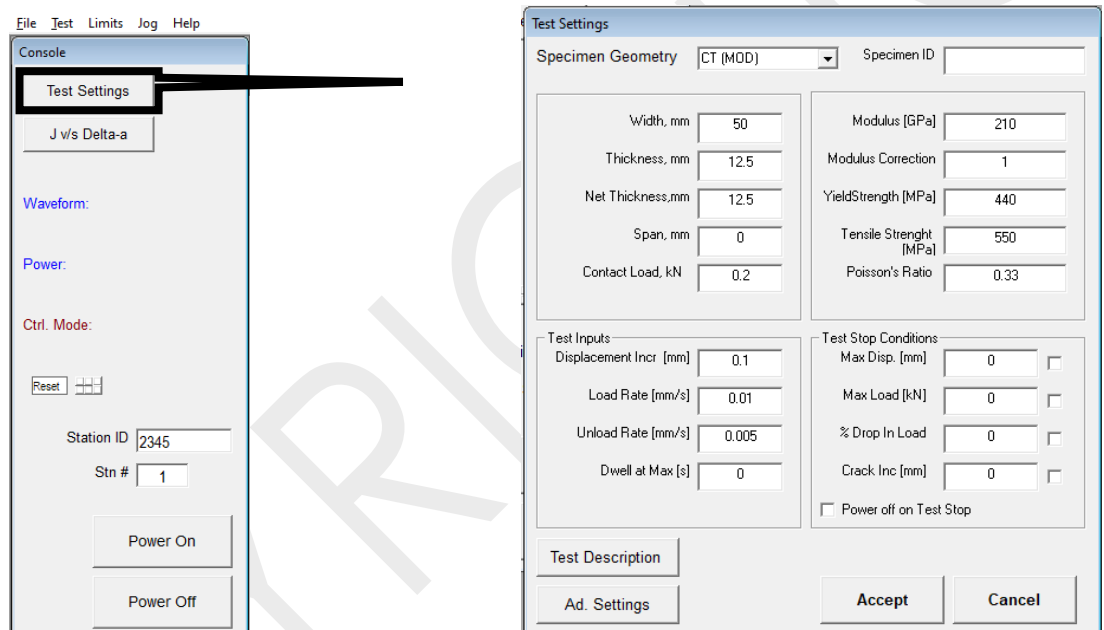


- Go to “File” menu to create new/open an existing file. To create new file, click save as & enter desired file name
- To open a previous file, click open to browse the required & save it with a different name (New file must be created for each test)

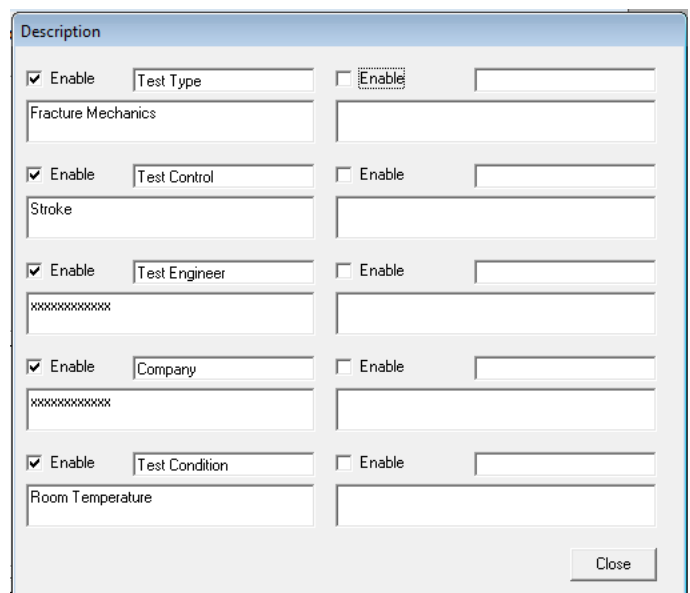
- Enter Site ID & station number match those entered in MTL32_2020 software. Go to “Test” menu & select “Go-online”. After going “Online” rest of the options will be highlighted



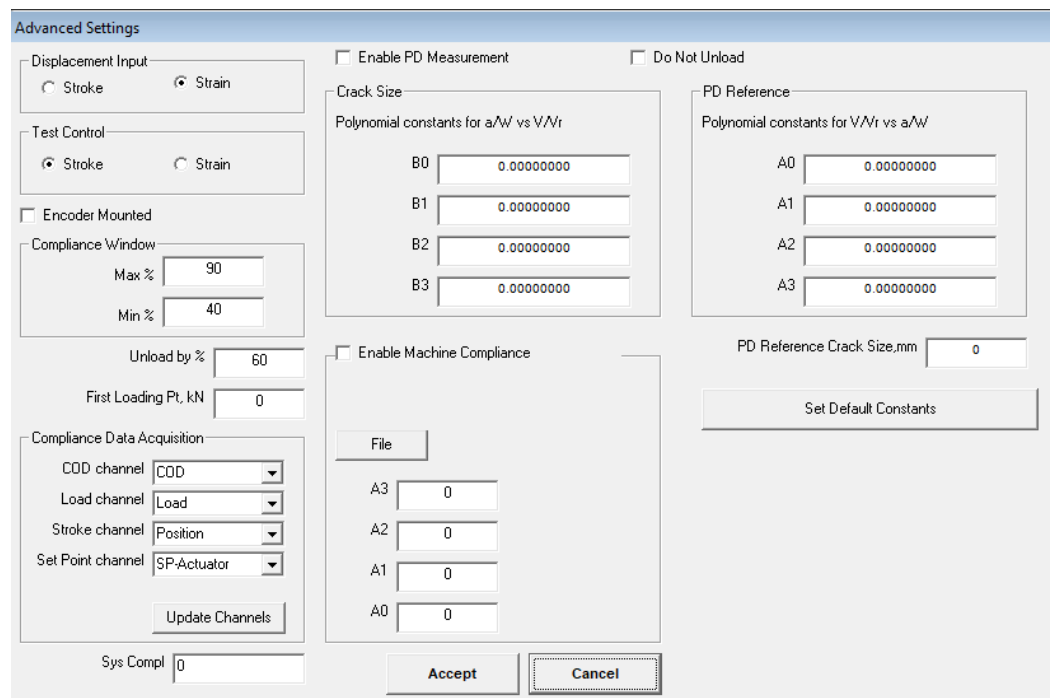
- Go to “Test Settings” page, select specimen type & ID. Enter specimen geometry, material properties, test inputs & test stop conditions



- Under “Test Stop Conditions”, enabling check box will activate the particular test condition & enable “Power off on test stop” to turn off the system at test stop or any limit hit
- Under “Test Description” page click the enable check box to enter the test details



- Under “Ad Setting” page, select displacement input & control mode. Enter “Compliance window” for compliance calculation, “Unload %” & “First Loading Pt, kN”
- Under “Compliance Data Acquisition”, select respective DAQ channels & click “Update Channels” & click “Accept”



Advanced Settings

Displacement Input: ☐ Stroke ☒ Strain

Test Control: ☒ Stroke ☐ Strain

☐ Encoder Mounted

Compliance Window: Max % Min %

Unload by % First Loading Pt, kN

Compliance Data Acquisition: CDD channel Load channel Stroke channel Set Point channel

Sys Compl

☐ Enable PD Measurement ☐ Do Not Unload

Crack Size: Polynomial constants for a/W vs V/V_i
 B0
 B1
 B2
 B3

☐ Enable Machine Compliance

File

A3
 A2
 A1
 A0

PD Reference: Polynomial constants for V/V_i vs a/W
 A0
 A1
 A2
 A3

PD Reference Crack Size, mm

- Go to “Limits” in menu, set Stroke, Load & COD limits



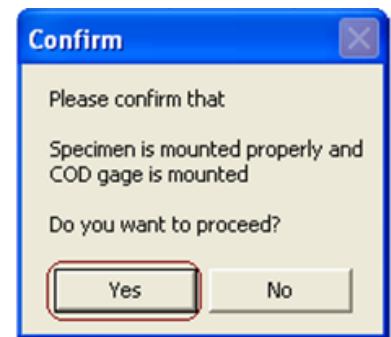
Limit Settings

☒ Position ☐ Load ☐ Extensometer

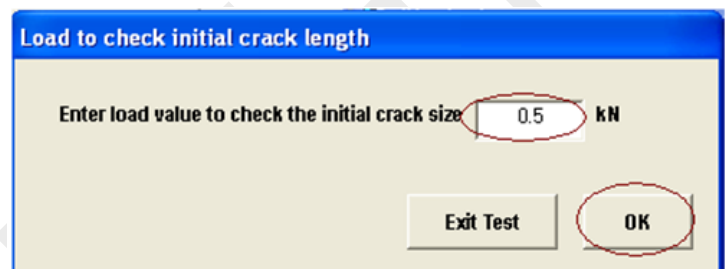
Hi Limit: ☒ Stop ☐ Hold ☒ Trip

Lo Limit: ☒ Stop ☐ Hold ☒ Trip

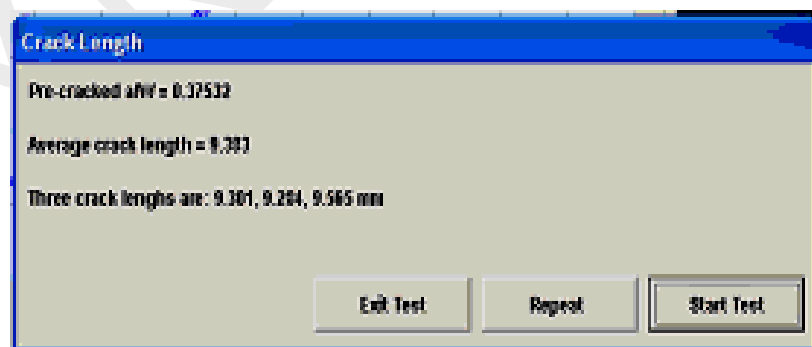
- Go to “Run” page, click “Start” to run the test. A pop-up message will appear, click “Yes” when prompted to confirm.



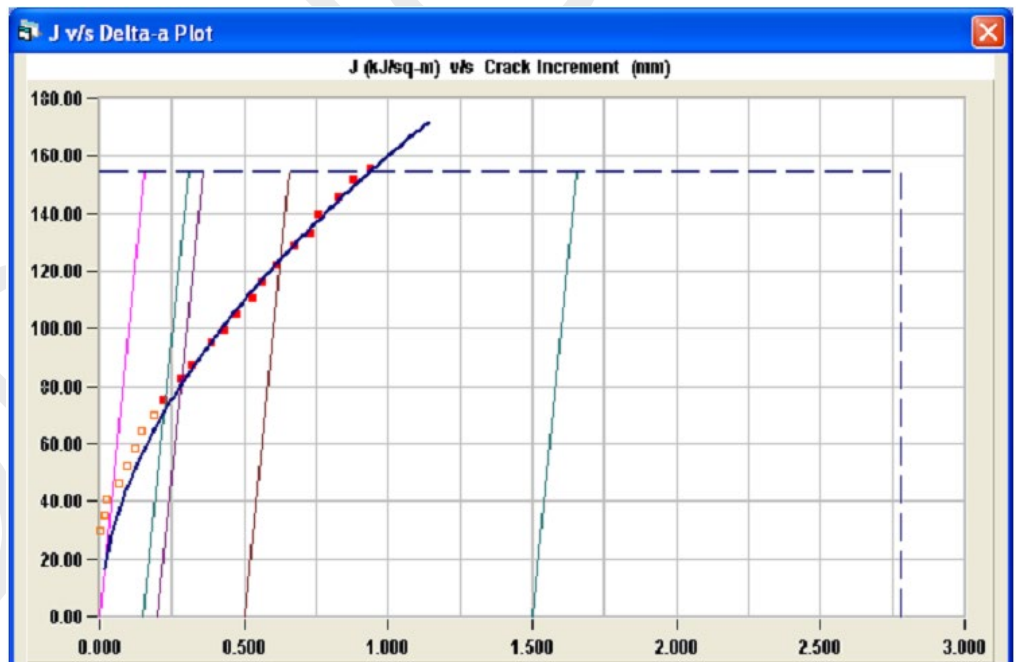
- A second pop-up message will appear requesting for load value to check initial crack size. Here, enter the final load value of VAFCP test & click OK



- The third pop-up message will appear with “Pre-cracked a/W”, “Average crack length” & “Three crack length” values. These values are the final parameters of the VAFCP test, to confirm click “Start Test” or to repeat the process click “Repeat” or to exit test click “Exit Test”

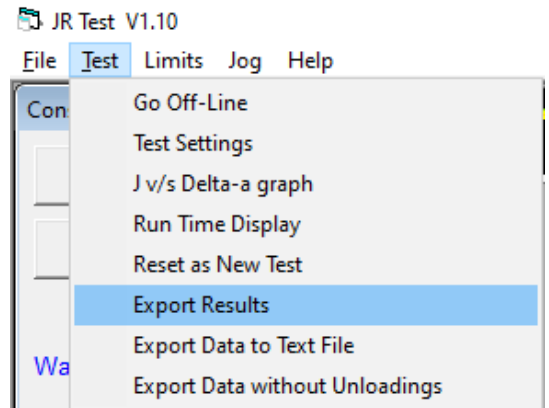


- Upon starting the test, a “Load, kN v/s Displacement, mm (COD)” graph will be displayed. To view online plot, click “J v/s Delta-a”

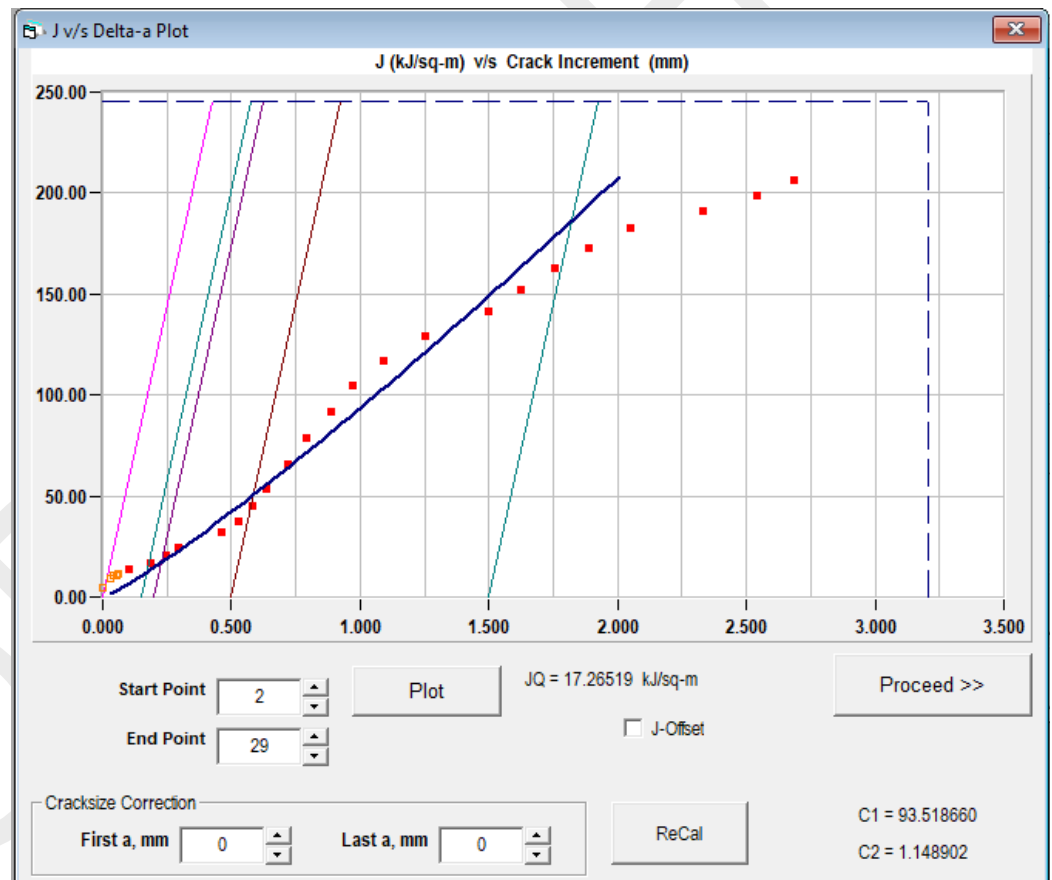


9.4.5 Exporting Data

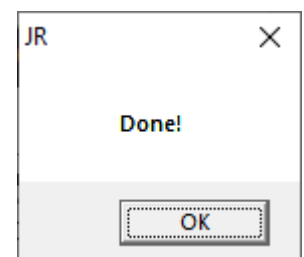
- Go to “Test” menu & select “Export Results” to go to post processing Application.



- Enter “Start Point” & “End Point”. Click “Plot” & then click “Proceed”



- After completing the post processing, the below message will pop up, click “OK” to view the report



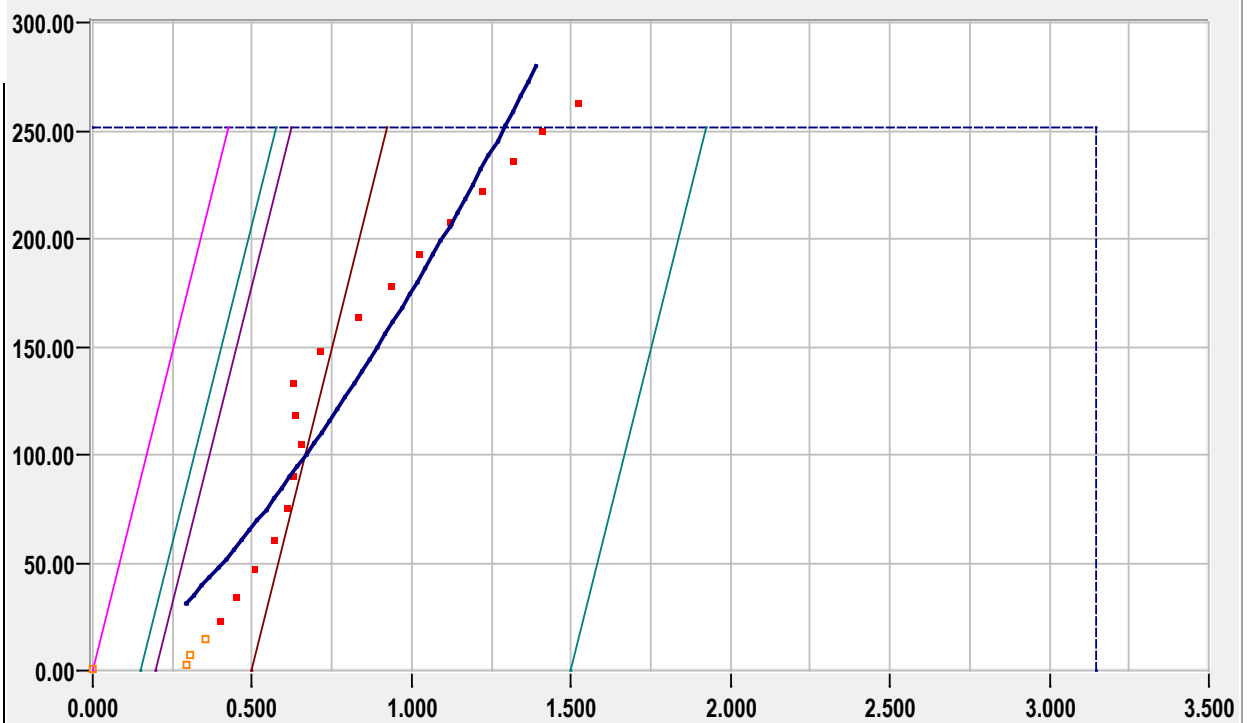
JR TEST REPORT

Width : 25.4 mm
Thickness : 6.38 mm
Net Thickness : 6.38 mm
Disp. Inc. : 0.1 mm
Loading Rate : 0.01 mm/sec
Unloading Rate : 0.005 mm/sec

Date : 12-07-2021
Time : 19:31:50
Modulus : 72 GPa
Modulus Correction : 1
Yield Strength : 270 MPa
Tensile Strength : 320 MPa
Poissons Ratio : 0.33
Pre-cracked a/W : 0.504

Specimen ID :

J (kJ/sq-m) v/s Crack Increment (mm)



JQ = 23.782 kJ/sq-m
Jlimit = 250.947 kJ/sq-m
 $\Delta a_Q = 0.240$ mm

JQ qualifies as J1c!

Validation Conditions

- 1) Thickness, $B > 25 \cdot JQ / \text{Yield Stress}$
- 2) Initial ligament, $b_0 > 25 \cdot JQ / \text{Yield Stress}$
- 3) Slope of power law regression line, dJ/da , evaluated at Δa_Q is less than Yield Stress

Thickness = 6.38 mm
 $25 \cdot JQ / \text{Yield Stress} = 2.015$
mm
 $b_0 = 12.594$ mm
 $dJ/da = 138.771$
MPa
Yield Stress = 295 MPa

C:\Users\ARIES GROUP\Documents\Integration\Test Data\VAFCP\CT\CT_AI_J1c.jrm

Note: If no other test is being conducted, its recommended to Turn OFF/Shut down the following devices

- a. System hydraulic pressure (In application)
- b. J1c application
- c. MTL32 basic software
- d. Controller
- e. Computer
- f. UPS
- g. UPS input
- h. Selector Switch
- i. Lastly HPS Mains.

9.5 Fracture Toughness- K1c Testing

TITLE

Standard test method for Measurement of Fracture Toughness

TEST REFERENCE

ASTM E 399 (K1c)

Requirements

- Grips/Fixtures: C(T) Cups/3Point Bend fixture
- Wedges :

Sl No	Available Cups (Thickness)	Specimen	
		Width	Thickness
1	6.35	25.4	6.35

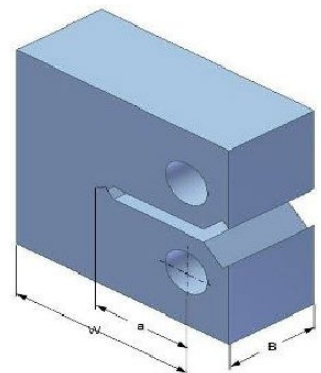
- COD: A 5 mm gage length extensometer is recommended
- Torque Wrench (M10 socket), M5 Allen key, Wooden blocks

Displacement/Strain-Controlled Test Procedure

- Specimen geometry
- Powering up the system
- Test setup
 - Mount grips
 - Insert suitable cups & mount specimen
 - Mount extensometer
- Run the test
- Export data

9.5.1 Specimen Geometry

- a. Measure specimen width, thickness & crack size
 - Instrument required - Vernier Calliper
- b. Take the average value of 3 readings



Refer to ASTM E 399 for more details on specimen

9.5.2 Powering up the system

- a. Please ensure to switch ON
 - System Mains



- Host Computer/Laptop



- b. Double click on MTL 32 icon on the desktop
- c. Click “Go-Online” button & wait for 15-30 seconds

9.5.3 Test Setup

❖ Mount grips & insert suitable cups

Top grip on the actuator rod end and bottom grip to the load cell will be already mounted only accessories pin for respective tests must be mounted (refer to the Machine & Maintenance manual)>> Universal Grips manual for more details on universal grips. Before that ensure PLI application on tab is open.

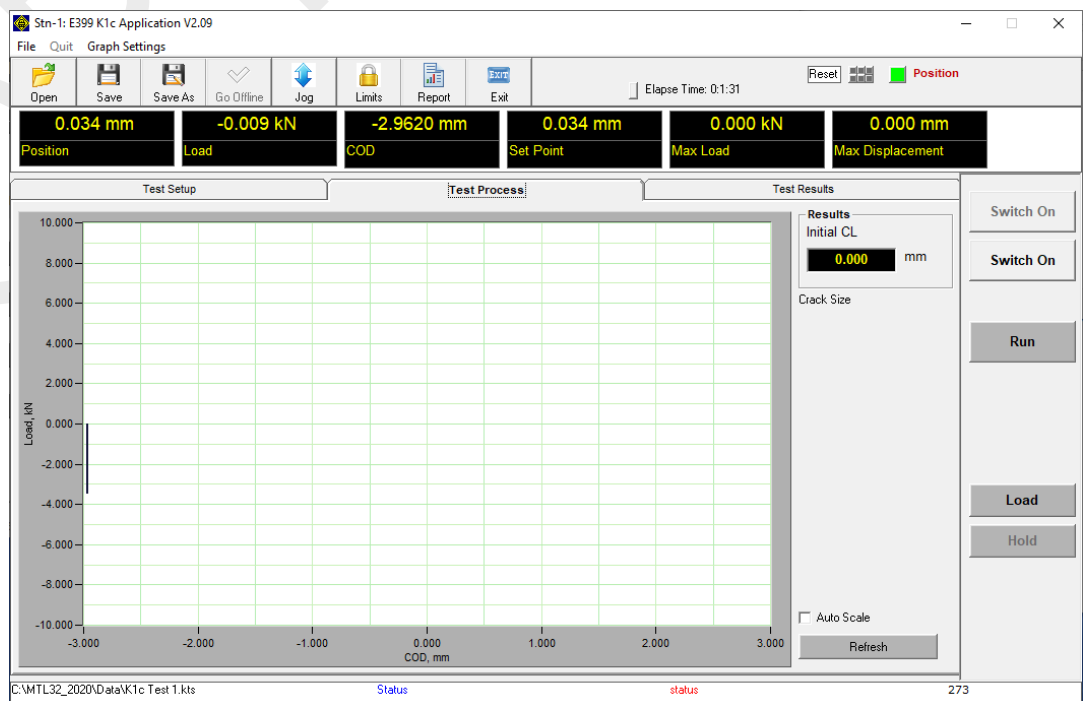
Refer to Tension testing (Mount grips) for details on cross head adjustment

❖ Insert suitable cups & mount specimen

Refer Fatigue crack propagation testing (test setup part) for K1c test setup & COD mounting which remains unchanged

9.5.4 Run a test

- Double click on the Application software **K1c** icon



- Go to “File” menu to create new/open an existing file. To create new file, click save as & enter desired file name
- To open previous file, click open to browse the required file
- Go to “Test Setup” page, enter Site ID & station number match those entered in MTL32_2020 software. On the menu bar select “Go-online”. After going “Online”, “Go-Online” button will change to “Go-Offline button”

- In “Test Setup” page, under “Specimen” page select specimen type. Enter specimen geometry (Modulus correction should not exceed >5% i.e., 0.95-1.05) & material properties
- Under “Test Description” page enter test details

- In “Test Setup” page, under “Test Control” page select “Control mode” & “Displacement input”. Enter “Test Inputs” & “Test Stop Conditions”
- Then select “Unit System” & “Channel Selection” (Select respective DAQ channels & click “Update”)

Stn-1: E399 K1c Application V2.09

File Quit Graph Settings

Open Save Save As Go Offline Jog Limits Report Exit Elapse Time: 0:0:53 Reset Position

0.033 mm Position	-0.009 kN Load	-2.9620 mm COD	0.033 mm Set Point	0.000 kN Max Load	0.000 mm Max Displacement
----------------------	-------------------	-------------------	-----------------------	----------------------	------------------------------

Test Setup

Specimen

Test Control

☒ Stroke ☐ Load ☐ COD

Disp Input

☐ COD ☒ Stroke

Machine Compliance: 0.0000

Test Inputs

Rate: 0.050 mm / sec

Total Displacement: 0.010 mm

Test Stop Conditions

Max Load, kN: 10.000 ☒

% Drop in Load: 5.000 ☐

Load Threshold, kN: 2.000

Max Disp, mm: 2.000 ☐

Unit System

☒ Metric ☐ British

Test Control

Channel Selection

Position: Stroke Ch

Load: Load Ch

COD: COD Ch

Update

Switch On

Switch On

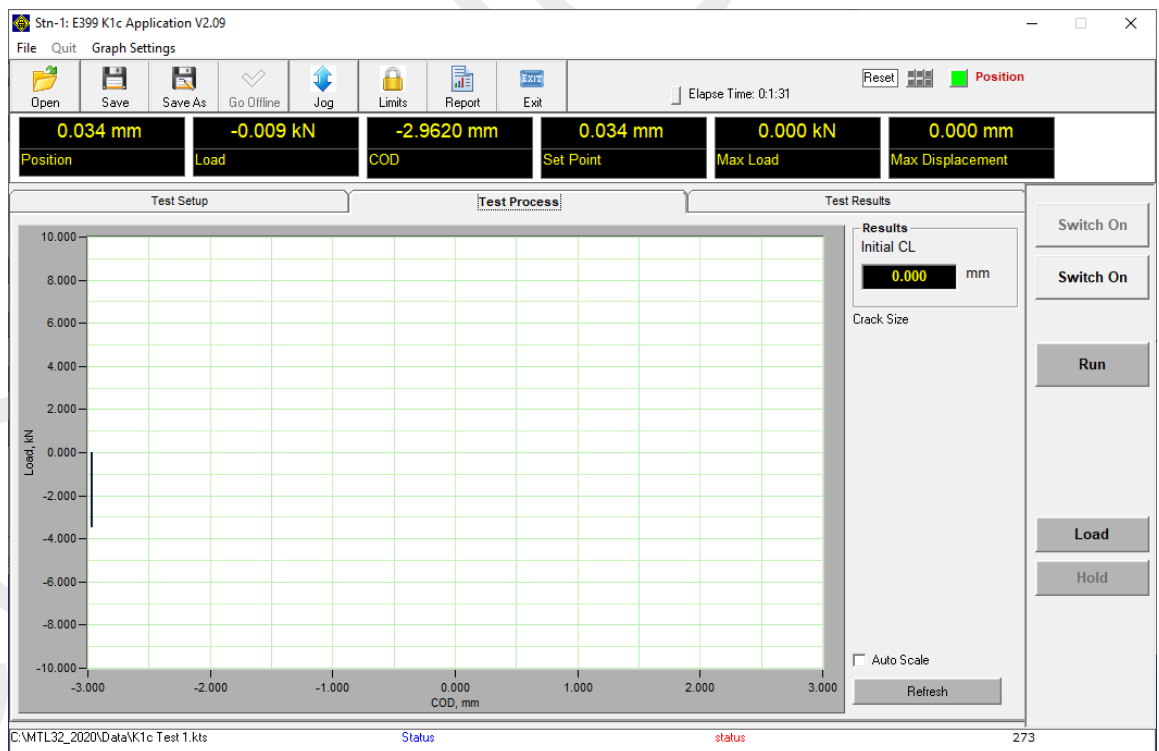
Run

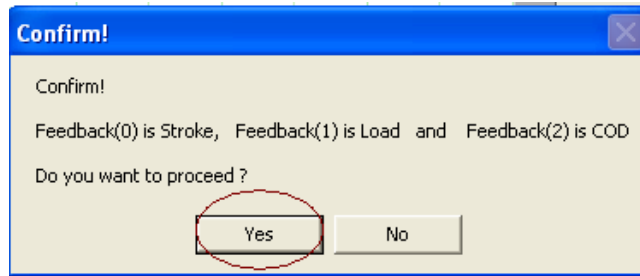
Load

Hold

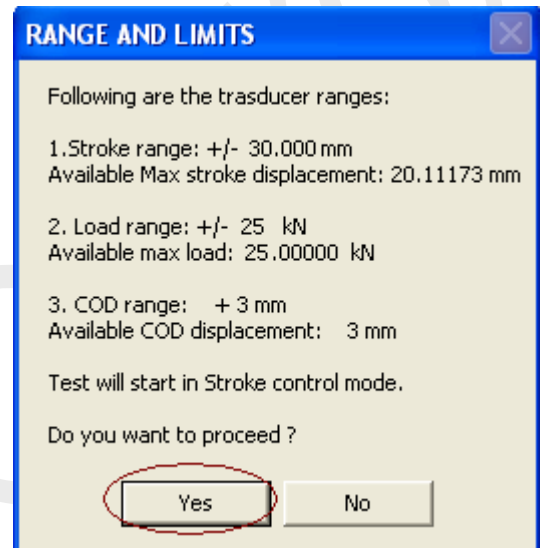
C:\MTL32_2020\Data\K1c Test 1.kts Status status 157

- Go to “Limits” in menu, set Stroke, Load & COD limits
- Go to “Test Process” page, click “Run” to start the test. A pop-up message will appear, click “Yes” when prompted to confirm.
-

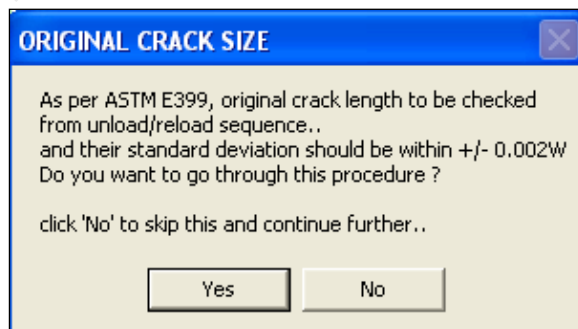




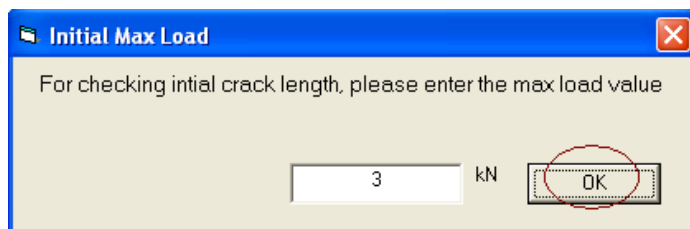
- A second pop-up message will appear displaying transducer ranges, click “Yes” to continue



- The third pop-up message will appear requesting for original crack length measuring procedure, click “Yes” to continue & “No” to skip

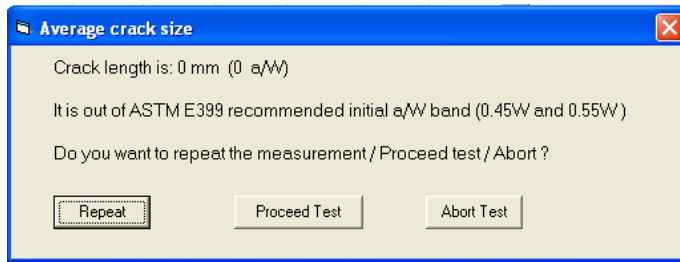


- If the user selects “Yes” in the above panel, a fourth pop-up message appears requesting for final load value of fatigue crack propagation test, enter the load value & click “OK”

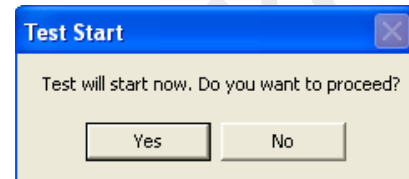


- After checking the initial crack size fifth pop-up message will appear, click “Yes” to continue with the test
- If the original crack is not correct, the user will get a message as shown below. Here user must take appropriate action accordingly

Repeat: Repeats the procedure
Proceed Test: Continues with the test
Abort Test: Stop the test

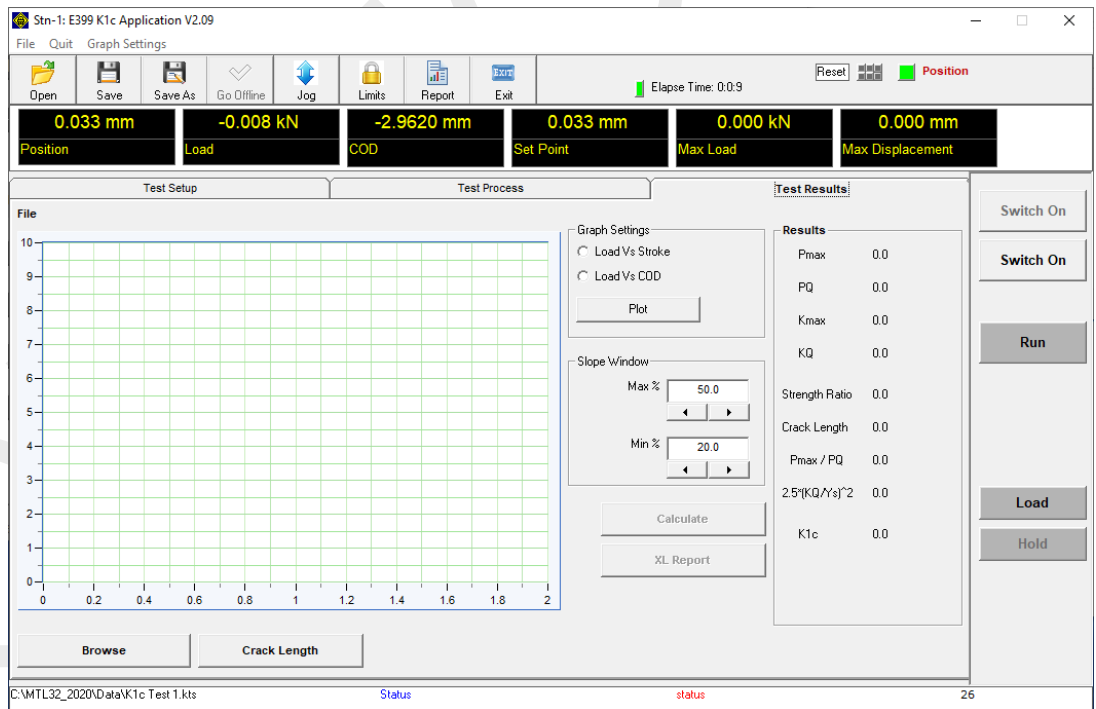


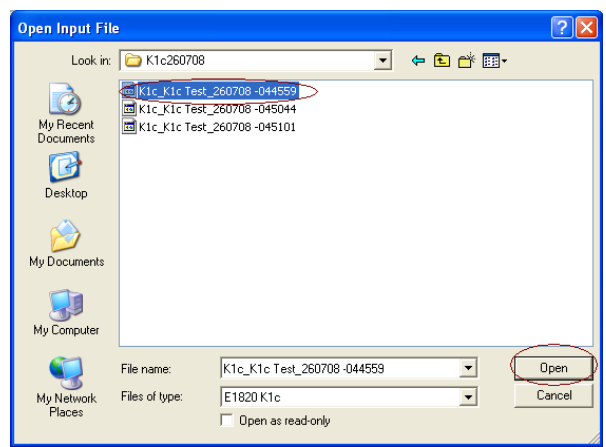
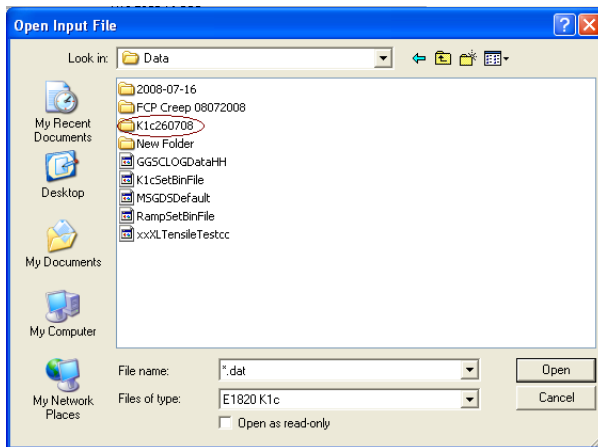
- If user clicks "Proceed Test" below message appears, click "Yes" to run the test



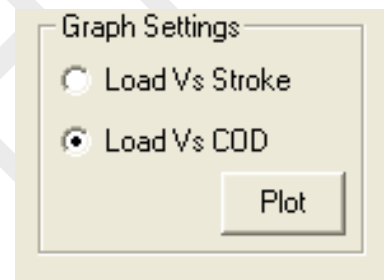
9.5.5 Exporting Data

- Go to "Test Results" menu & click "Browse" to browse the test file

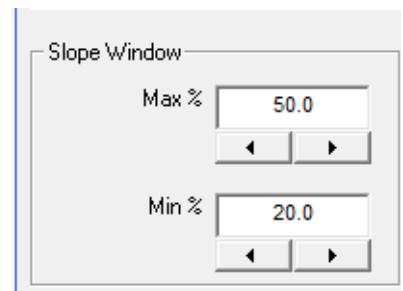




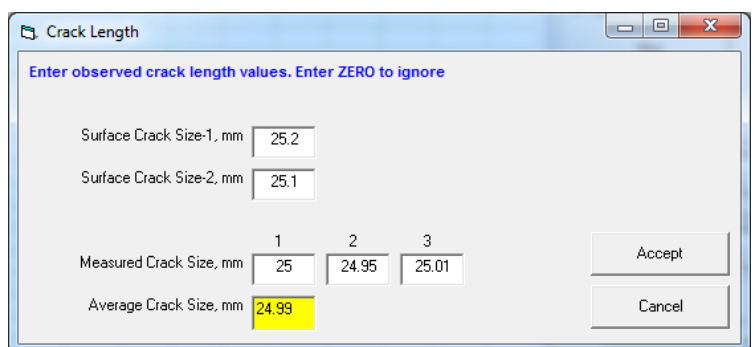
- Select required “Graph Settings” & click “plot”.



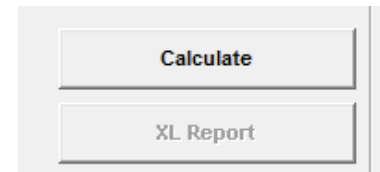
- Calculate the slope of curve within the Hookean region or young’s modulus by taking percentage of max and min loads



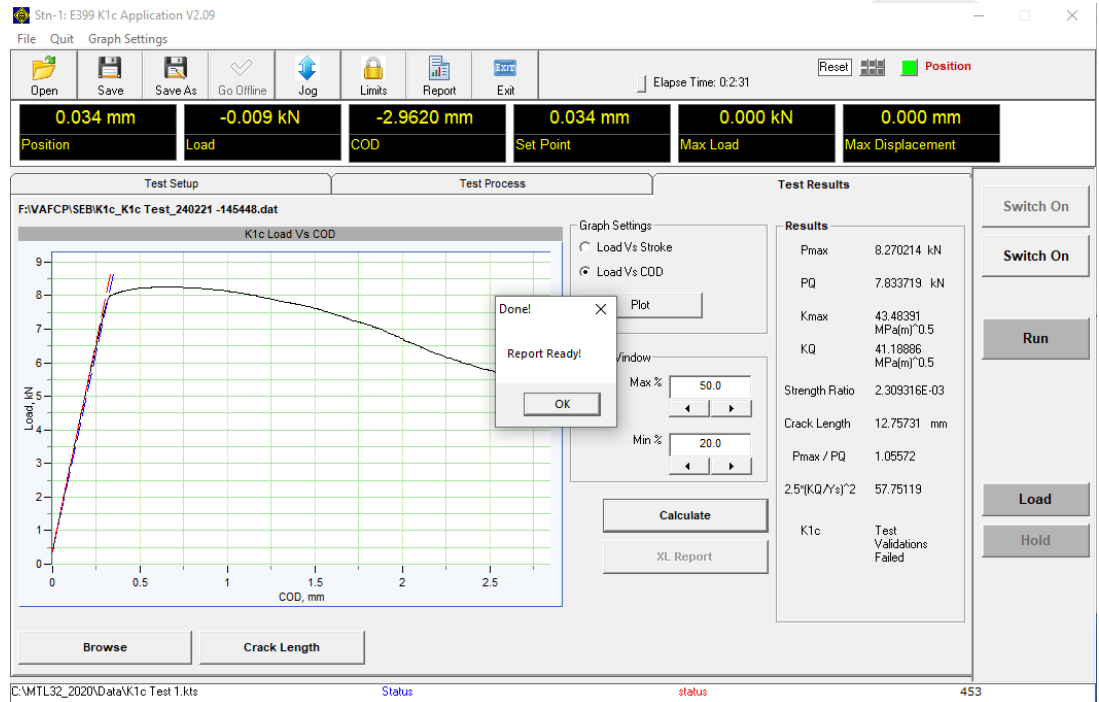
- Click on “Crack length” button to enter observed pre cracked crack length to calculate the “K1C” or enter “zero” value to ignore this procedure



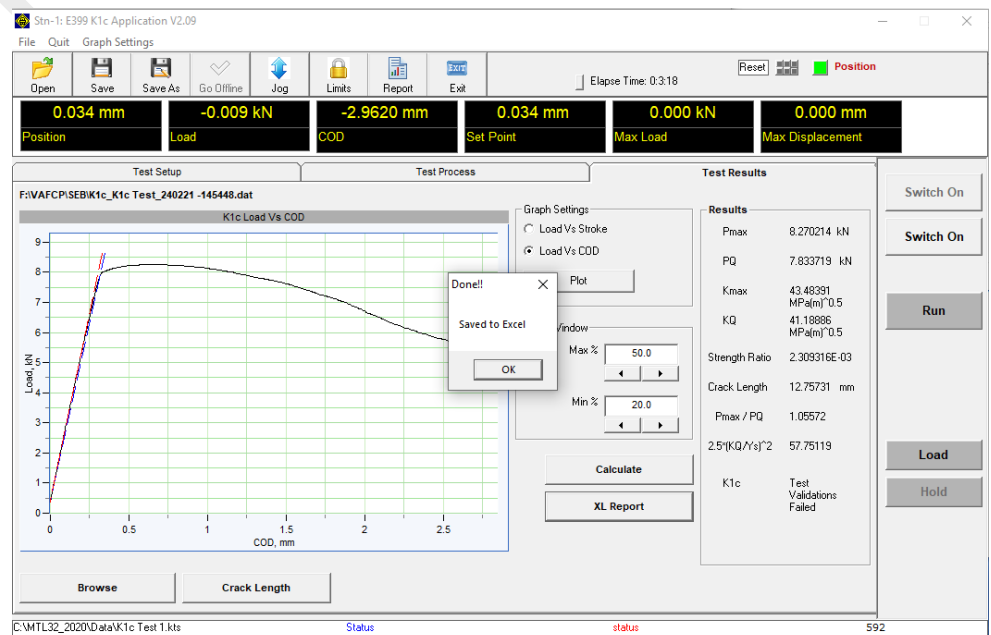
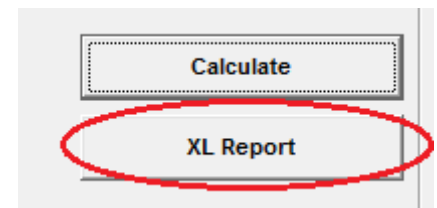
- After entering the slope window click “Calculate”, user will get slope on the graph window



- Upon “Report Ready” pop-up, click “OK”



- Click “XL report” to export the data into an excel sheet



- Below is the sample result sheet

K1c Test Results

Specimen ID 16

Geometry SE(B)

Test Control Stroke

Displacement 10 mm

Rate 0.05 mm /sec

Thickness 12.7 mm

Net Thickness 12.7 mm

Width 25.5 mm

Date 7/2/2021

Time 16:42:37

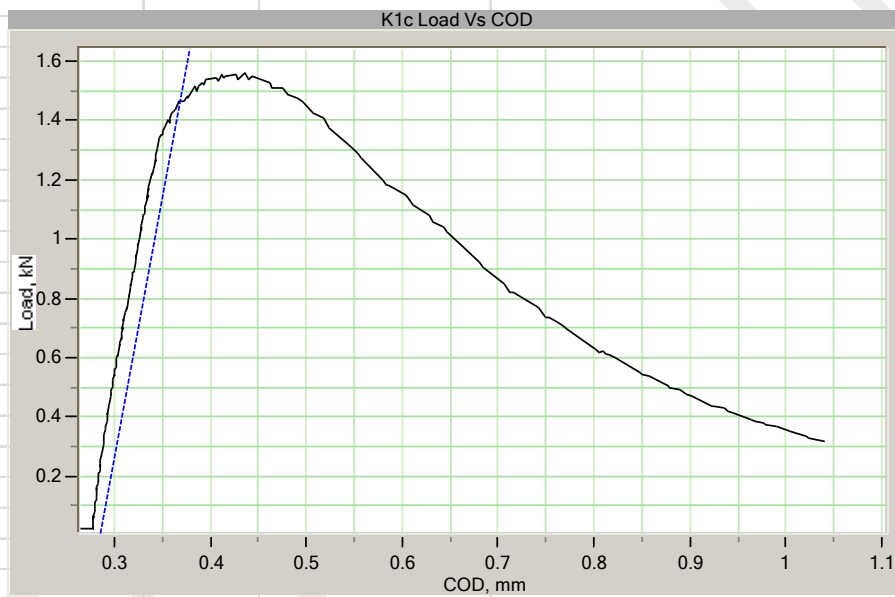
Modulus 71 GPa

Yield Stress 105 MPa

Tensile Strength 130 MPa

Poissons Ratio 0.3

Initial a/W 0.527



Pmax 1.561 kN

PQ 1.464 kN

Kmax 8.773 MPa m ^{0.5}

KQ 8.227 MPa m ^{0.5}

Specimen Strength Ratio 0.001

Crack Length 13.439 mm

Pmax / PQ 1.066

2.5*(KQ / YS) ² 12.13

K1c 8.773 MPa m ^{0.5}

Elastic Region Slope 17.720 kN/mm

Validation Conditions

Pmax / PQ Should be Less Than or Equal to 1.1

Thickness Should be Greater Than 2.5*(KQ / YS) ²

Note: If no other test is being conducted, its recommended to Turn OFF/Shut down the following devices

- a. System hydraulic pressure (In application)
- b. K1c application
- c. MTL32 basic software
- d. Controller
- e. Computer
- f. UPS
- g. UPS input
- h. Selector Switch
- i. Lastly HPS Mains.

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9.6 Low Cycle Fatigue Testing

TITLE	Standard test method for Strain-Controlled Fatigue Testing
TEST REFERENCE	ASTM E 606

Requirements

- Grips/Fixtures: Mechanical LCF
- Wedges :

SI No	Available wedges	Specimen Dia
1	M8	M8
2	M10	M10
3	M12	M12

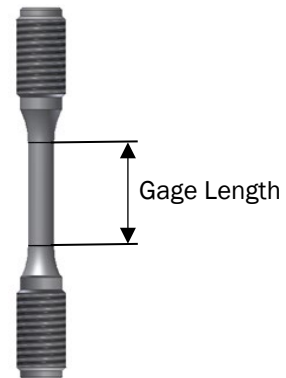
- Extensometer: Recommended 12.5 mm gage length extensometer.
- Torque Wrench (M10 socket), M5 Allen key, Wooden blocks

Displacement/strain-controlled test procedure

- Specimen geometry
- Powering up the system
- Test setup
 - Mount grips
 - Insert suitable cups & mount specimen
 - Mount extensometer
- Run the test
- Export data

9.6.1 Specimen Geometry

- a. Measure specimen diameter on the gauge section of specimen (3 times)
 - Instrument required - Vernier Calliper
- b. Take the average value of 3 readings



Refer to ASTM E 606 for more details on specimen

- c. Mark the gauge length on the reduced section of the specimen as shown above

Specimen Parameters Table

Parameters	Software Configuration Options	Default Setting
Geometry	Standard Geometry options include Circular and Irregular. Irregular is chosen as the default so the user will be prompted to enter the cross-sectional Area (nominal) as per the required standard. Note: Dimensions reported in the Results Table must be updated if the geometry is changed.	Irregular (Area)
Specimen Length	Specimen Length should represent the gauge length of the specimen, not the total length. This is the distance between the grips for machined specimens and this length will be used to determine strain from actuator travel after extensometer removal.	-
Initial Gauge Marks	Initial Gauge Marks represents the distance between the marks that are placed on the specimen for purposes of measuring Elongation After Fracture. These marks are often made with a marker but can be made other ways.	-

9.6.2 Powering up the system

- a. Please ensure to switch ON

- System Mains

- Host Computer/Laptop



- b. Double click on MTL 32 icon on the desktop
- c. Click "Go-Online" button & wait for 15-30 seconds



9.6.3 Test Setup

❖ Mount grips & insert suitable cups

Top grip on the actuator rod end and bottom grip to the load cell will be already mounted only accessories pin for respective tests must be mounted (refer to the Machine & Maintenance manual>> Universal Grips manual for more details on universal grips). Before that ensure PLI application on tab is open.

Refer to Tension testing (Mount grips) for details on cross head adjustment

❖ Insert suitable cups & mount specimen

1. Apply 30N-m torque to the M12X55 Allen bolt in a clockwise direction to open the grip.

IDEALLY THE GRIP ID
SHOULD NOT MATCH
WITH THE PIN OD

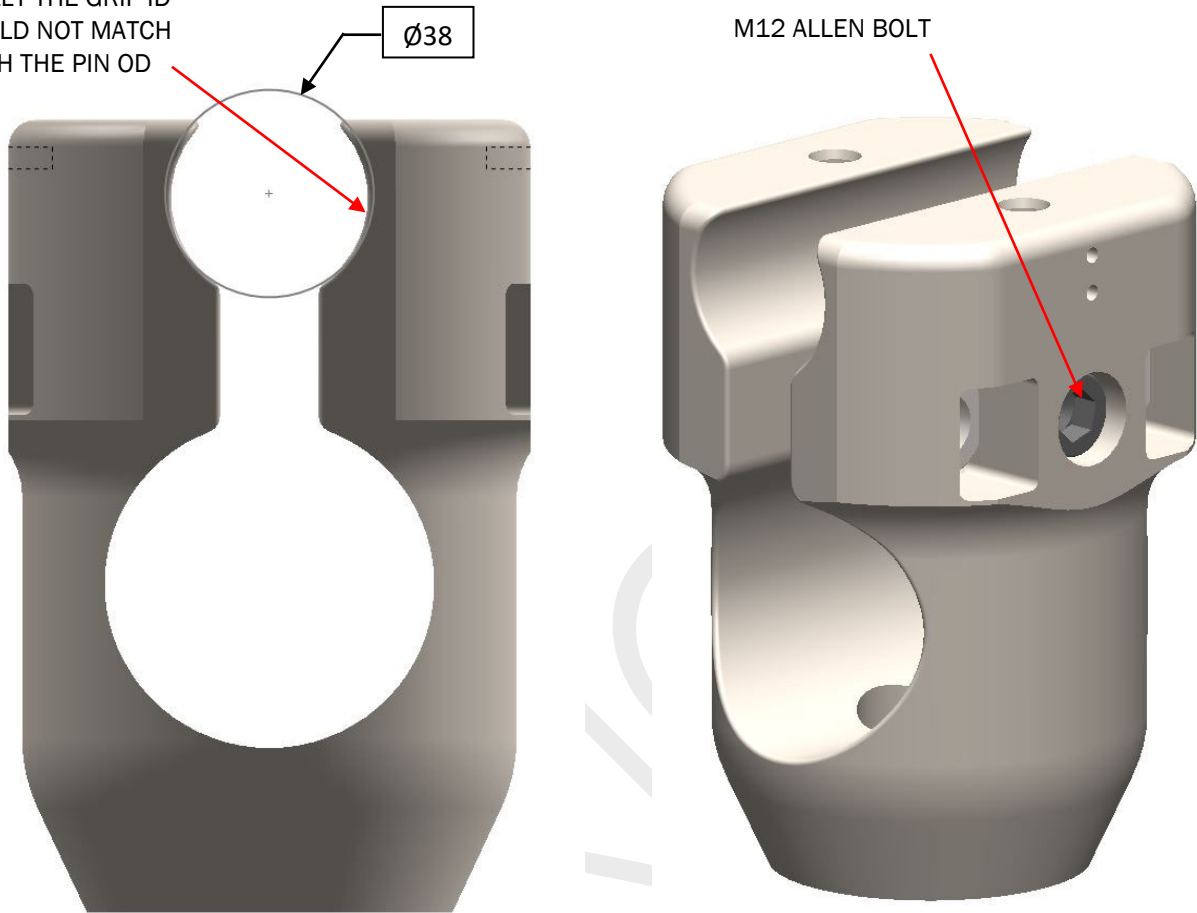


Figure 1: Condition for grip opening

Table 1: Bolt details for Grip opening

SI No.	BOLT SIZE	QUANTITY	RECOMMENDED TORQUE
1	M12X55	1 No	30-35 Nm

2. Insert slit discs with appropriate specimen (Threaded or Smooth) into the groove provided in Ø38mm pin and fasten using M5X10 Allen bolts.

NOTE 2:

- Ensure not to fasten the M5 Allen bolts tightly, as degree of freedom is to be allowed for axial alignment.
- Ensure that the same set of slit discs are used on either side of the specimen i.e. (1-1) and (2-2)

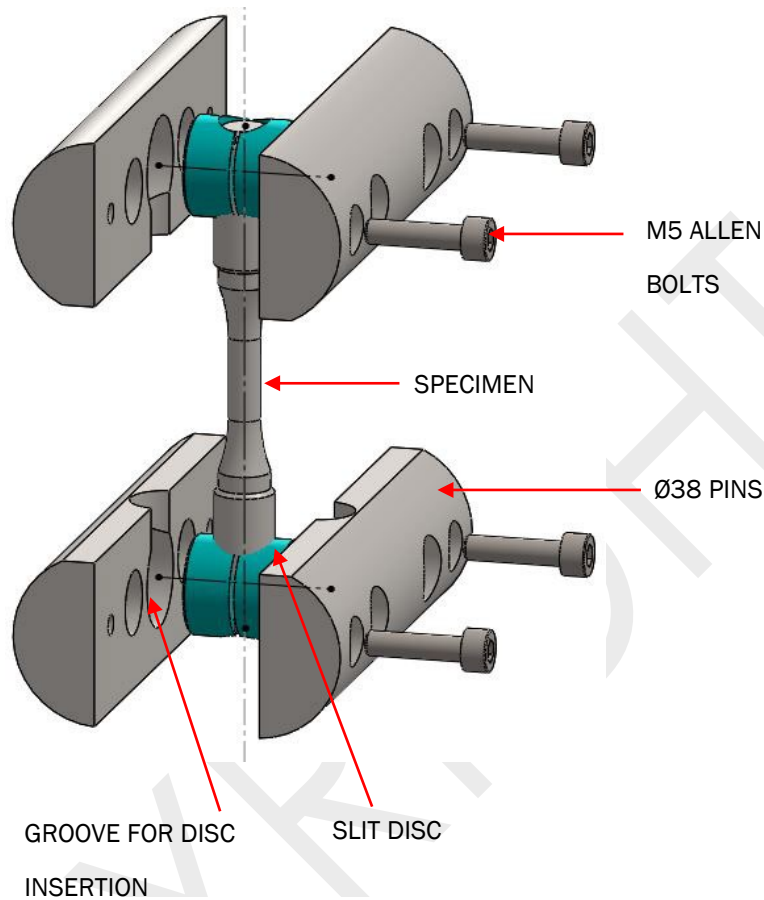
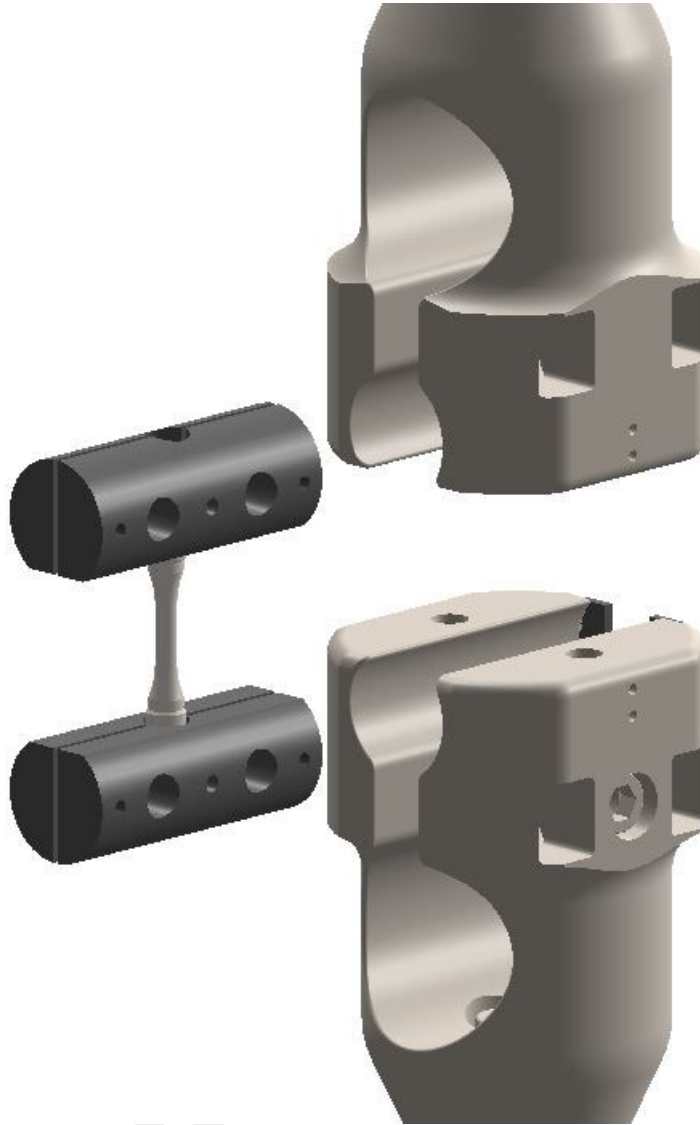


Figure 2: Dia 38mm Pin and specimen assembly

Table 2: Bolt details for Dia 38mm Pin and specimen assembly

SI No.	BOLT SIZE	QUANTITY	RECOMMENDED TORQUE
1	M5X10	4 No's	-

-
3. Insert Dia 38 pin and specimen assembly into top and bottom grip. Utilize actuator stroke to slide the dia 38 pin and specimen assembly into top and bottom grip. Ensure actuator is in stroke/position control.



-
4. Butt Dia 38 pin and specimen assembly to the stopper provided in top and bottom grip to attain axial alignment.

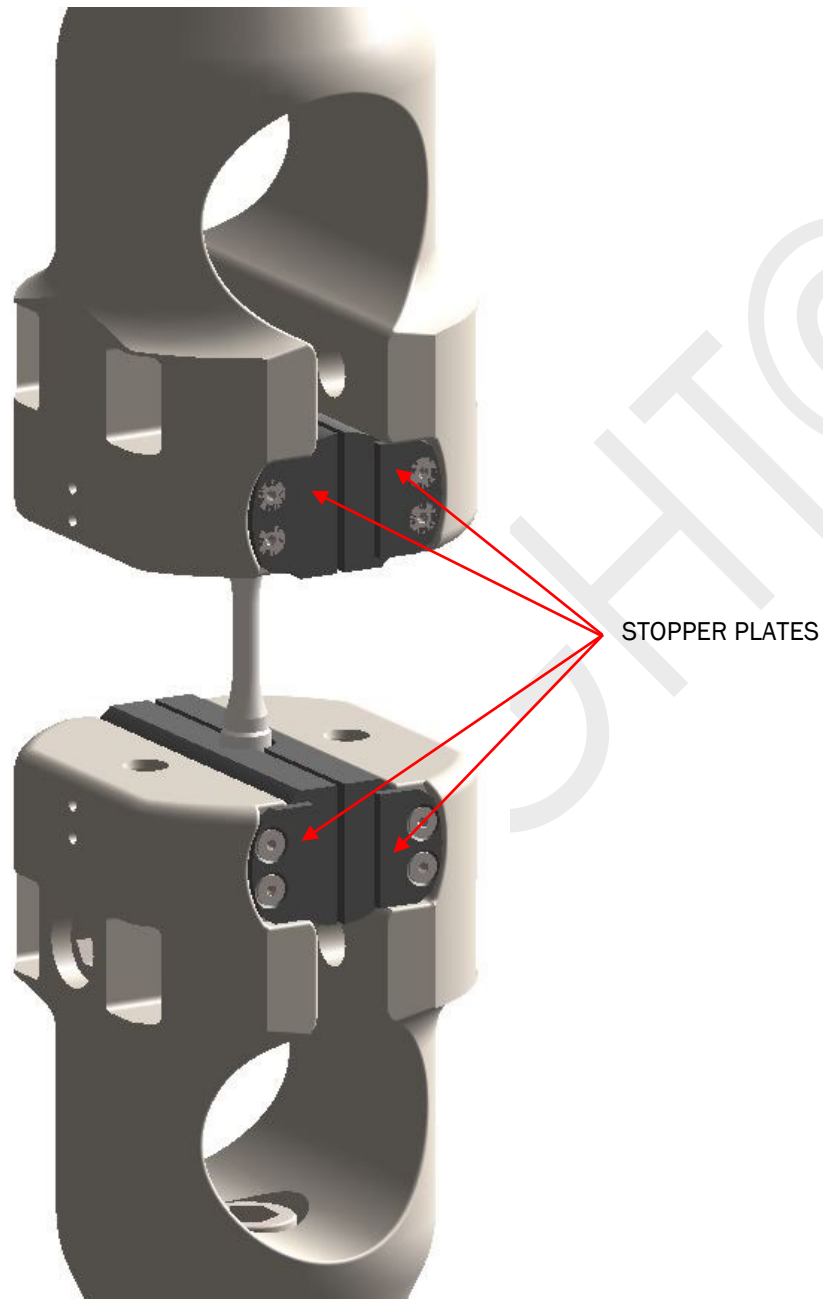


Figure 4: Axial alignment

-
5. Release the torque on M12X55 bolt to clamp the specimen assembly.

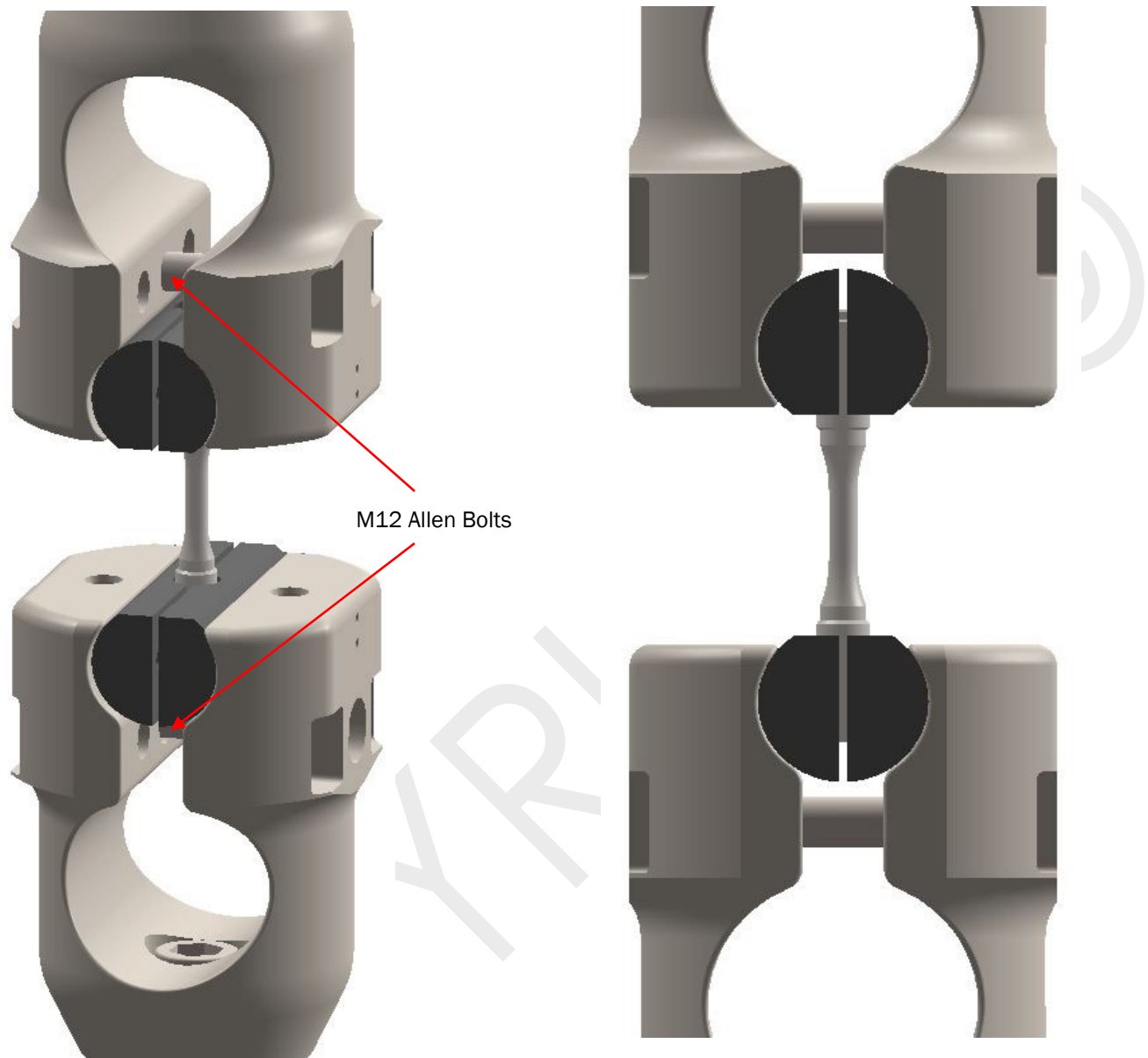


Figure 5: Grip in closed condition.

Note 3: Before applying clamping force ensure to release the torque on the M12 X 55 Allen bolt. Not doing so will induce the clamping force onto the bolt subsequently damaging it.

6. Switch to load control mode and set the load to zero before proceeding to the next step.
7. Apply a clamping force using M12X70 bolts with curved washer. Ensure to match the profile of curved washer with the profile in the grip.

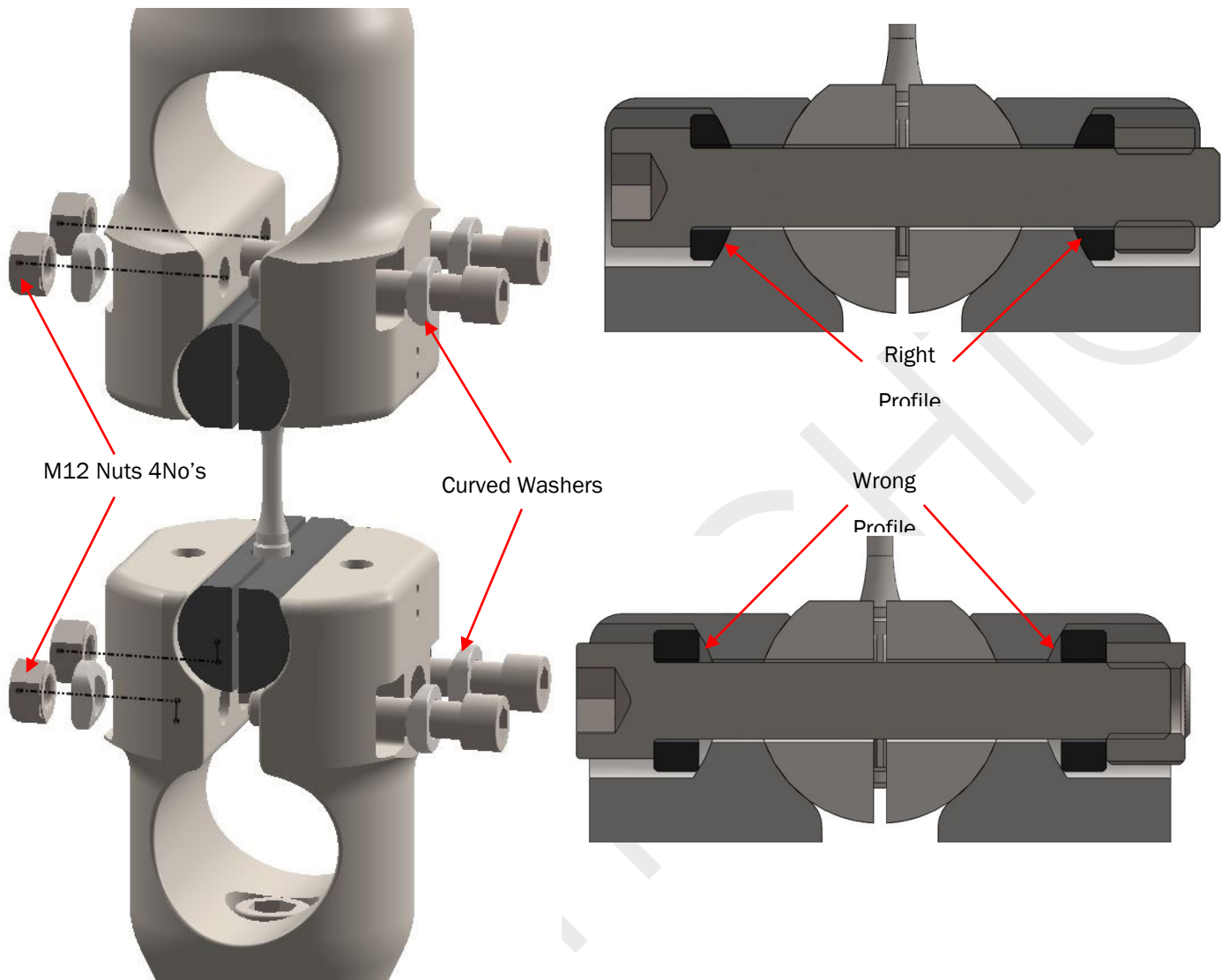


Figure 6: Applying clamping force to the grip.

Table 3: Bolt Details for applying clamping force

SI No.	BOLT SIZE	QUANTITY	RECOMMENDED TORQUE
1	M12X70	4 No's	50 Nm/BOLT
2	M12 Nut	4 No'S	-

Note 4: Ensure to release the applied clamping force on the clamping bolts before any torque is applied to activate the grip opening bolt (M12X55 Allen bolt) during specimen change over.

-
8. Grip is ready for low cycle fatigue testing.

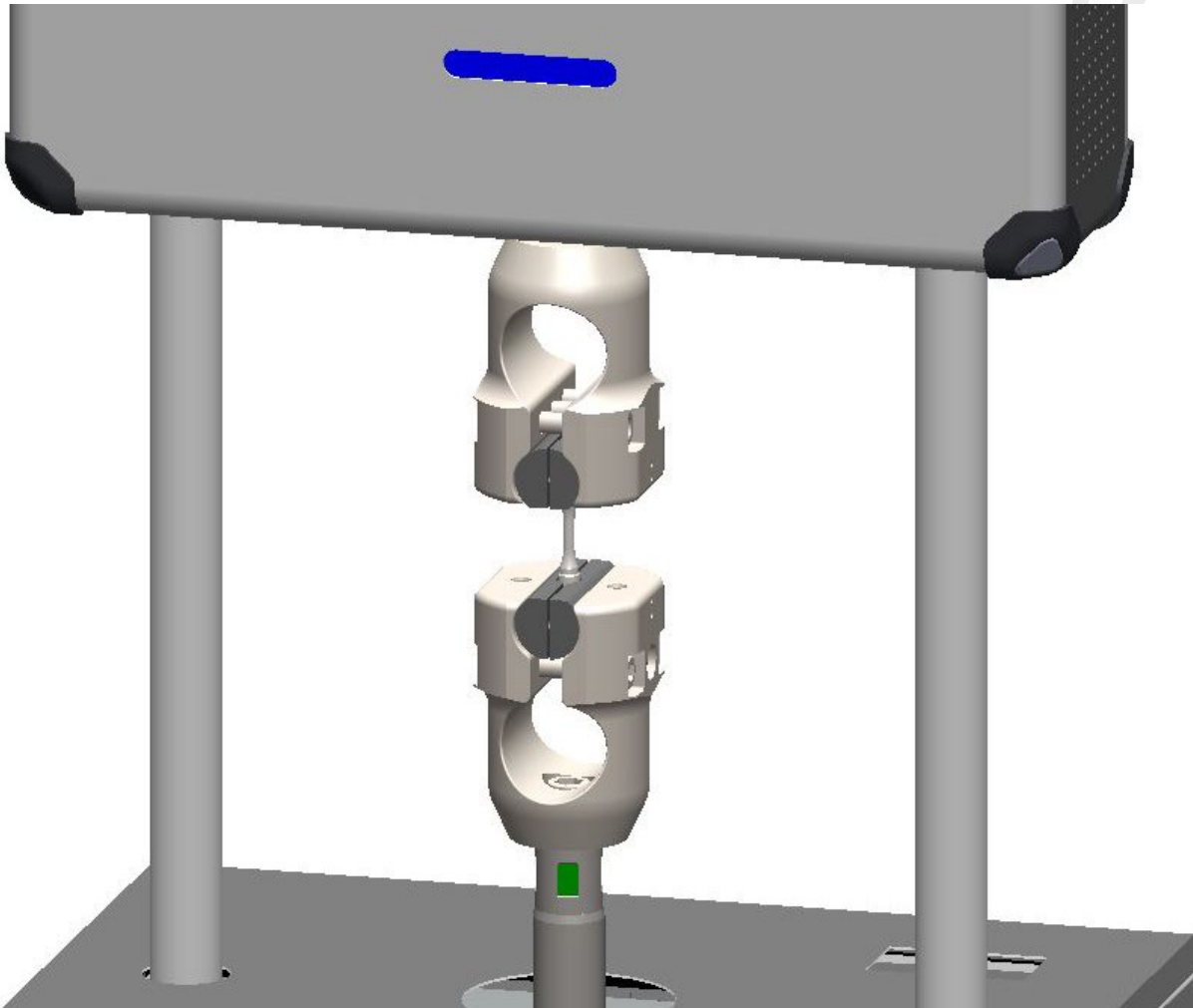


Figure 7: Universal grip with threaded Specimen.

SMOOTH SPECIMEN LCF TEST

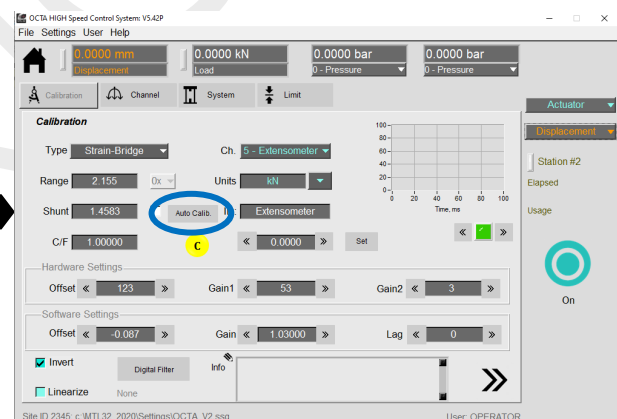
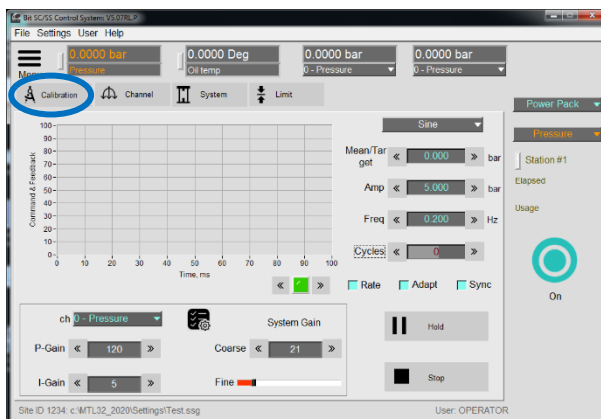
Follow steps 1 to 8 from threaded specimen test setup, to mount Smooth Sample

❖ Extensometer Mounting

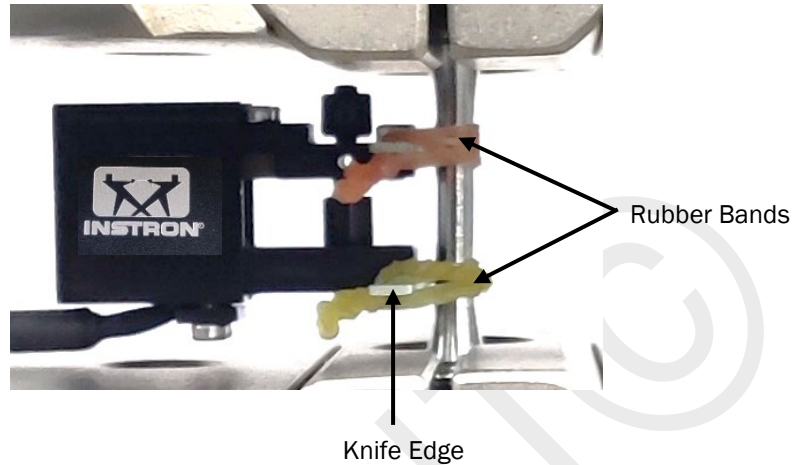
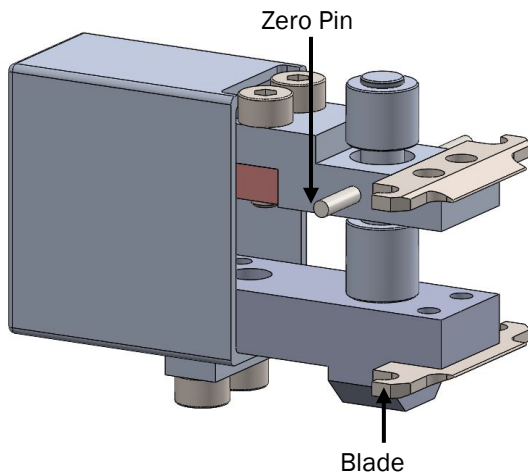
- Locate strain 1 channel (Load Frame) & connect extensometer to it as shown below



- Before mounting the extensometer, **ensure that the control mode is not in Extensometer (select stroke)** & it is recommended to auto calibrate it.
- To auto calibrate, go to MTL32 “Calibration” page & click on “Auto calibrate” button & wait for 30-45s for auto calibration to complete



- Without removing the zero pin mount the extensometer on the specimen by rubber band, so that the blades coincide with the lines marked on specimen. Mount rubber band gently, do not over tight



- Auto calibrate again & remove the zero pin slowly so that the blades position is not disturbed. Ensure that the extensometer read out on system is close to zero, if not remove the extensometer and mount it again
- Set the Extensometer limits in the limit panel



*Zero pin should be inserted while mounting/removing the extensometer to/from the specimen
Zero pin should be removed only when conducting a test.*

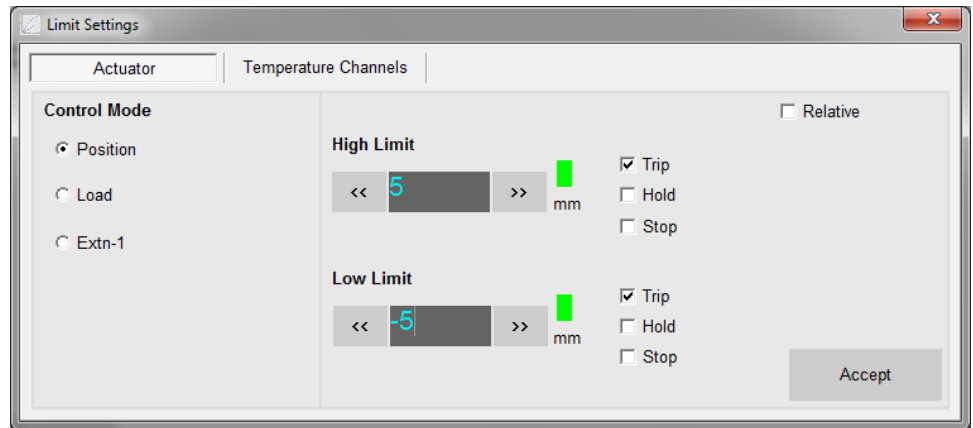
9.6.4 Run a test

- Double click on the Application software **LCF** icon

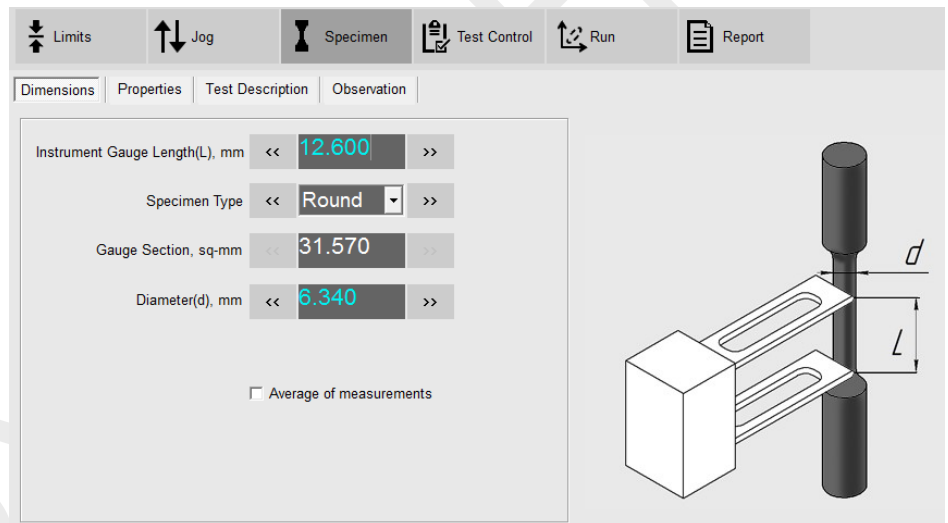


- Click “Open” button to open an existing Settings file. The results will be logged into this file without overwriting existing results
- To create a new settings/result file, click on “Save As” button

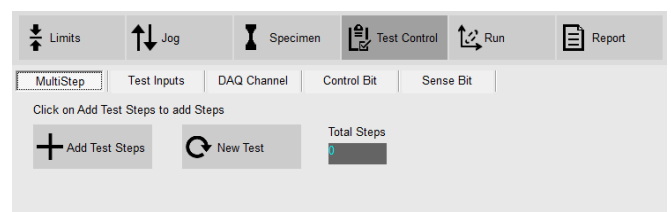
- Go to “Limits” tab, set high & low limits of Position, Load & Extensometer depending on the current readout



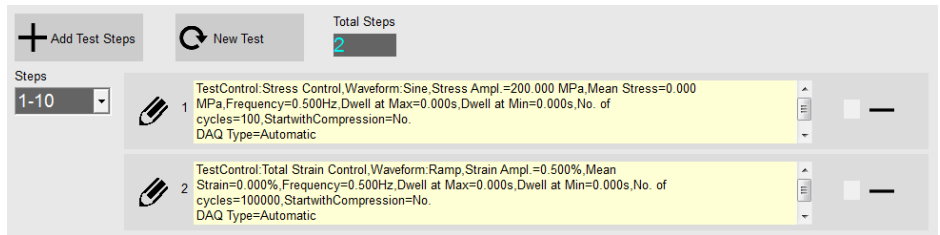
- Select appropriate action to be taken if limits are hit
- Click Accept button to set the limits
- Go to specimen page



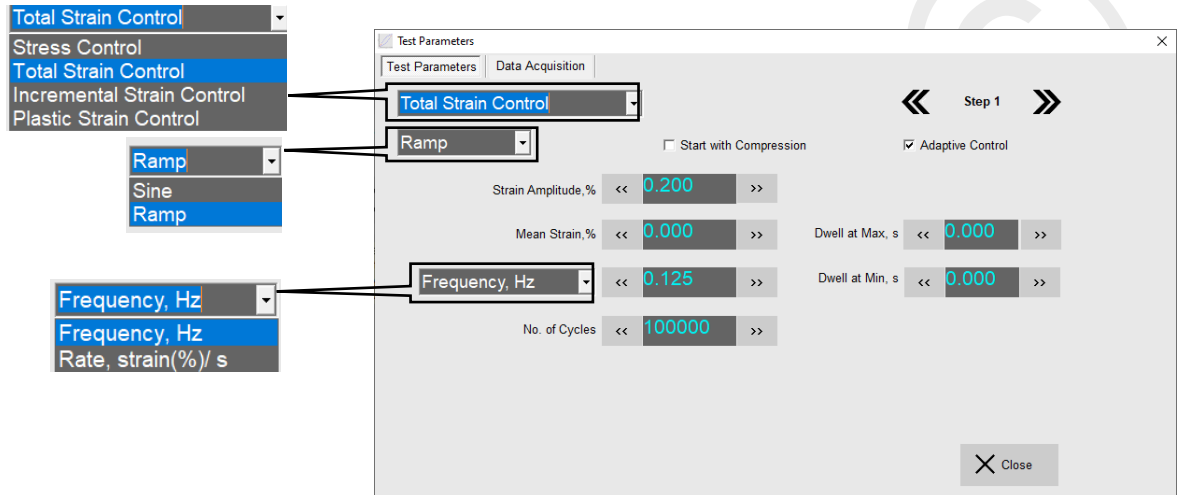
- Under “Dimensions” tab select specimen type from drop down menu & enter “Gage Length”, “Gage Section” & “Specimen Diameter”
- Under “Properties” tab, enter material properties of the specimen
 - Modulus Window: For example, if Stress at start point of unloading area is 300MPa, then modulus window will be points having stress value between 100MPa to 290MPa, based on the values shown above.
 - Modulus (GPa): Enter value depending upon the material
 - Poisson’s Ratio: Remains constant for metal sample. Other than metal sample user must enter the known value.
- Under “Test Description” tab, enter details which will be displayed in reports generated in post processing application
- Go to “Test Control” page



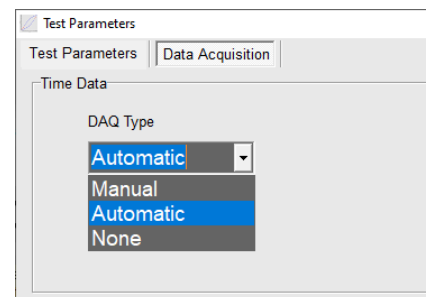
- Click on “Add Test Steps” to add one step at a time. Maximum limit is 100. Alternatively, enter required number of steps under “Total Steps”



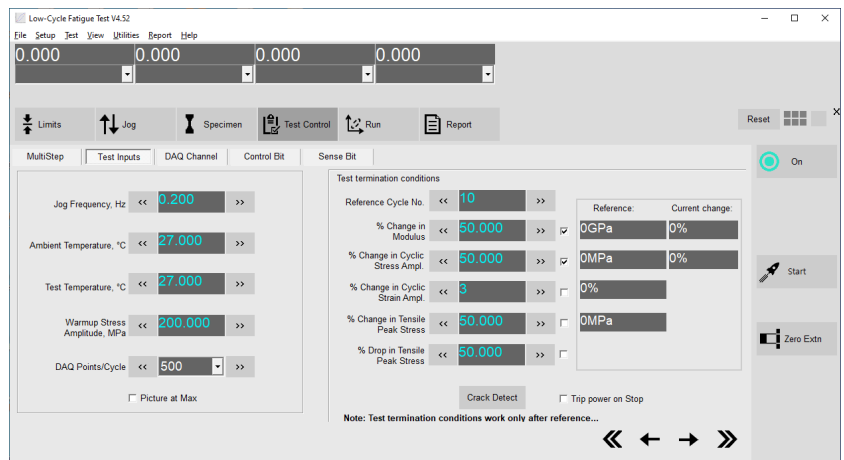
- Click on edit  button to open “Test Parameters” panel



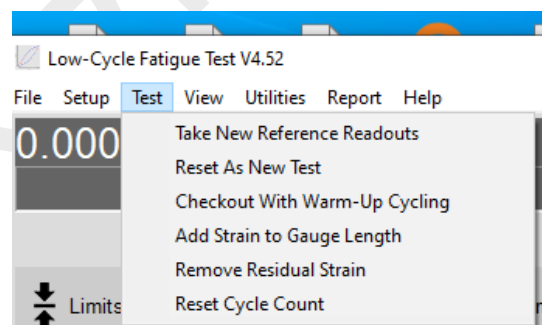
- Select type of control, wave form & rate of cycling from the drop-down menu
- Cycling will start from maximum amplitude by default. If “Start with Compression” parameter is enabled, then cycling will start from minimum amplitude
- Enter required Amplitude and Mean depending on the control.
- Enter number of cycles to run under selected control.
- Enter dwell parameters if required.
- Adaptive control, if enabled, will only be applied on non-dwell sections. E.g., if dwell at max is 10s, then adaptive control will be applied only for minimum and vice versa if dwell at min is set. If dwell is used for both sides, then adaptive control option will not be available.
- Under “Data Acquisition”, select type of recording
 - Manual: “Record” button will be displayed to start/stop recording anytime
 - Automatic: Enables recording automatically at the test start
 - None: No recording



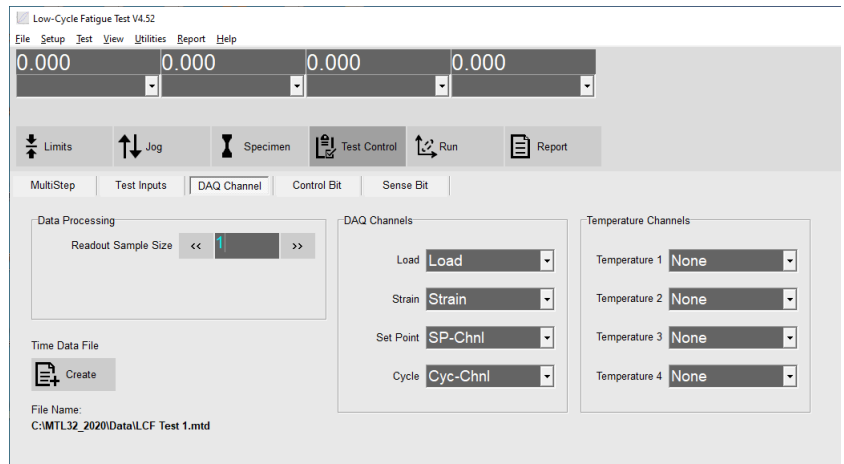
- Under “Test Inputs”, enter “Test termination conditions”



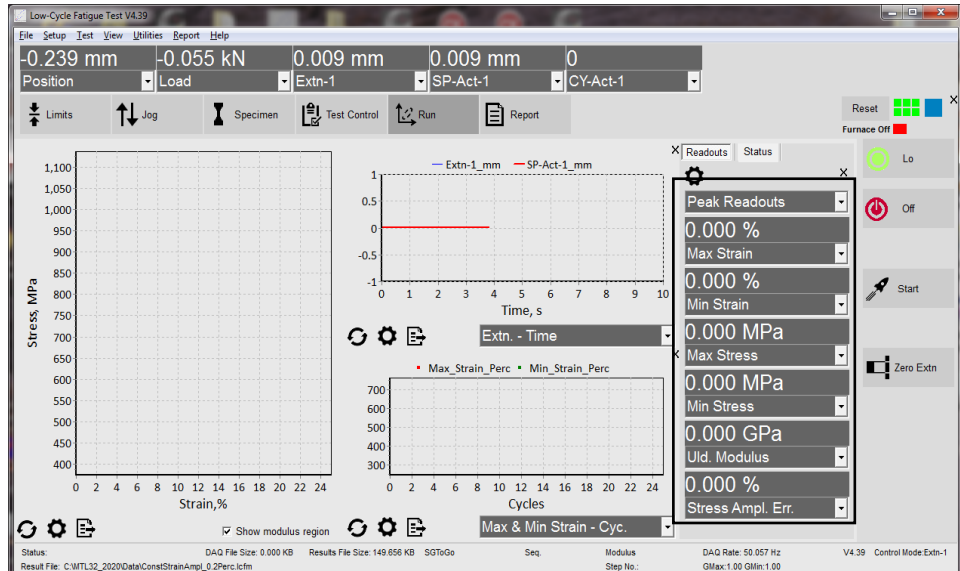
- To use these conditions, enter a valid Reference Cycle No. i.e., within the cycles of the test.
- Enable required terminating condition(s)
- To switch off power on termination, check “Trip power on Stop” option
- References and current change in % will be displayed across respective condition. For Peak Stress, there are 2 options. One being change and the other being drop. Use only one of them according to requirement
- To change references during test, click on “Take New Reference Readouts” option



- Under “DAQ channel”, Readout Sample Size: To get averaged results of multiple cycles, change the default value 1. Eg if 5 is entered. Then, the results are averaged after every 5 cycles and recorded into the data file



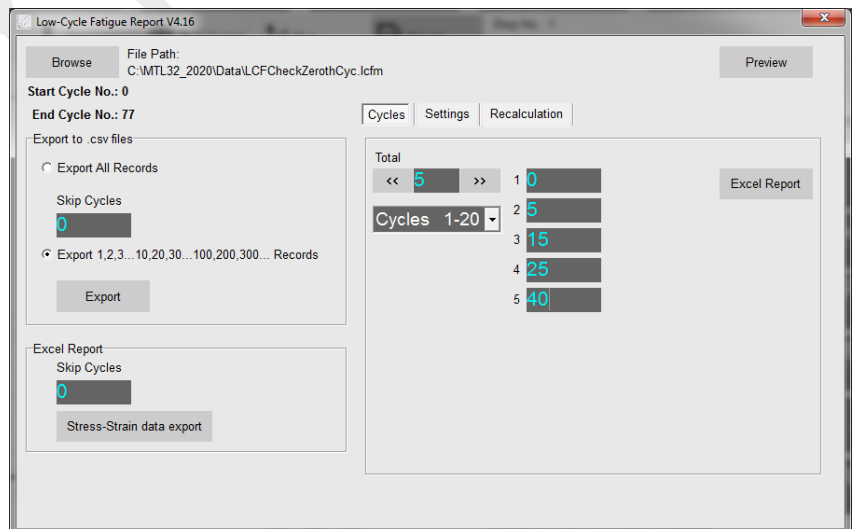
- Time Data File: Create a file (.mtd) to record log channel data.
- Select respective DAQ channels
- Go to “Run” page & select required readouts to be displayed



- Use “Zero Extn” button to reset the Extensometer channel to zero. Avoid using this unless necessary
- Check all test inputs and channel assignment before the start of the test.
- Click on Start button to start the test. The test can be stopped at any time by clicking on “Stop” button

9.6.5 Exporting Data

- Go to “Report” page, by default this panel opens with latest test file (Click on *Browse* to select different)



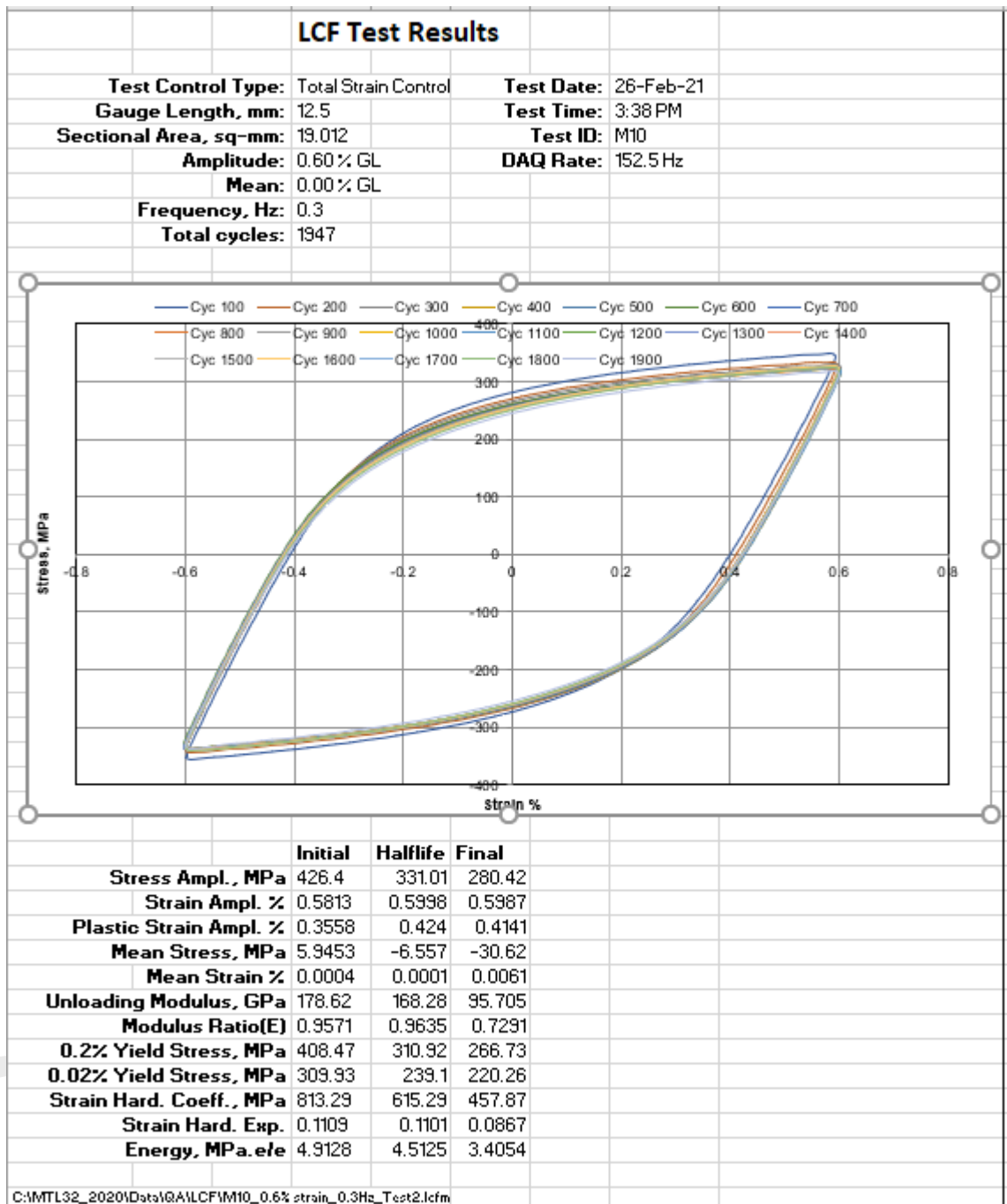
- Export to .csv files: To get results of all cycles, use Export All Records option with Skip Cycles as 0. Use Skip Cycles to reduce number of cycles. Eg, if it is 1, then cycles exported would be 0,2,4,6,8....
- To further reduce cycles, use Export 1, 2, 3...10, 20, 30...100, 200, 300...Records option. Say there are 500 cycles, then only 1, 2, 3, 10, 20, 30, 100, 200, 300 cycles are exported.
- Excel Report: Use this option to export stress-strain data of all the cycles in a single excel file.
- Use Skip Cycles option to evenly reduce the number of cycles in report.
- Excel report of selected cycles:
- Under “Cycles” tab, enter total number of cycles required. A max of 50 cycles can be entered, depending on the cycles available in the data file.

- Enter cycle numbers in ascending order. Refer to the Start and End Cycle number. Further, use Preview option to view Stress-Strain curve and results per cycle.
- At a time, a max of 20 cycles can be shown. If number of cycles exceed 20, use the drop-down menu to view or edit the cycles.
- Under “Settings” Set the Excel format depending on the excel version available in the system.

	From	To	Interval
☒	0	100	10
☒	100	500	50
☒	500	1000	100
☒	1000	5000	500
☒	5000	10000	1000

Cycles at Intervals:

- Use this to export cycle ranges at required intervals. For example, 1-100 at interval 1, 100-1000 at interval 100, 1000 to 10000 at interval 1000, and so on.
- Enable the options required and enter From, To and Interval according to the cycles available in the data file.
- Crack Initiation cycle, Ni and No. of Cycles around Ni: To get cycles around Ni, enter Ni and the required number around it. Say, Ni is 300 and to get cycles from 298 to 302, enter No. of Cycles around Ni as 2.
- No. of last cycles: To get cycles at the end, enter required number well within the available cycle range.
- Click on Export to CSV to get the Stress-Strain data of the cycle intervals in a CSV file.
- Click Excel Report button under Cycles section to get report in excel file.
- Shown below is a sample of single report sheet in excel.



Note: If no other test is being conducted, its recommended to Turn OFF/Shut down the following devices

- System hydraulic pressure (In application)
- LCF application
- MTL32 basic software
- Controller
- Computer
- UPS
- UPS input
- Selector Switch
- Lastly HPS